

Venndir Book

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Contents

Preface	5
Does the world need another Venn tool?	5
Supporting Tools	7
1 What is a Venn diagram?	9
1.1 Problem Statement	10
2 Data Import	13
2.1 Common Data Input Types	14
2.2 Specialized Input Types	31
3 Venndir Basics	37
3.1 Default Venndir	38
3.2 Venn Set Colors	43
3.3 Venndir Labels	48
3.4 Item Labels	58
3.5 Fonts and Font Sizes	67
3.6 Venn Legends	76
3.7 Figure Scale and Position	90
4 Deeper Customizations	97
4.1 Venndir Borders	97
4.2 Modify Venn Overlaps	101
4.3 Customize Any Label	104
4.4 Automatic Text Contrast	105
4.5 Highlight Venn Overlaps	108
4.6 Rotate the Venndir	110
4.7 Nudge the Legend	111
4.8 Hidden Overlaps	112
4.9 Nudge specific labels	115
4.10 Venndir Markdown Support	117
4.11 Patchwork with Venndir	122
4.12 Custom Signs	123

4.13 Venndir Graphics Objects	131
5 Venndir Gallery	139
5.1 Figure Boosting	139
5.2 Venndir Case Studies	148
6 What's New by Version	153
7 Glossary	155

Preface

This book describes the `venndir` R package, intended for the display of Venn diagrams with directionality using R. These visualizations reflect many years developing Venn diagrams, refining features like directional arrows, and integrating the thoughtful suggestions and shared wisdom of many brilliant colleagues.

Does the world need another Venn tool?

You may think the world has enough Venn. Venndir boldly suggests there is room for more, with arrows, labels, and points to make.

Venndir is customizable

In fact, Venndir enables some of the most customizable Venn and Euler diagrams available, short of using a graphics design tool. Set labels, fonts, borders, colors, placement, highlights, and item labels, are all customizable when needed. Text labels even support Markdown, and with robust Unicode character rendering.

That said, Venndir aims to deliver publication-ready figures, as closely as possible, using default settings. But still, they're still just Venn diagrams.



Directionality is important

This book focuses on the techniques used to create Venn diagrams, including the use of directionality where appropriate. The need to indicate directionality depends on the questions asked and the conclusions one hopes to draw from the figure.

5.2.1 Seehawer Kmt2 Gene Venns Examples such as Section 5 illustrate a common pattern: They tested only the items showing **concordance**, agreement in direction, then interpreted the findings without evaluating discordance at all.

Without also evaluating the discordance, they lacked the proper context to interpret the results.

Conclusions drawn when testing only concordance, without also assessing discordance, are inherently biased, with limited validity of interpretation.

In the study of genes and their effects, change is more than just up or down -- it can signify sickness or health, disease or cure, even life or death.

The arrows aren't just decorative, they're insightful

It is quietly satisfying when Venndir illustrates that although two sets may overlap, they are moving in different directions -- revealing inconsistencies that may otherwise be missed. Clarity doesn't just come from what overlaps, but from how things move together.

Visualize with purpose. Annotate with direction.

One of the most challenging components in Venndir is placing the labels. Who knew?

Venndir automates the job of placing labels inside or outside the Venn circles, then allows you to nudge, style, highlight, and customize.

Together with **marquee** (Pedersen and Mitáš, 2025), Venndir provides robust support for custom fonts across all graphics devices, with Unicode character substitution, Markdown and inline styling, and ability to embed images or other graphics.

Venndir figures use **grid** graphics, are compatible with **patchwork** (Pedersen, 2024) for multi-panel figures, and can be customized down to the level of **grid** graphics objects (grobs).

Put ‘*What are those?*’ to rest, one label at a time.

One of the most unexpectedly fun features in Venndir is the ability to display item names inside the figure. The best data visualizations anticipate and answer that "next question."

For a Venn diagram it is most often:

‘What are those?’

Inspired by curiosity, designed for clarity, sharpened by use.

Venndir exists mostly because I thoroughly enjoyed creating it. It touches many interesting aspects of algorithm development, problem solving, data visualization design, and the practical driver of publication. It has received substantial feedback, and I always love listening to new ideas and suggestions.

Yes, there are a lot of labels.

Yes, the figures can become very busy.

And yes, I still think the results are useful and necessary.

All that said, Venndir with directionality (see “each” in **Overlap Type**) is a **lot** to look at, and may be most useful *during analysis*. The final published figure may or may not be simplified for visual clarity.

My philosophy is to make the figure useful for me before anyone else, then for my colleagues next, finally polishing it to present to others. Venndir tries to accomplish that in one step, but failing that it tries to make it useful for your.

The typical Venndir figure often follows the same progression: complicated details for me, optimized improvements for colleagues, then clear and simplified for others.

This book roughly follow the same progression:

- Introduction
- Data Import
- Venndir Basics
- Advanced Customization
- Venn Figures and Case Studies

Supporting Tools

As with most **R packages**, none of this capability would be possible without the many foundational libraries upon which Venndir was built.

Graphics

The graphics system used by Venndir is **grid**, a core R package that empowers reliable calculation of figure units, metrics, font placement, and dynamic updating of a figure upon resizing.

The supporting extension **gridGeometry** (Murrell and Wong, 2025) enables inner and outer borders which will be discussed at length in the advanced

customization topics.

Similarly **gridExtra** (Auguie, 2017) brings the ability to create a nice table legend, which has become an essential default feature of Venndir.

Venndir manipulates polygons primarily through **polycclip** (Johnson and Baddeley, 2024), which enables geometry calculations including **intersection**, **union**, **subtraction**, and **buffer region**.

Proportional **Euler diagrams** are made possible by **eulerr** (Larsson, 2024), which performs the hard work of predicting the best available alignment between Venn counts and geometry.

Fonts and Labels

A large part of Venndir focuses on labeling, which necessarily involves detailed control over fonts, glyphs, and assembling multiple label components together into a group.

The **marquee** (Pedersen and Mitáš, 2025) package provides methods to place text, replacing previous methods in part due to its use of **systemfonts** (Pedersen et al., 2025) for font and glyph substitutions. Without these features, Venndir encountered too many scenarios where the font, symbol, or glyph would be incorrect due to the specific computing architecture, character locale, or output graphics device. Most such scenarios are resolved by using **systemfonts**, while **marquee** adds substantial flexibility with its support of markdown formatting.

Venndir count labels are often comprised of several pieces, and **gtable** (Wickham and Pedersen, 2024) enables grouping them into one graphical object.

Venndir is a refined set of methods that fuses these amazing tools together into one uniquely powerful package.

Chapter 1

What is a Venn diagram?

A Venn diagram is a deceptively simple figure, intended to compare two or more sets in a visually intuitive way.

Each set is represented by a circle, drawn with a label beside it. Two circles are shown below for sets A and B, and these circles are drawn so they partially overlap.

The overlapping region represents the *intersection* of sets A and B. Using mathematical symbols, the intersect is written $A \cap B$, but we use **A&B** for convenience since the symbol $\cap$ does not exist on most computer keyboards.

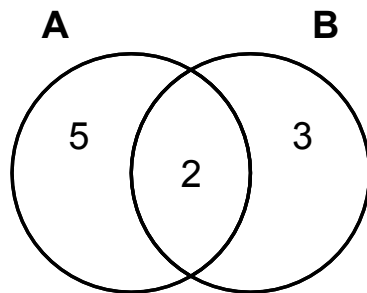


Figure 1.1. A simple Venn diagram is shown with two sets: A and B.

The numbers in the Venn diagram represent the number of items in each logical grouping:

- '5' is the number of items in 'A' and 'not B'.
- '3' is the number of items in 'B' and 'not A'.
- '2' is the number of items in 'A' and 'B'.

1.1 Problem Statement

"What problem are you trying to solve?" -- Lazar Arulnayagam

When comparing biological effects on genes in two experiments, two important questions arise:

1. Are the affected genes shared?
2. Are the changes in direction **concordant**?

A standard Venn diagram can address shared gene membership, but without directionality they fall short of answering question 2.

Venndir proposes a simple yet powerful extension to Venn diagrams: placing **signed counts** beside the **main counts**. The signed counts summarize directionality, indicating whether directions of change were up or down. Direction is indicated using arrows and color cues, red for changes up, and blue for changes down.

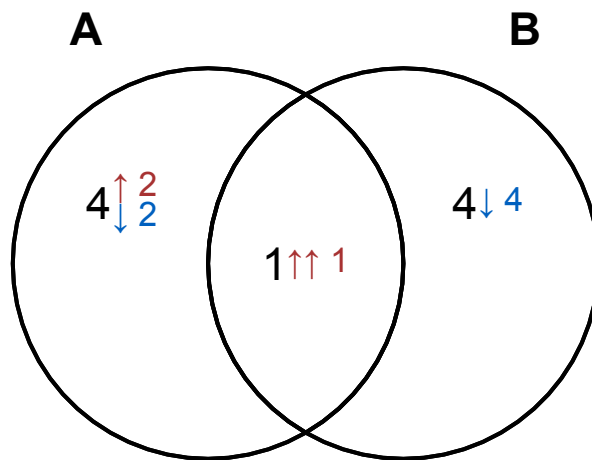


Figure 1.2. Venndir introduces an extension to a Venn diagram, with addition of directional counts beside each Venn overlap count. The direction is indicated by arrows up and down, and colors red and blue, respectively.

For the simple example shown in Figure 1.2, the numbers are described as follows:

- There are 4 items in 'A' and 'not B'.
 - 2 of these items are UP.
 - 2 of these items are DOWN.
- There are 4 items in 'B' and 'not A'.

- All **4** of these items are DOWN.
- There is **1** item shared by 'A' and 'B'.
 - This **1** item is UP.

Venndir also strives to answer *the very next question*:
"What are those items?"

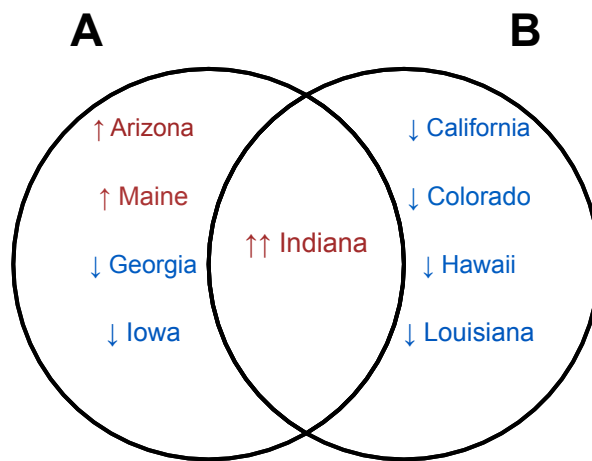


Figure 1.3. Venndir displays the items inside the diagram.

Chapter 2

Data Import

The first key step in using Venndir is to convert data into a recognized format. The two native formats are **List of sets** and **List of signed sets**.

There are several formats available:

1. **List of sets**, referred to as a **setlist**
 - a list of character item vectors.
 - sets are represented by `names(setlist)`
2. **List of signed sets**, referred to as a signed **setlist**
 - a list of *signed* numeric vectors.
 - signed vectors use values -1 or 1, and store items in `names()`.
 - sets are represented by `names(setlist)`
3. **Incidence matrix**
 - a numeric matrix with items as `rownames()`, sets as `colnames()`.
 - items are indicated by the value 1 in the appropriate row and column.
4. **Signed incidence matrix**
 - a numeric matrix with items as `rownames()`, sets as `colnames()`.
 - items are indicated by -1 or 1 in the appropriate row and column.
5. **Overlap counts**
 - a numeric vector with the number of items in each overlap.
 - the overlap name is stored in `names()`, in format: 'A', 'B', 'A&B'.
6. **Signed overlap counts**
 - a list of numeric vectors, named by overlap set.
 - each overlap is also a list named by the directionality.
7. **Other formats**
 - **Items**
 - **Overlap list**
 - **Signed overlap list**
 - **List of data frames**

8. Specialized formats

- `limma`
- `DESeq2`

2.1 Common Data Input Types

2.1.1 List of sets

The simplest input is a List of `sets`, where each set is represented as an R vector of items.

```
setlist <- make_venn_test()
str(setlist)
```

```
## List of 3
## $ set_A: chr [1:35] "item_014" "item_195" "item_170" "item_050" ...
## $ set_B: chr [1:18] "item_155" "item_188" "item_053" "item_135" ...
## $ set_C: chr [1:79] "item_041" "item_175" "item_090" "item_060" ...
```

Each vector is not considered 'signed', because each vector has no `names()`.

This `setlist` can be used directly with `venndir()`. In Figure 2.1 there are no signed counts displayed.

```
venndir(setlist)
```

2.1.2 List of signed sets

The next common input is a list of `signed sets`, represented as `numeric` vectors with items stored as `names()`.

An example is shown below, showing only the first six elements of each vector. The vector names are "item_014", "item_195", while the values are '-1', '1'.

```
setlist <- make_venn_test(do_signed=TRUE)
lapply(setlist, head, 6)
```

```
## $set_A
## item_014 item_195 item_170 item_050 item_118 item_043
##      -1      -1        1        1       -1       -1
##
## $set_B
## item_155 item_188 item_053 item_135 item_198 item_200
##      -1        1        1       -1        1        1
##
## $set_C
```

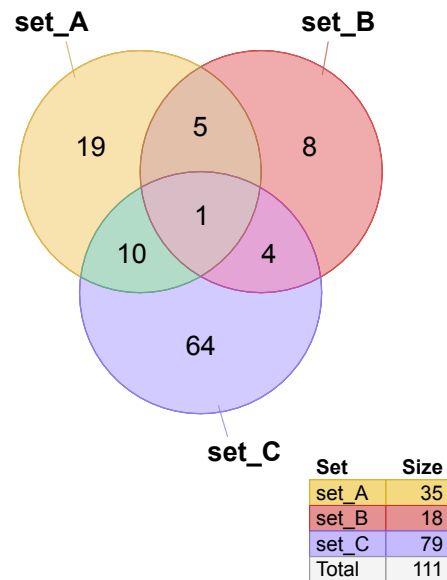


Figure 2.1. Default Venn diagram with three sets.

```
## item_041 item_175 item_090 item_060 item_016 item_116
##          1          1          -1          -1          -1          -1
```

Figure 2.2 illustrates the output from `venndir()`, where **signed counts** are displayed using `overlap_type='concordance'` by default for a **signed setlist**.

```
venndir(setlist)
```

The output is described below:

- The region "set_A" contains 19 items that are not present in any other set for this Venn diagram.
 - 10 of these 19 items are "up", indicated by the red label which includes the up arrow ↑ 10.
 - 9 of these 19 items are "down", indicated by the blue label which includes the down arrow ↓ 9.
- The region of overlap between 'set_A' and 'set_B' contains 5 items which are only present in these two sets, and not present in 'set_C'.
 - 3 of these 5 items are "up" in both 'set_A' and 'set_B', indicated by the red label with two up arrows: ↑↑ 3.
 - 2 of these 5 items are discordant in direction, which means the sign in 'set_A' disagrees with 'set_B'. The label is indicated in grey color and large 'X' X2. The purpose of using one 'X' is to avoid describing all possible combinations of "up" and "down".

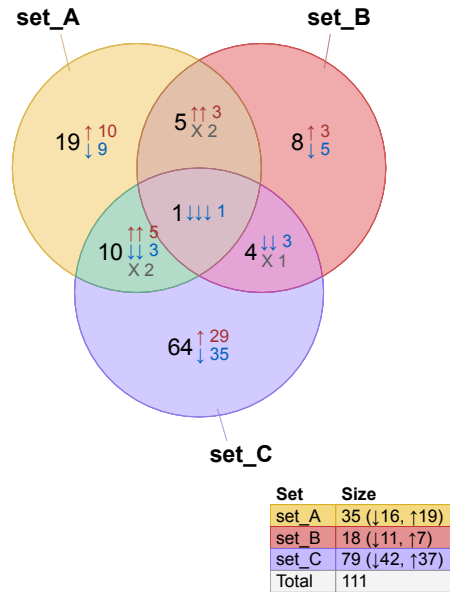


Figure 2.2. Default Venn diagram with three signed sets. The overlap counts are each tabulated by directional sign.

The default summary for a **signed setlist** is `overlap_type="concordance"`, which summarizes directional discordance in one category 'X', and tabulates counts in each subset that involves only one direction.

For other approaches to summarize directional counts, see the summary in Table 3.2, in the section **Overlap Type**.

2.1.3 Incidence matrix

An **incidence matrix** (im) is another common input format, a **matrix** whose **rownames** are items, and **colnames** are sets. Any non-zero, non-empty value in the matrix indicates the item (row) exists in the set (column).

```
setlist <- make_venn_test()
im <- list2im_opt(setlist)
head(im, 10)
```

```
##          set_A set_B set_C
## item_014     1     0     0
## item_195     1     0     1
## item_170     1     0     0
## item_050     1     0     1
```

```
## item_118    1    0    1
## item_043    1    0    0
## item_200    1    1    0
## item_196    1    0    1
## item_153    1    1    1
## item_090    1    0    1
```

Figure 2.3 shows the Venn diagram created by `venndir()` accepts an incidence matrix as input data. The input data is converted to a `setlist` within the function.

```
venndir(im)
```

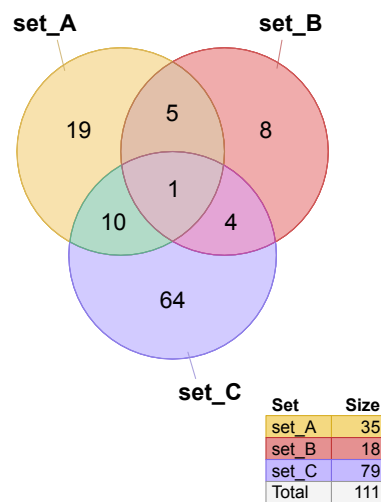


Figure 2.3. Default Venn diagram with three sets, using an incidence matrix as input data.

When the `incidence matrix` only contains positive values, it is assumed to be non-directional. This assumption can be changed by using the argument `overlap_type`.

2.1.4 Signed incidence matrix

A `signed incidence matrix` is similar to an incidence matrix, except that it may include positive and negative values.

```
setlist <- make_venn_test(do_signed=TRUE)
im <- list2im_value(setlist)
head(im, 10)
```

```
##          set_A set_B set_C
## item_014    -1     0     0
## item_195    -1     0     1
## item_170     1     0     0
## item_050     1     0     1
## item_118    -1     0    -1
## item_043    -1     0     0
## item_200     1     1     0
## item_196    -1     0    -1
## item_153    -1    -1    -1
## item_090    -1     0    -1
```

Figure 2.4 shows the signed Venn diagram created by `venndir()` also accepts a **signed incidence matrix** as input data, and subsequently displays signed counts.

```
venndir(im)
```

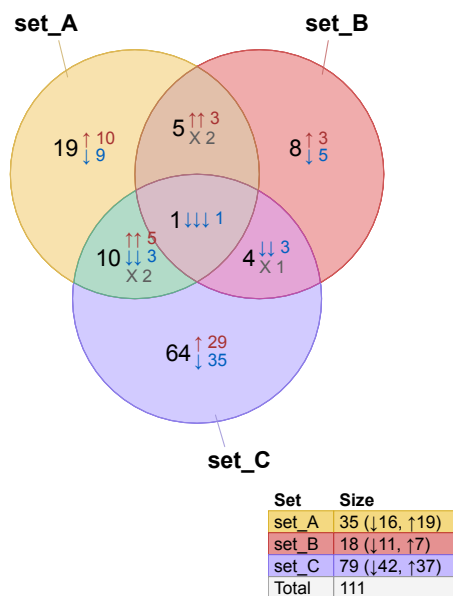


Figure 2.4. Default Venn diagram with three sets, using a signed incidence matrix as input data.

2.1.5 Overlap counts

If the **Venn overlap** counts are already known, they can also be used to re-create the corresponding Venn diagram.

The function `counts2setlist()` accepts Venn counts, and returns the corresponding `setlist`. The counts should be named by the **Venn overlap**, by using the name of each set involved, separated by ampersand '&'. The example below uses sets "A" and "B", and the corresponding overlap between 'A' and 'B' is named 'A&B'.

Note that the overlap name should be defined in quotes in R code, so the ampersand '&' is stored properly.

```
counts <- c(
  "A"=12,
  "B"=9,
  "B&A"=15)
setlist <- counts2setlist(counts)
str(setlist)
```

```
## List of 2
## $ A: chr [1:27] "A_1" "A_2" "A_3" "A_4" ...
## $ B: chr [1:24] "B_1" "B_2" "B_3" "B_4" ...
```

Figure 2.5 shows the output of `venndir()` using this `setlist` as input.

```
venndir(setlist)
```

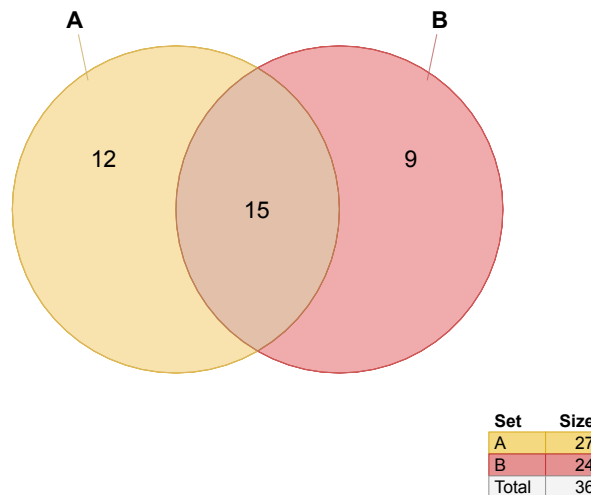


Figure 2.5. Venn diagram created by using overlap counts as input data.

Tip:

When starting with overlap counts, it is recommended to define set names with single characters, such as 'A', 'B', and 'C'.

- The set names can be adjusted afterwards by editing `names(setlist)`.
- However, the preferred approach is to use `venndir()` arguments `setlist_labels` and `legend_labels`, also described in [Custom legend labels](#).
- Figure 2.6 illustrates this process.

The complete combination of sets and overlaps can be defined by calling `make_venn_combn_df()` using a vector of set names.

For example `rownames(make_venn_combn_df(LETTERS[1:3]))` produces the following:

'A', 'B', 'C', 'A&B', 'A&C', 'B&C', 'A&B&C'

```
venndir(setlist,
  setlist_labels=c("Set A:\ncontrol state",
    "Set B:\ntest state"),
  legend_labels=c("A: control state",
    "B: test state"))
```

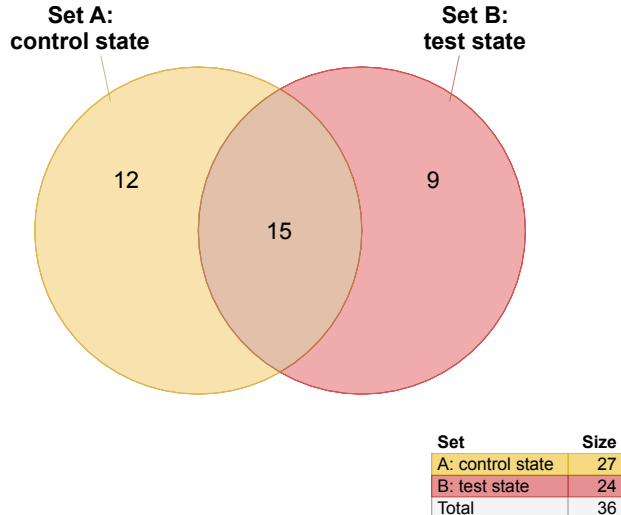


Figure 2.6. Venn diagram using custom set names using `setlist_labels` and `legend_labels`.

2.1.6 Signed overlap counts

Similar to providing overlap counts as above, this approach defines counts for each directional overlap, using `signed_counts2setlist()`. This input is quite complex, also the least common.

The input is a `list` of integer count vectors. Each vector is named by the overlap, for example "A" or "A&B". Each vector is named by the direction, delimited by underscore "_", for example "1" for 'up', or "1_-1" for 'up_down'.

Notice the format of the input data:

```
signed_counts <- list(
  "A"=c(
    "1"=80,
    "-1"=95),
  "B"=c(
    "1"=15,
    "-1"=30),
  "A&B"=c(
    "1_1"=100,
    "1_-1"=3,
    "-1_1"=4,
    "-1_-1"=125))
signed_counts
```

```
## $A
## 1 -1
## 80 95
##
## $B
## 1 -1
## 15 30
##
## $`A&B`
## 1_1 1_-1 -1_1 -1_-1
## 100 3 4 125
```

This input format is complicated for 2-way data, and certainly even more complicated for 3-way data. However, sometimes it is the most practical way to produce a given figure.

Counts are converted to a signed setlist with `signed_counts2setlist()`. Item names are generated only to create the `setlist` and are not otherwise useful.

```
setlist <- signed_counts2setlist(signed_counts)
lapply(setlist, jamba::middle, 5)
```

```
## $A
##      A_1_1      A_-1_22      A&B_1_1_29      A&B_-1_-1_24      A&B_-1_-1_125
##      "1"        "-1"        "1"            "-1"            "-1"
##
## $B
##      B_1_1      A&B_1_1_25      A&B_1_1_94      A&B_-1_-1_56      A&B_-1_-1_125
##      "1"        "1"            "1"            "-1"            "-1"
```

Figure 2.7 shows the `setlist` visualized with `venndir()`.

```
venndir(setlist, overlap_type="each")
```

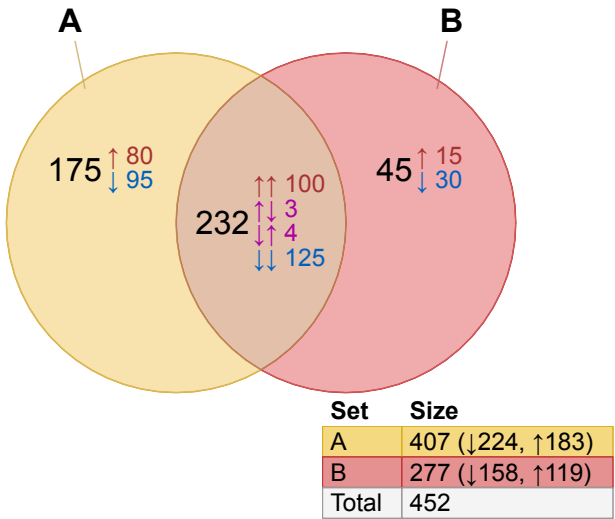


Figure 2.7. Venndir showing signed counts summarized using `overlap_type="each"`, which includes counts for each combination of signs.

2.1.7 Overlap list

This approach is conceptually similar to **Overlap counts** which starts with the Venn counts as input, except in this case instead of supplying the integer count in each **Venn overlap**, the data contains the actual items.

Consider a simple example:

- two sets: 'A' and 'B'
- three total items, one only present in 'A', one unique to 'B', and one shared by both groups which is assigned to 'A&B'.

```
overlaplist <- list(
  A=c("Christina"),
  B=c("James"),
  "A&B"=c("Jillian", "Zander")
)
str(overlaplist)
```

```
## List of 3
## $ A : chr "Christina"
## $ B : chr "James"
## $ A&B: chr [1:2] "Jillian" "Zander"
```

This `overlaplist` is converted into `setlist` using `overlaplist2setlist()`, which can be input to `venndir()`.

Figure 2.8 shows the default Venn diagram (left), and a variation which displays the item labels (right) using argument `show_labels="Ni"`. The `show_labels` argument is described in **Label Content**.

```
setlist <- overlaplist2setlist(overlaplist)
venndir(setlist)
venndir(setlist, keep_item_order=TRUE,
  item_cex_factor=c(0.6, 0.7, 0.9),
  show_labels="Ni")
```

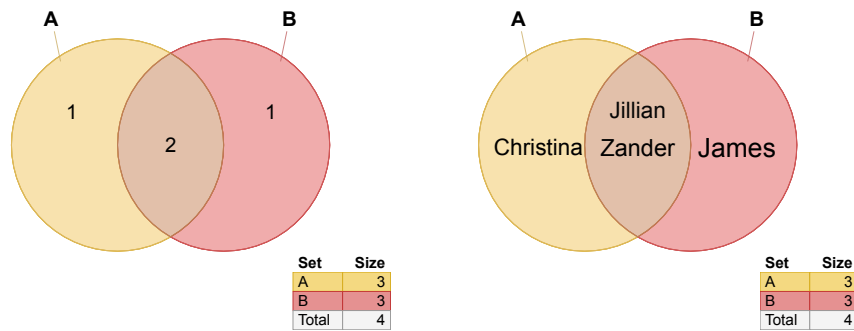
The argument `item_cex_factor` is used to adjust the item label font size, and is described in **Item Labels**.

2.1.8 Signed overlap list

A powerful but complex import option is a signed overlap list, similar to the previous section **Overlap list** with the addition of directional sign.

The format is similar to overlap list:

- list named by the **Venn overlap**: 'A', 'B', 'A&B'
- Each list element is also a `list`, named by the directional sign.



(a) Overlap counts.

(b) Overlap item labels.

Figure 2.8. Venn diagram after converting an overlap list to `setlist`, showing overlap counts (left), and overlap item labels (right).

- Each sign is defined using '1' or '-1' separated by a space.
- Signs involving one set: "1", "-1"
- Signs involving two sets: "1 1", "1 -1", etc.
- Signs involving three sets: "1 1 1", "1 -1 1", etc.

```
signed_overlaplist <- list(
  A=list(
    "-1"=c("Item_A1"),
    "1"=c("Item_A2")),
  B=list(
    "1"=c("Item_B1", "ItemB2"),
    "-1"=c("ItemB3")),
  "A&B"=list(
    "1 1"=c("Item_AB2"),
    "1 -1"=c("Item_AB4"),
    "-1 1"=c("Item_AB1"),
    "-1 -1"=c("Item_AB3"))
)
str(signed_overlaplist)
```

```
## List of 3
## $ A :List of 2
## ..$ -1: chr "Item_A1"
## ..$ 1 : chr "Item_A2"
## $ B :List of 2
```

```
## ..$ 1 : chr [1:2] "Item_B1" "ItemB2"
## ..$ -1: chr "ItemB3"
## $ A&B:List of 4
## ..$ 1 1 : chr "Item_AB2"
## ..$ 1 -1 : chr "Item_AB4"
## ..$ -1 1 : chr "Item_AB1"
## ..$ -1 -1: chr "Item_AB3"
```

The `signed_overlaplist` is converted to `setlist` using `overlaplist2setlist()`.

Figure 2.9 shows the resulting Venn diagram with item labels enabled. Notice the item labels include a directional arrow, and are colored by the sign.

```
setlist <- overlaplist2setlist(signed_overlaplist)
v <- venndir(setlist,
  show_labels="Ni",
  item_cex_factor=0.6,
  xyratio=1.5)
```

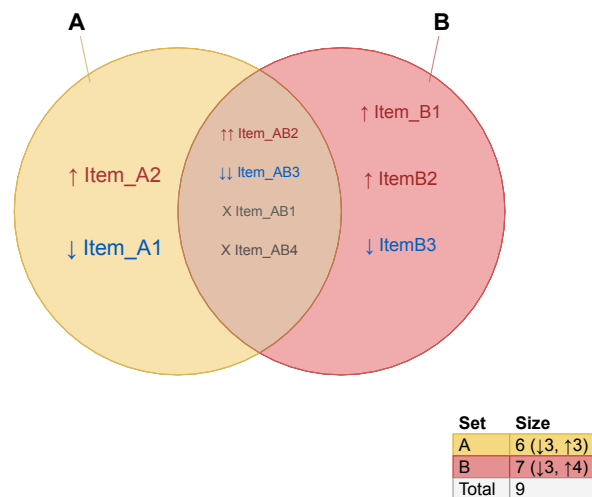


Figure 2.9. Venn diagram derived from a signed overlap list. The figure shows item labels which are colored by sign, and placed beside a directional arrow.

This input format can also be generated from the `Venndir` object itself using `overlaplist(v)`. However, it requires that `venndir()` was called with argument `overlap_type="each"` to preserve each sign. The default `overlap_type="concordance"` does not preserve signs for **discordant** overlaps.

2.1.8.1 Alternative signed overlap list

An alternative input format is shown below, which may be more convenient to produce in some circumstances.

The alternate format is shown below:

```
alt_signed_overlaplist <- list(
  A=c("Item_A1"="-1",
      "item_A2"="1"),
  B=c("Item_B1"="1",
      "ItemB2"="1",
      "ItemB3"="-1"),
  "A&B"=c("Item_AB2"="1 1",
          "Item_AB4"="1 -1",
          "Item_AB1"="-1 1",
          "Item_AB3"="-1 -1"))
alt_signed_overlaplist
```

```
## $A
## Item_A1 item_A2
##    "-1"      "1"
##
## $B
## Item_B1 ItemB2 ItemB3
##    "1"    "1"   "-1"
##
## $`A&B`
## Item_AB2 Item_AB4 Item_AB1 Item_AB3
##    "1 1"   "1 -1"  "-1 1"  "-1 -1"
```

First, the alternative format is converted to the expected format for `overlaplist2setlist()`. Then it is converted to `setlist`.

```
signed_overlaplist <- lapply(alt_signed_overlaplist, function(i){
  split(names(i), i)
})
setlist <- overlaplist2setlist(signed_overlaplist)
```

The resulting `setlist` can be used with `venndir()` as in Figure 2.9.

2.1.9 Items

A streamlined approach to convert items from a `vector` or `list` into `setlist` is provided by `venn_meme()`, as described [Venn Memes](#).

The process assumes the input data is provided in a specific order. The process is quite similar to that described in [Overlap list](#), except that this

process assumes the order matches the output of `make_venn_combn_df()`.

The assumptions:

- Input must be provided in the order defined by `make_venn_combn_df(n)` where `n` is the number of sets, or `make_venn_combn_df(x)` where `x` is a vector of set names.
- Input is provided without using overlap names, they are assigned based upon the length of the input data.
- Duplicate items are not permitted.

To illustrate the process, each example below uses a vector of character `LETTERS`, showing how the letters are populated into the Venn diagram.

Figure 2.10 illustrates the process by showing the output of `venn_meme()`, with each panel using a slightly longer vector of items.

- For 1 item: the output includes one set (not shown).
- For 2 or 3 items: A 2-way Venn diagram is created (a).
- For 3 to 7 items: A 3-way Venn diagram is created (b).
- For 8 to 15 items: A 4-way Venn diagram is created (c).
- For 16 to 31 items: A 5-way Venn diagram is created (d).

```
vm3 <- venn_meme(head(LETTERS, 3))
vm7 <- venn_meme(head(LETTERS, 7))
vm15 <- venn_meme(head(LETTERS, 15))
vm31 <- venn_meme(1:31)
```

Tip:

A useful by-product of calling `venn_meme()` is the creation of a `setlist` or `overlaplist`, without needing to provide all the **Venn overlap** labels upfront. For example, use `setlist(vm3)` to obtain the setlist, or `overlaplist(vm3)` to obtain the overlap list, created by the call to `venn_meme()`.

```
vm3 <- venn_meme(head(LETTERS, 3), do_plot=FALSE)
setlist(vm3)
```

```
## $A
## [1] "A" "C"
##
## $B
## [1] "B" "C"
```

```
overlaplist(vm3)
```

```
## $A
```

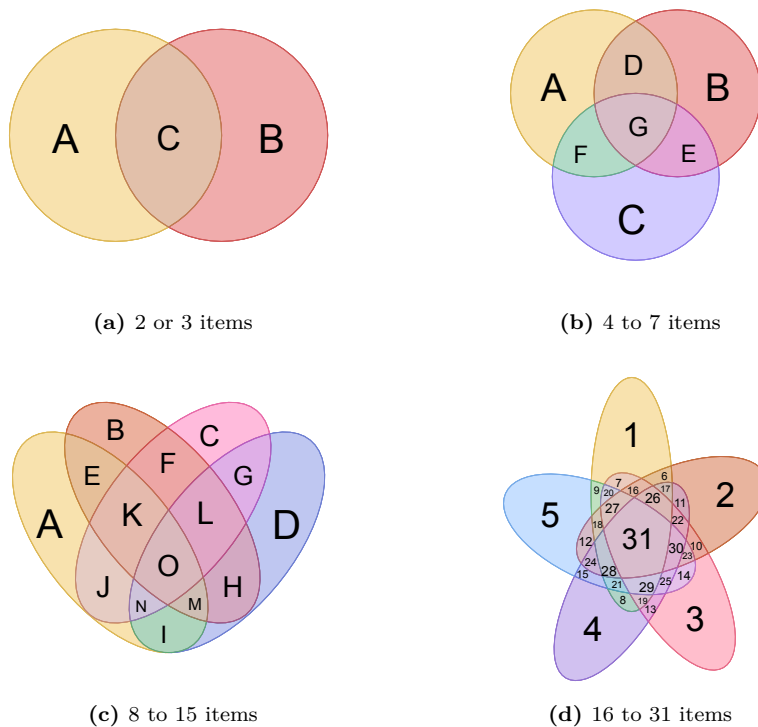


Figure 2.10. Four Venn diagrams showing the output from `venn_meme()` when providing a certain different number of items.

```
## [1] "A"
##
## $B
## [1] "B"
##
## $`A&B`
## [1] "C"
```

2.1.10 List of data frames

A common starting point is a `data.frame` for each set, with a column of numeric values, and column of item names. Very often this input is a "table of stats" which may also need to be filtered for statistical significance.

The general guidance is to filter each `data.frame`, then convert the result into a vector. The vector can either be un-signed, or signed.

1. **Setlist**: a `character` vector of items.

2. **Signed setlist**: vector using `sign()`, named by item.

To illustrate, consider the test data generated below, as two `data.frame` objects with P-values and log2 fold changes.

```
# define some test data
set.seed(123)
df1 <- data.frame(
  item=paste0("item", jamba::padInteger(1:20)),
  pvalue=round(digits=3, stats::runif(20) / 5),
  log2FoldChange=round(digits=3, rnorm(20) * 3))

df2 <- data.frame(
  item=paste0("item", jamba::padInteger(1:20)),
  pvalue=round(digits=3, stats::runif(20) / 5),
  log2FoldChange=round(digits=3, rnorm(20) * 3))

head(df1)
```

item	pvalue	log2FoldChange
item01	0.058	3.672
item02	0.158	1.079
item03	0.082	1.202
item04	0.177	0.332
item05	0.188	-1.668
item06	0.009	5.361

Option 1. Setlist

One convenient approach is to create a list of `data.frame` objects, apply a P-value filter to each entry, and return column 'item'.

This output does not have directional sign.

```
# create a list of data.frame objects
dflist <- list(Dataset1=df1,
  Dataset2=df2)

# iterate the list
setlist <- lapply(dflist, function(df){
  # apply stat threshold
  dfsub <- subset(df, pvalue < 0.1);
  # return the item name
  dfsub$item
})
str(setlist)
```

```
## List of 2
## $ Dataset1: chr [1:9] "item01" "item03" "item06" "item10" ...
## $ Dataset2: chr [1:11] "item02" "item03" "item04" "item06" ...
```

Option 2. Signed setlist

The other common alternative is to extend Option 1 to create a signed list, using the sign of 'log2FoldChange'. The vector should be the sign, then *named* by 'item'.

```
# create a list of data.frame objects
dflist <- list(Dataset1=df1,
              Dataset2=df2)

# iterate the list
setlist_signed <- lapply(dflist, function(df){
  # apply stat threshold
  dfsb <- subset(df, pvalue < 0.1);
  # create vector using sign()
  x <- sign(dfsb$log2FoldChange);
  # add item names
  names(x) <- dfsb$item;
  x
})
setlist_signed

## $Dataset1
## item01 item03 item06 item10 item12 item15 item17 item18 item19
##      1      1      1     -1     -1     -1      1      1     -1
##
## $Dataset2
## item02 item03 item04 item06 item10 item14 item15 item16 item17 item19 item20
##     -1     -1      1     -1     -1      1     -1      1     -1      1      1
```

Figure 2.11 shows the resulting Venn diagram without sign (left) and with sign (right).

```
venndir(setlist)
venndir(setlist_signed)
```

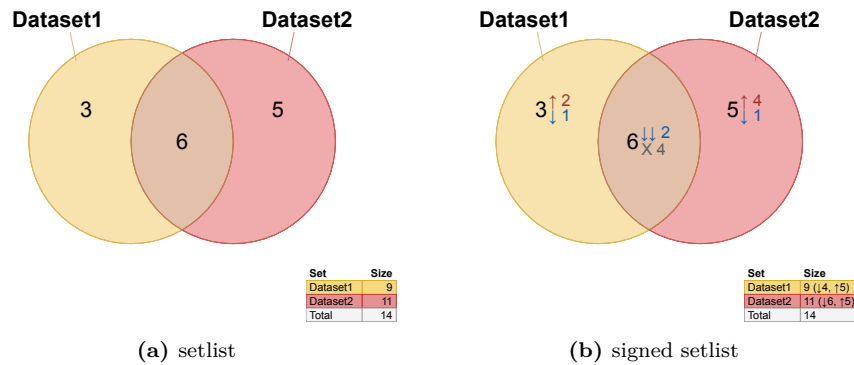


Figure 2.11. Venn diagram using a list of data frames to a **setlist** (left) and signed **setlist** (right).

2.2 Specialized Input Types

The driving motivation for Venndir involved gene expression data, and there are several common methods to import differentially expressed genes.

2.2.1 limma

The **limma** (Smyth et al., 2025) package covers an amazing array of gene and Omics expression analysis methods, spanning transcript and proteomics microarray data, transcriptomics RNA sequencing (RNA-seq) data, and a broad range of other data types including proteomics, lipidomics, metabolomics, and DNA methylation.

The common paradigm, beginning with **x** as expression data. See `limma::getEAWP()` for a description of all recognized data object types.

```
library(limma)

# fit linear model
fit <- lmFit(x, design=design)

# apply contrasts
fit2 <- contrasts.fit(fit, contrasts=contrasts)

# Optional Empirical Bayes moderated t-statistic
fit3 <- eBayes(fit)
```

Now one would use `fit3`, or `fit2` if `eBayes` was not necessary, to summarize the statistical contrasts.

Two common approaches are used here, both can be valid:

1. Extract each contrast into a `data.frame` using `limma::topTable()`, then follow [List of data frames](#).
2. Use `limma::decideTests()` to filter all contrasts together. This function returns `limma::TestResults`-class which is equivalent to [Signed incidence matrix](#).

For example, to filter by adjusted P-value 0.05, and 1.5-fold change:

```
im <- limma::decideTests(fit3,
  p.value=0.05,
  lfc=log2(1.5))
```

The resulting [incidence matrix](#) `im` contains one column per contrast.

When using only a subset of contrasts, use the argument `sets` as described in [Consistent Set Colors](#). The example below chooses `sets=c(1, 3)`.

```
venndir(im, sets=c(1, 3))
```

2.2.2 DESeq2

DESeq2 ([Love et al., 2025](#)) is widely used and excellent for RNA-seq analysis. The example below is adapted from `DESeq2::DESeq()` to use three groups: 'A', 'B', 'C'.

The following code sets up some test data.

```
suppressPackageStartupMessages(library(DESeq2))
set.seed(123)
cnts <- matrix(rnbinom(n=1500, mu=100, size=1/0.5), ncol=15)
rownames(cnts) <- paste0("row", jamba::padInteger(1:100))
cond <- factor(rep(LETTERS[1:3], each=5))
colnames(cnts) <- jamba::makeNames(cond, suffix="")
cnts[1, 1:5 + 0] <- cnts[1, 1:5] + 275;
cnts[2, 1:5 + 5] <- cnts[2, 1:5 + 5] + 355;
cnts[3, 1:5 + 10] <- cnts[3, 1:5 + 10] + 425;
cnts[4, 1:5 + 10] <- cnts[4, 1:5 + 10] + 655;

# object construction
suppressMessages(
  dds <- DESeqDataSetFromMatrix(cnts, DataFrame(cond), ~ cond)
)

# standard analysis
suppressMessages(
```

Table 2.1. The first 6 rows following DESeq2 statistical analysis.

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
row001	176.5	-1.48	0.63	-2.36	0.0181392	0.3023208
row002	188.8	-0.24	0.65	-0.37	0.7096112	0.9355705
row003	262.0	1.85	0.45	4.08	0.0000453	0.0022649
row004	294.1	3.11	0.59	5.30	0.0000001	0.0000118
row005	82.5	-0.47	0.50	-0.93	0.3502942	0.7961232
row006	87.5	-0.16	0.67	-0.24	0.8121139	0.9443185

```
dds <- DESeq(dds)
)
```

The object `dds` can be used to extract statistical results for each contrast, for example see `DESeq2::resultsNames(dds)`.

The next steps extract a `DESeqResults` object for each contrast of interest. Note: Use whichever method you typically use at this point, there are too many possible workflows to cover here.

```
# Contrast: 'B - A'
resBA <- results(dds, contrast=c("cond", "B", "A"))

# Contrast: 'C - A'
resCA <- results(dds, contrast=c("cond", "C", "A"))

# Print the first few rows of the "C - A" contrast results
head(data.frame(check.names=FALSE, resCA))
```

(fig:deseq2-example2-real) The first 6 rows following DESeq2 statistical analysis.

Two steps are applied, intentionally lenient for this example:

1. Filter by statistical thresholds
 - adjusted P-value `padj < 0.20`
 - fold change `abs(log2FoldChange) > log2(1)`
2. Convert to **signed set**, assigning names using the DESeq2 result rownames.

```
# filter statistics
resBA_hitdf <- subset(resBA,
  padj <= 0.2 &
  abs(log2FoldChange) > log2(1))
```

```
# generate named vector
resBA_hits <- setNames(
  sign(resBA_hitdf$log2FoldChange),
  rownames(resBA_hitdf))

resBA_hits

## row001 row002 row022 row023 row040 row045 row096
##      -1      1      1      1     -1      1      1
```

The same steps are performed for contrast 'C - A'.

```
resCA_hitdf <- subset(resCA,
  padj <= 0.2 &
  abs(log2FoldChange) > log2(1))
resCA_hits <- setNames(
  sign(resCA_hitdf$log2FoldChange),
  rownames(resCA_hitdf))

resCA_hits
```

```
## row003 row004 row023 row045 row056
##      1      1      1      1      1
```

The two **signed sets** are combined into a list then visualized with `venndir()` in Figure 2.12.

```
setlist <- list(
  "B-A"=resBA_hits,
  "C-A"=resCA_hits)
v <- venndir(setlist,
  legend_headers=c(Set="Contrast", Size="Differential Genes"))
```

To view the genes in the Venn diagram, skip ahead to **Item labels** and **Signed items**, or to extract the genes use `overlaplist(v)`.

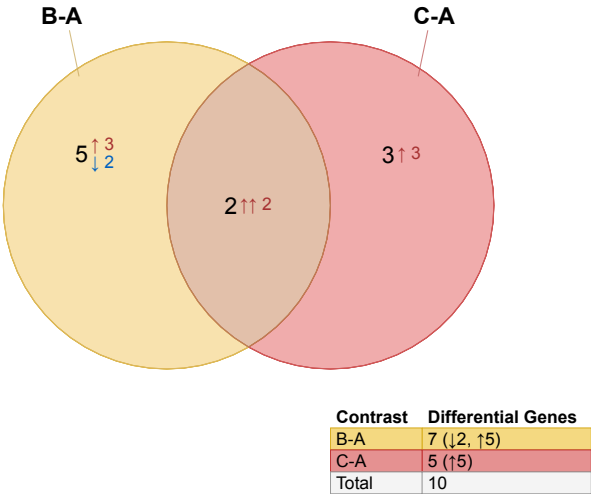


Figure 2.12. Directional Venn diagram comparing statistical hits from two contrasts analyzed using DESeq2.

Chapter 3

Venndir Basics

A Venn diagram seems relatively simple and straightforward. It uses circles (or sometimes ellipses) to represent the shared and distinct elements of each set.

Proportional **Euler diagrams** use areas proportional to the number of elements of each set, and are somewhat unpredictable.

An important feature of any good Venn diagram is clear labeling. Venndir is as much about labeling as it is about the diagram. Adding *signed labels* only heightens the challenge.

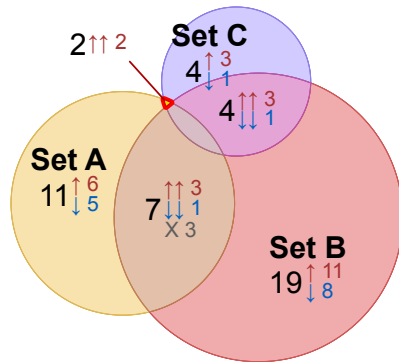


Figure 3.1. Three-way proportional Euler diagram, showing the difficulty of labeling each region. Here, the overlap of set_A and set_C has 2 items. The label is placed outside, connected by a line segment.

Venndir employs a variety of strategies to automate labeling, and adds approaches to allow for manual adjustments. Figure 3.1 demonstrates that sometimes there is no other option but to place a label outside.

Most of this chapter describes basic features of Venndir. Although some techniques may feel advanced, I promise it becomes more "advanced" in the next chapter!

3.1 Default Venndir

The following steps begin with a `setlist`, which contains varying examples for the purpose of demonstrating various features. To create a `setlist`, refer to [Data Import](#) for detailed examples.

3.1.1 Data for Testing

Most examples use `make_venn_test()` to create a `setlist`. This function creates simple sets by default, or signed sets with argument `do_signed=TRUE`.

The arguments help customize useful characteristics:

- `n_items=200`: Overall number of items in the "universe".
- `n_sets=3`: Number of sets.
- `do_signed=FALSE`: Whether to create signed sets.
- `concordance=0.5`: The directional concordance on a scale of -1 to 1.
- `sizes`: Specific sizes for each set, optional.
- `min_size,max_size`: Minimum and maximum set size, when `sizes` is NULL.
- `items`: The full universe of items to use, optional.
- `set_names`: Specific set names, optional.

```
setlist <- make_venn_test()
```

3.1.2 Create the Venn

Given a `setlist`, a Venn diagram is straightforward with `venndir()`.

```
venndir(setlist)
```

The output of `venndir()` is an object with class `Venndir`, returned invisibly. Returning the object invisibly means that the object is not printed to screen, but can be assigned to a variable if desired.

The `Venndir` object contains all the supporting data required to make a figure, in addition to other relevant data that can be exported for review, or edited to customize the figure.

A `Venndir` object can be printed which shows a brief summary.

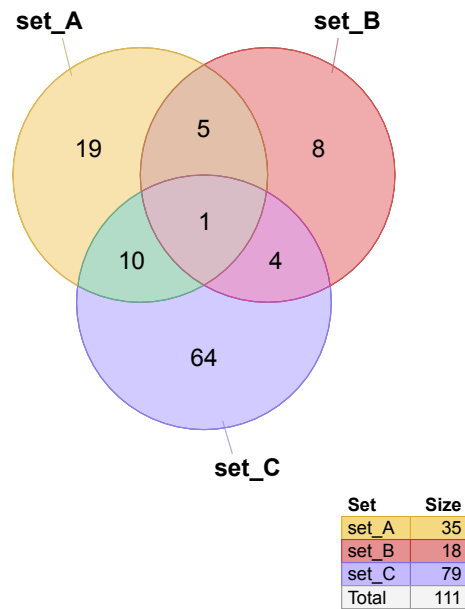


Figure 3.2. Default Venn diagram with three sets.

```
v <- venndir(setlist, do_plot=FALSE)
v
```

```
## class: Venndir
## number of sets: 3
## number of polygons: 3 sets, 7 overlaps
## overlap_type: 'overlap'
## sets:
##   set_name setlist_labels legend_labels set_colors size
## 1   set_A      set_A      set_A      #EEC12E    35
## 2   set_B      set_B      set_B      #D54848    18
## 3   set_C      set_C      set_C      #9F8DFF    79
```

There are many customization options, which will be explored in detail throughout this user guide.

3.1.3 Create the Euler

To make a proportional **Euler diagram**, use argument `proportional=TRUE`, Figure 3.3.

```
venndir(setlist, proportional=TRUE)
```

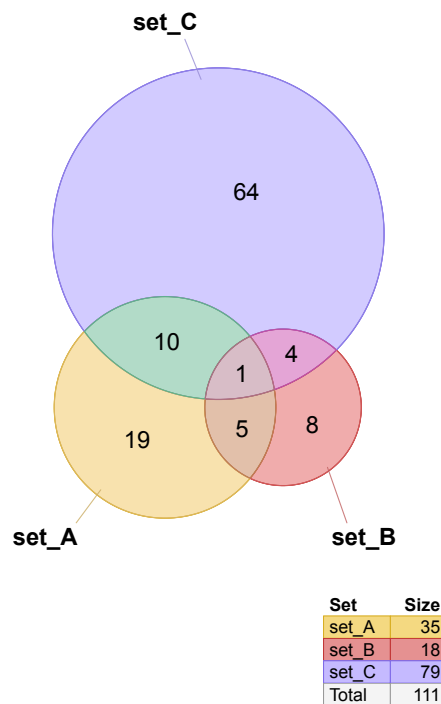


Figure 3.3. Default proportional Euler diagram with three sets.

3.1.3.1 Euler with ellipse

Venndir uses R package **eulerr** (Larsson, 2024) to define the model Euler diagram, which uses circles by default. However, it can be coerced to use ellipse shapes with argument `shape="ellipse"`.

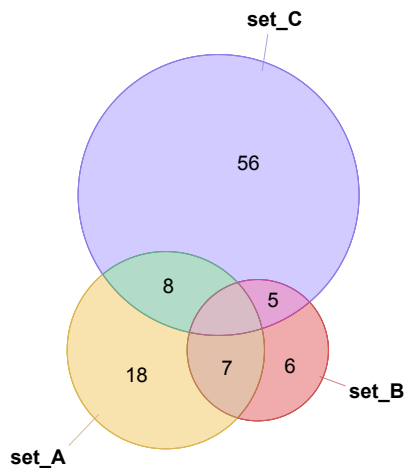
Figure 3.4 compares circular and elliptical shapes, and illustrates that neither option is perfect. The circular option shows one empty set

```
setlist <- make_venn_test(sizes=c(33, 18, 69))
v1 <- venndir(setlist,
  vector_method="label",
```

```

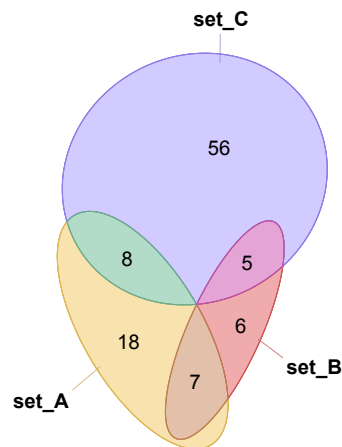
proportional=TRUE)
v2 <- venndir(setlist,
  proportional=TRUE,
  vector_method="label",
  shape="ellipse")

```



Set	Size
set_A	33
set_B	18
set_C	69
Total	100

(a) Circular sets.



Set	Size
set_A	33
set_B	18
set_C	69
Total	100

(b) Elliptical sets.

Figure 3.4. Proportional Euler diagram using circle (left) and ellipse (right).

Euler diagrams are imperfect with three or more sets, which means there is no guaranteed geometry (with circles or ellipses) that guarantees the overlap regions will be exactly proportional to the number of items in each region.

Now is a good time to mention that Euler diagrams are useful but imperfect.

Key Points:

1. Euler diagrams are only *approximately proportional* for three or more sets.
2. Some overlap regions may be shown which have no items.
3. Some overlap regions may not be shown which contain items.

A diagram with any **hidden overlaps** will display a small footnote symbol † in the bottom-right corner.

Points **1** and **2** test one's coping skills: "When is it *okay* not to be perfect?"

Point **3** is more concerning, and is discussed in detail in the section **Hidden Overlaps**. Briefly:

- Hidden overlaps are indicated by a footnote.
- Hidden overlaps are stored in the `Venndir` object.
- Information about "hidden overlaps" can be reviewed using: `print(v)`, `footnotes(v)`, `warnings(v)`, or `overlaplist(v)`.

The footnote is rendered on the bottom-left corner by default, and can be controlled using arguments in `render_venndir_footnotes()`. Notably, argument `footnote_style` can be:

- `'symbol'`: (default) only the footnote symbol
- `'footnote'`: one line for each footnote
- `'header'`: Simple summary of footnote marks.

The intention is for the footnote to be subtle, but visible, so there is clear indication when something should be reviewed in detail.

```
setlist3 <- make_venn_test(500, n_sets=4)
v3 <- venndir(setlist3,
  draw_legend=FALSE,
  proportional=TRUE)
v3b <- venndir(setlist3,
  footnote_style="footnote",
  footnote_fontsize=12,
  draw_legend=FALSE,
  proportional=TRUE)
```

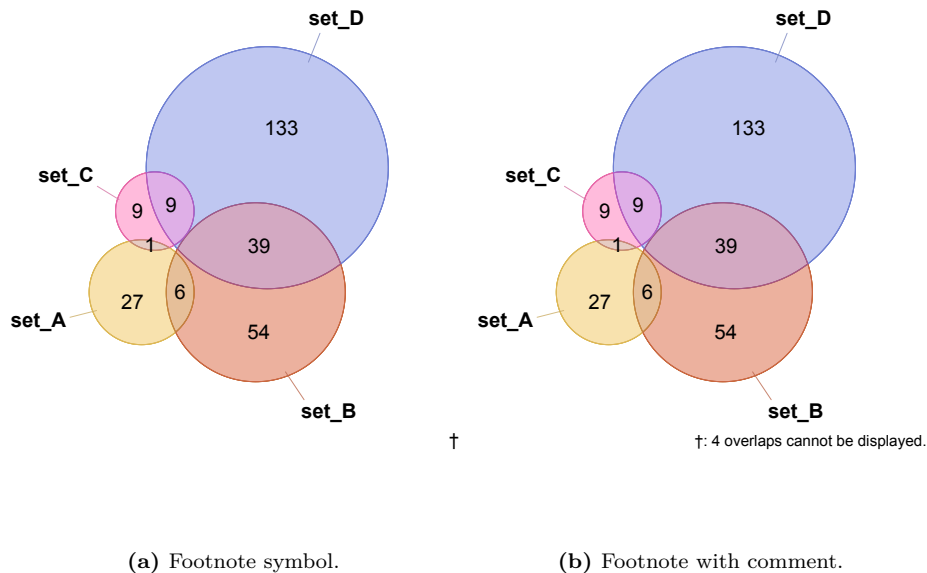



Figure 3.5. Euler diagrams showing a footnote symbol (left panel), and footnote comment (right panel).

3.2 Venn Set Colors

The argument `set_colors` is used to define colors for each set. When it is `NULL`, as by default, it assigns rainbow categorical colors by calling the **colorjam** (Ward, 2025) R package.

Figure 3.6 demonstrates user-defined colors (left), and monochrome transparent colors (right).

```
venndir(setlist,
  set_colors=c("firebrick", "orange", "royalblue"))
# blue transparent colors
venndir(setlist,
  poly_alpha=0.4,
  set_colors=c("royalblue", "royalblue", "royalblue"))
```

The background fill color is also adjusted by `poly_alpha` to control the alpha transparency. Values range from 0 (fully transparent) to 1 (fully opaque). The default `poly_alpha=0.6` is mostly opaque.

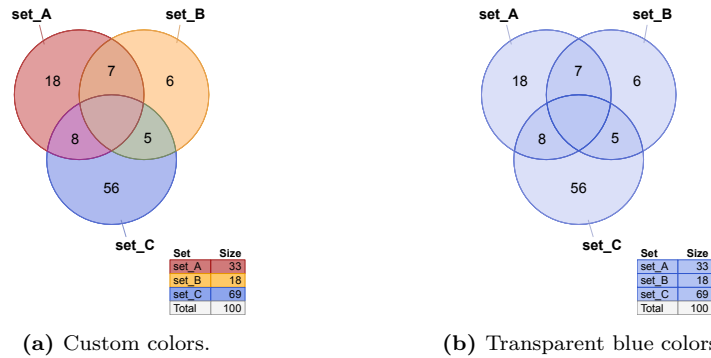


Figure 3.6. Venn diagram showing custom set colors: red, orange, blue.

Figure 3.7 shows the effect of using low transparency, which allows some background colors to become dark. Notice some text labels become white to maintain visual contrast.

```
venndir(setlist,
  set_colors=c("firebrick", "orange", "royalblue"),
  poly_alpha=0.9)
```

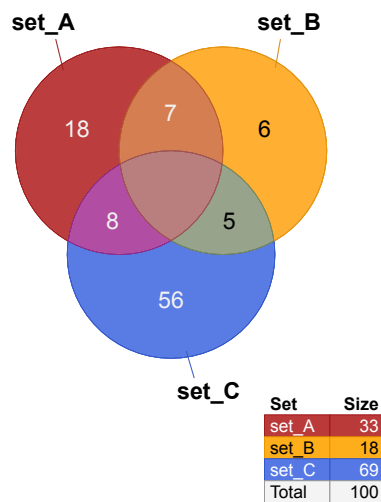


Figure 3.7. Venn diagram using opaque colors, which also causes some text labels to become white for visual contrast.

Note:

Some R graphics devices do not support alpha transparency, which can be reviewed with `dev.capabilities()`. It should be run when the specific graphics device is open, for example after using `cairo_pdf()` to open a PDF file for output.

Colors can also be customized in any section of the figure, using techniques described in [Modify Venn Overlaps](#).

3.2.1 Consistent Set Colors

When the argument `setlist` is passed to `venndir()`, and no custom colors are defined with argument `set_colors`, categorical colors are assigned to each set.

In a situation where `setlist` contains five sets, it is recommended to provide all five sets even for a figure that uses only a few sets. The following steps make sure colors are used consistently for each set name.

1. Supply `venndir()` with the complete `setlist`.
2. Use argument `sets` to indicate which entries in `setlist` should be used.
3. Optionally define `set_colors`, `setlist_labels`, and `legend_labels` for all sets in `setlist`.

Figure 3.8 represents five sets with specific categorical colors assigned to each set. When making Venn diagrams with two sets, they should use consistent colors.

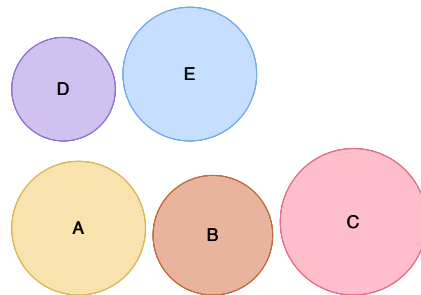


Figure 3.8. Five sets are depicted as circles, each with a specific categorical color.

Figure 3.9 shows the effects in a Venn diagram with sets A and E, using the respective colors.

```
setlist <- make_venn_test(n_sets=5, n_items=70, set_names=LETTERS[1:5])
v2 <- venndir(setlist,
  sets=c(1, 5))
```

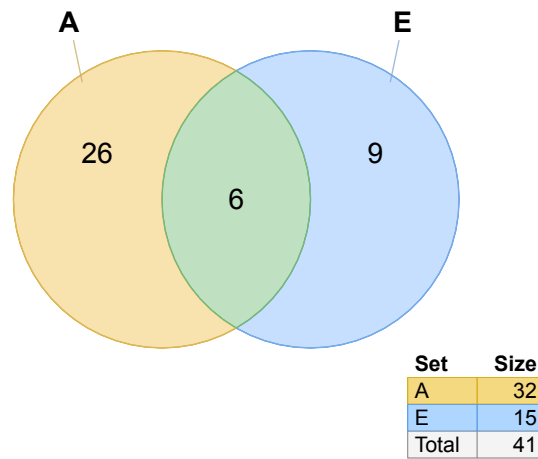


Figure 3.9. Venn diagram comparing sets A and E. The colors are taken from the five-color palette, using yellow for A, and blue for E.

Figure 3.10 shows the desired outcome, several Venn diagrams using colors consistent for each set.

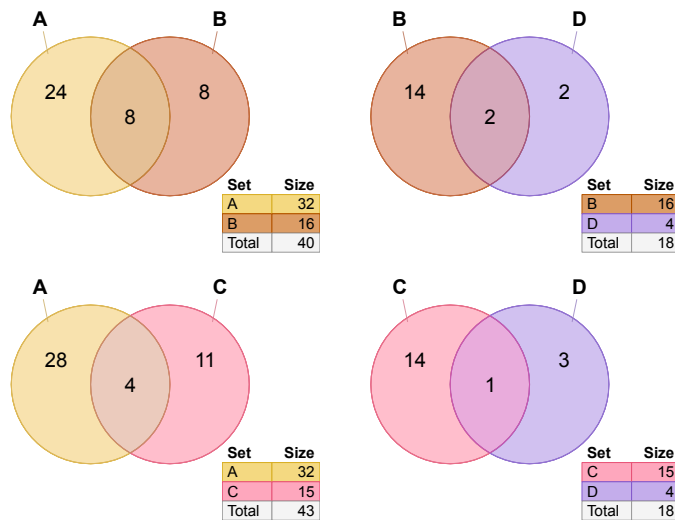


Figure 3.10. Four Venn diagrams using consistent colors for each set.

Figure 3.11 shows the problem to avoid, several Venn diagram with colors not consistent for each set.

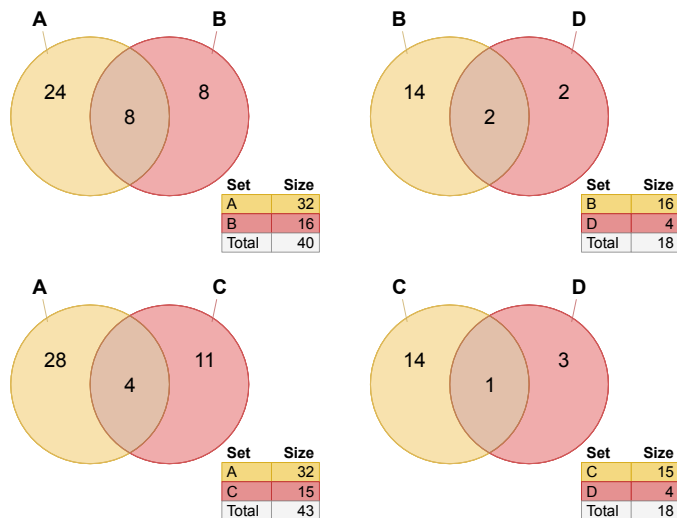


Figure 3.11. Four Venn diagrams using the default colors, illustrating the problem to avoid: Colors are not consistent for each set.

Other alternatives to avoid mis-using color:

- Supply argument `set_colors` to `venndir()` with specific colors per set.
- Create Venn with no color, and no background fill.

3.2.2 Venndir Without Color

It is possible to create a Venndir without color. For example, some publications require black-and-white output.

Figure 3.12 demonstrates Venndir output with no color.

```
v <- venndir(make_venn_test(),
  show_segments=FALSE,
  poly_alpha=0,
  border="black",
  border.lwd=2,
  legend_color_style=c("noborder", "nofill"))
```

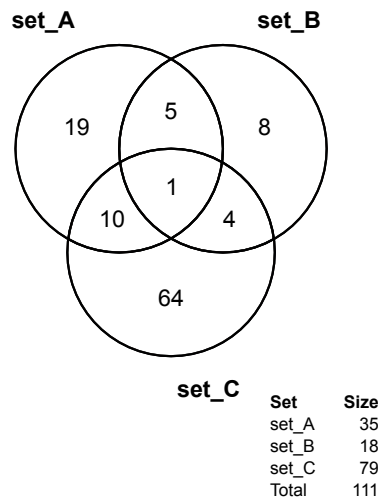


Figure 3.12. Venndir figure with no color.

Tip:

Using `poly_alpha=0` also makes signed labels not colored.

To maintain signed colors, use `poly_alpha=0.01`.

For more control over signed label colors, see [Custom Sign Label Colors](#).

3.3 Venndir Labels

3.3.1 Label Content

The term 'label' may refer to many elements of a Venndir figure. Of course, the sets have labels, and Venn overlap counts have labels. When using signed data, there are also signed count labels. There are also optional labels such as percent overlap, and even individual item labels. In future, there may be additional labels such as Jaccard overlap, Kruskal's directional concordance, or z-score of directionality.

There are so many types of labels, it became unwieldy to assign a separate function argument for each label, which is why labels are controlled by one argument: `show_labels`.

Table 3.1 describes each type of label, and the letter assigned to each.

(fig:label-table) Summary of recognized options for the argument show labels.

Including the respective letter will enable that label. Using UPPERCASE

Table 3.1. Summary of recognized options for the argument `show_labels`.

Label	Letter	Notes
Name	'N' or 'n'	The set name provided in 'setlist'.
Count	'C' or 'c'	The count of the number of items in each Venn region. This label is sometimes referred to as 'main count' or 'overlap count'.
Signed	'S' or 's'	The count of the number of items by each observed combination of signs.
Percent	'P' or 'p'	The percent of items in each Venn region, related to the total items represented in the diagram.
Item	'i'	The item labels represented in each Venn region. Items can only be displayed inside the Venn diagram.

places the label outside, and using lowercase places the label inside the Venn diagram.

The default `show_labels="Ncs"` places the Name outside, then counts and **signed counts** inside.

Tip:

A good default to include the percentage is `show_labels="Ncsp"`. The percentage is always located with the **main counts**, unless **main counts** are not shown.

Currently, item labels can only be placed inside. It is possible to place item labels in a table outside the Venn diagram, however it involves advanced techniques.

Figure 3.13 illustrates common examples for `show_label`, showing how the components of each label are grouped together.

3.3.2 Overlap Type

The argument `overlap_type` provides different approaches to summarize directional overlaps. This option determines which count labels will be displayed in the Venndir figure.

Table 3.2 summarizes the options for `overlap_type`.

(fig:overlap-type-table) Summary of recognized overlap types, with a description of the associated output in a Venndir figure.

- When the input `setlist` is not signed, `overlap_type='overlap'` is the

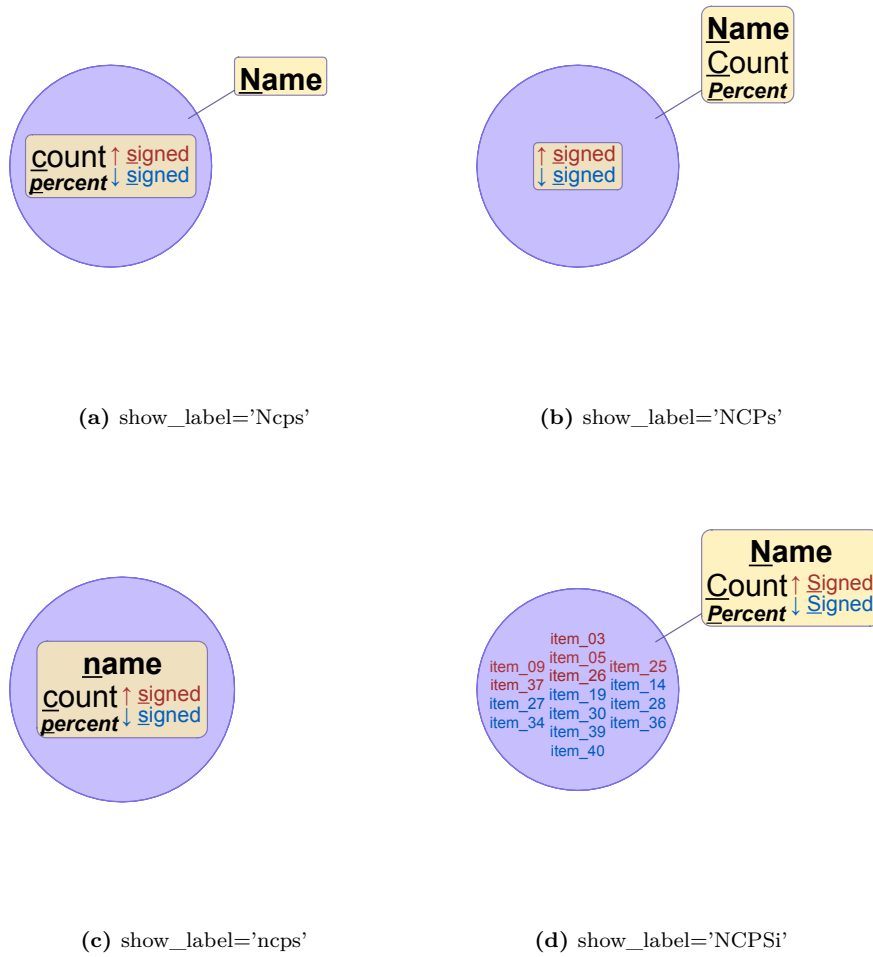


Figure 3.13. Four examples of various Venndir labels, placed inside or outside each figure. The label components are grouped by location, then organized in a defined way.

Table 3.2. Summary of recognized overlap types, with a description of the associated output in a Venndir figure.

overlap_type	Description
'overlap'	Only the summary overlap counts are displayed.
'concordance'	The summary overlap counts are displayed. Signed counts are tabulated for overlaps involving one direction. All other signed counts are summarized with 'X' for discordance.
'each'	The summary overlap counts are displayed. Signed counts are tabulated for each combination of signs observed.
'agreement'	The summary overlap counts are displayed. Signed counts are tabulated based upon agreement or disagreement of the directional sign.

default.

- When the input `setlist` is signed, `overlap_type='concordance'` is the default.

Figure 3.14 illustrates the four values for `overlap_type`.

3.3.3 Signed Label Placement

Signed count labels are placed beside main count labels by default, however they can be placed below main count labels.

The argument `template` controls the signed label placement, and there are two options, illustrated in Figure 3.15.

1. `template='wide'` (default) labels **signed counts** beside main overlap counts.
2. `template='tall'` labels **signed counts** below main overlap counts.

```
setlist3 <- make_venn_test(n_sets=3, do_signed=TRUE)
vt1 <- venndir(setlist3)
vt2 <- venndir(setlist3, template="tall")
```

3.3.4 Visual Styles

By default, Venn count and set labels are drawn without any particular color shading or border. The argument `label_style` is used to enable background color fill, and optional border.

The recognized keywords for `label_style`:

- "lite" - light background

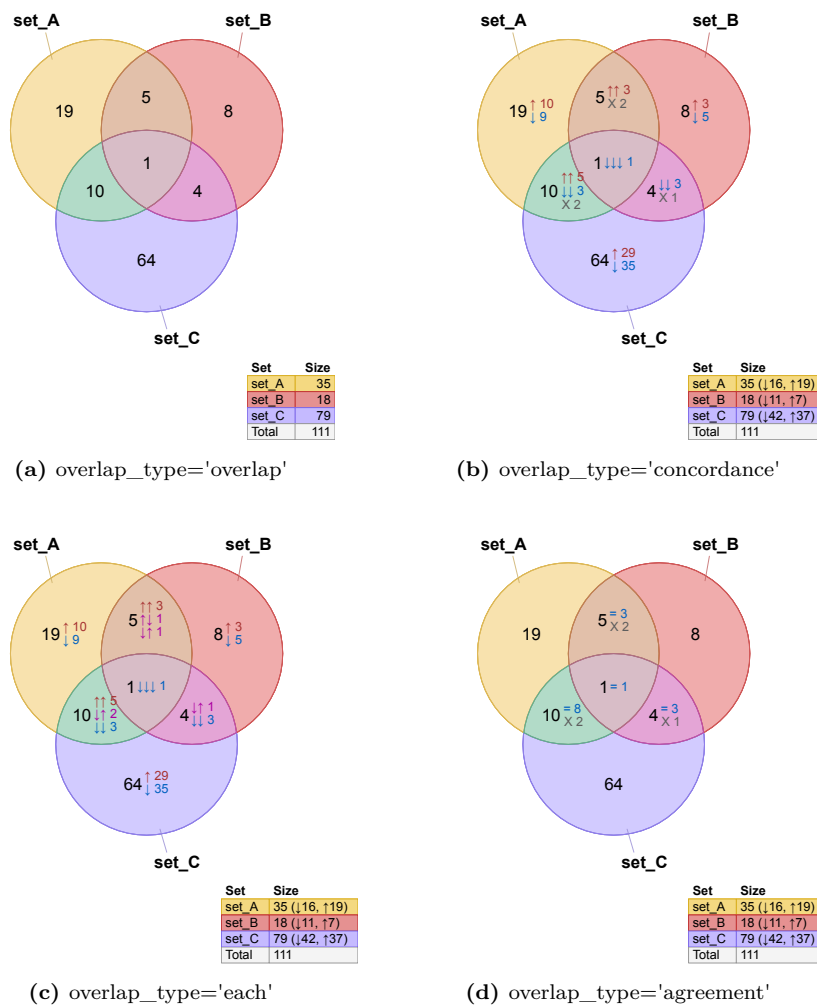


Figure 3.14. Venn diagrams showing four overlap types recognized: 'overlap' (default for non-signed data), 'concordance' (default for signed data), 'each', and 'agreement'.

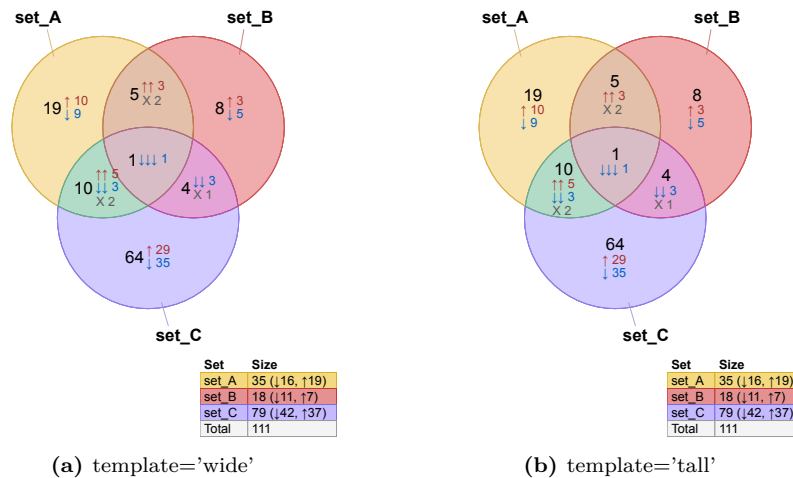


Figure 3.15. Venn diagram with signed counts labeled beside overlap counts (left) using `template='wide'`; and below overlap counts (right) using `template='tall'`.

- "shaded" - semi-transparent shaded color background
- "fill" - full color background
- "box" - draw a box as a border around the label

A straightforward example is shown in Figure 3.16 using `label_style="lite box"` with signed counts.

```
setlistS <- make_venn_test(do_signed=TRUE)
vS <- venndir(setlistS, label_style="lite box")
```

Figure 3.17 illustrates several possible styles. In general, when using a fill color with any of the options 'lite', 'shaded', 'fill', it may be preferred to add 'box' to include a visual border. For example, try 'lite box' or 'shaded box'.

3.3.5 Nudge a Venn Label

To nudge, that is to reposition, one label in a Venndir diagram, use `nudge_venndir_label()`.

Note that `venndir()` will define a label position for every Venn overlap, and defines a position 'inside' and 'outside' the Venn region.

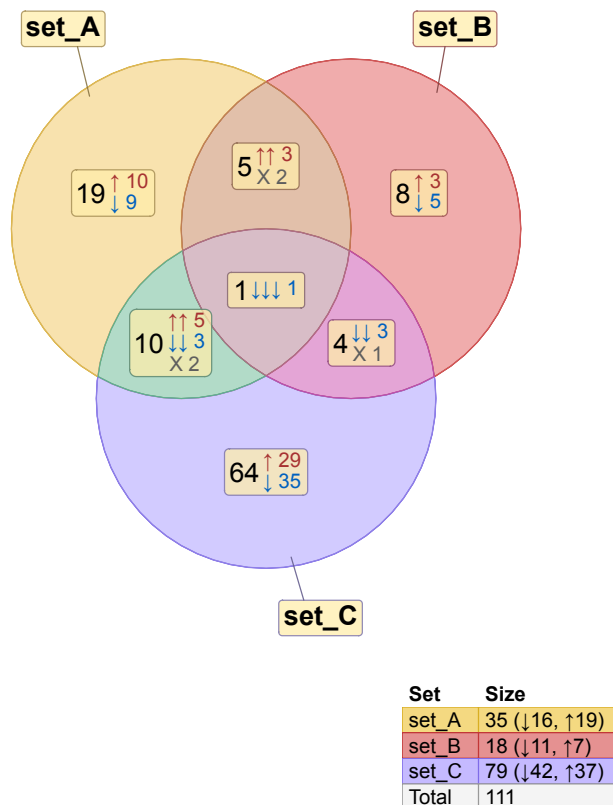


Figure 3.16. Venn diagram showing signed counts, using label style 'lite box'.

Exception:

Sometimes a proportional Euler diagram cannot represent every overlap, due to geometric constraints. In this case, not every overlap will be assigned a label position.

For a discussion of that issue, see Section 4.8 [Hidden Overlaps](#).

To nudge a label, define the label to nudge using these two arguments:

- `set` - the overlapping region to nudge
- `label_location` - the inside or outside label associated with that set

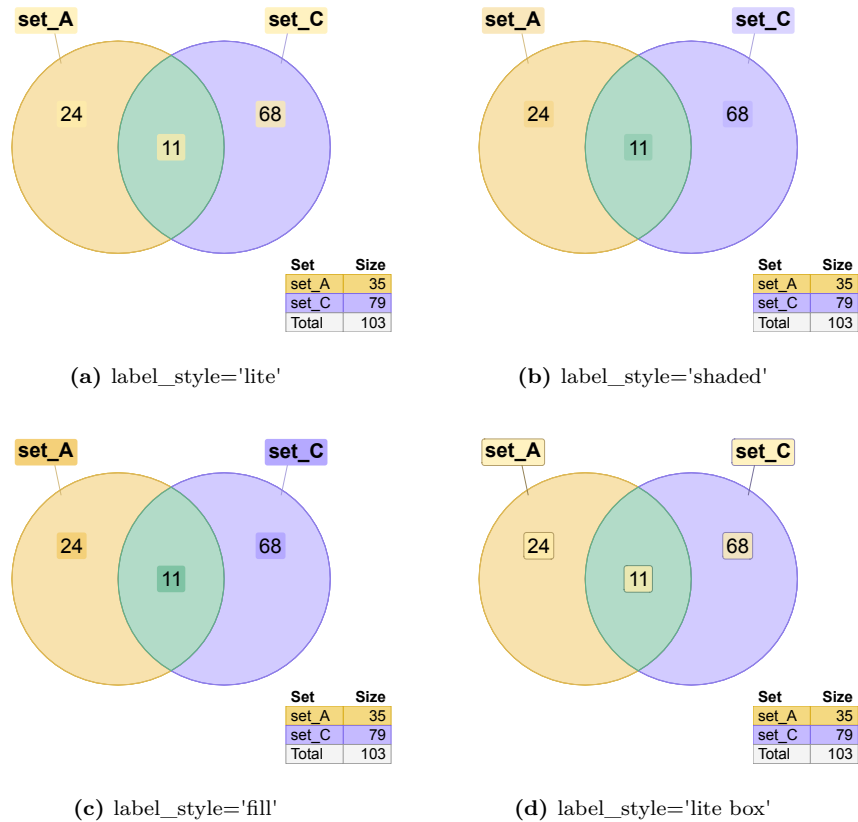


Figure 3.17. Venn diagrams showing the different label style options in each of four panels.

The label adjustment uses two coordinates. Units are proportional to the overall Venndir plot region, where 1 is the full width or height of the plot, whichever is larger.

- `x_offset`
- `y_offset`

The process requires an existing `Venndir` object. In the example below, a simple 3-way Venn diagram is stored in variable `v`. The label for `'set_C'` is moved left (decreasing the x coordinate), and up (increasing the y coordinate).

```
# default Venn
setlist <- make_venn_test()
v <- venndir(setlist)
```

```
# nudge 'set_C' up-and-left
v2 <- nudge_venndir_label(v,
  x_offset=-0.45, y_offset=0.25,
  set="set_C",
  label_location="outside")
plot(v2)
```

Figure 3.18 shows the Venn with default labels (left), and with the label 'set_C' adjusted (right).

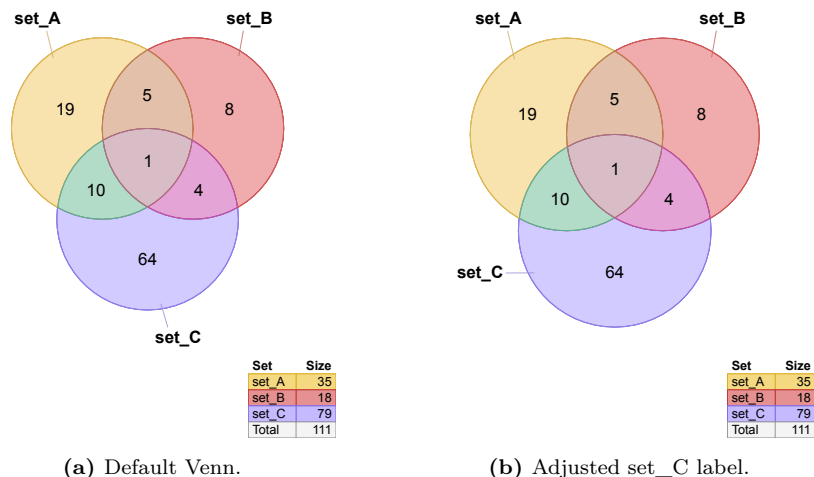


Figure 3.18. Default 3-way Venn diagram (left), with label 'set_C' moved up and to the left (right).

Notice that after the label is adjusted, the new `Venndir` object `v2` can be visualized using `plot()`.

3.3.6 Label Placement Rules

For each `Venndir` figure, sets are each represented by a circle or ellipse. It should be "easy" to label each circle, right? Experience has shown this to be more difficult than expected.

In Figure 3.19, where should the Set label be placed? Where should the line segments connect?

Specific rules for placing labels in the Venn figure:

1. **Set labels** are directed to the most specific region in the figure.

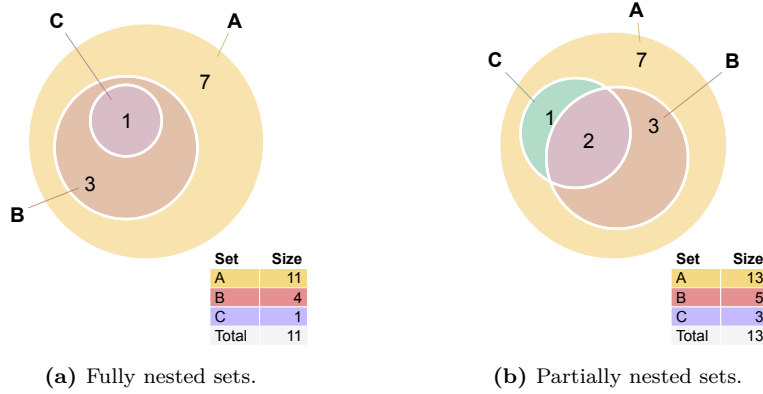


Figure 3.19. Two examples of with very challenging labels.

- For Venn diagrams, Set labels refer to the region "unique to set *A*", with no other overlapping sets.
 - For Euler diagrams, Set labels refer to the region "unique to set *A*" *if it exists*. Otherwise the Set refers to a region containing "*set_A*" with the fewest overlapping sets.
 - In Figure @ref(fig.label-rules) (left) Set '*C*' refers to the region inside B and C.
 - In Figure @ref(fig.label-rules) (right) Sets '*B*' and '*C*' each refer to regions as unique to B and C as possible, respectively.
2. **Count labels** will only be associated with the specific overlap region.
 - If the specific overlap region does not exist in the figure, the count label is not shown. Even though it may seem obvious, sometimes a count label cannot be shown. See [Hidden Overlaps](#).
 - If a count label is placed outside, a line segment will connect the label to the corresponding region.
 - The argument `inside_percent_threshold` may be used to place a label outside when the region is small.
 3. **Set and Count labels** may be grouped together when both are inside or outside.
 - When placed inside, a Set label will be grouped with the count label to which the Set label refers, as described in Rule 1.
 - When placed outside, a Set label will only appear with a count label that is specific to that Set.
 4. **Percentage labels** are shown together with **main counts** when **main counts** are shown.
 - Percentages are calculated directly from **main counts**, and as a

result it is most intuitive for them appear together.

- However, if **main counts** are hidden, the percentage may be placed inside or outside.

5. **Items labels** are only displayed inside, see **Item Labels**.

- Count labels are "moved outside" if they were enabled inside and items are to be displayed.
- When the number of items exceeds `max_items` then items are hidden, and the count labels may be displayed in their place.
- Set names may be displayed inside, together with item labels, however no effort is made to prevent overlapping labels. Set labels can be manually nudged, see **Nudge a Venn Label**.

3.4 Item Labels

To display items inside a Venn diagram, add "i" to the argument `show_labels`. For example `show_labels="Ni"` will display the set Name outside, and items inside.

I never thought I'd be able to place item labels in a Venn diagram, much less expect it, and eventually *need* it. Item labeling has become one of my favorite features.

Adding item labels also ticks the box, so to speak, of an important **data visualization** paradigm: Answering the very next question. For a Venn diagram:

"What are those?"

Figure 3.20 illustrates an example inspired by a published figure¹ (Salybekov et al., 2021).

(What is the very next question?)

"Which genes are in each region?"

Figure 3.21 represents the first example thus far of **Figure Boosting**, wherein the goal is to re-create published figures using Venndir.

3.4.1 Overview of item labeling

There is much utility in adding item labels into a Venn diagram, but most involve the critical word '*if*'. For example, "It would be helpful to add labels *if* they fit." Or "*if* the labels are legible."

And so there are many options associated with item labels.

Before making changes, it is helpful to understand the process.

¹https://www.researchgate.net/figure/Venn-diagram-of-genes-associated-with-susceptibility-to-PAD-CAD-and-stroke-identified_fig2_356459049

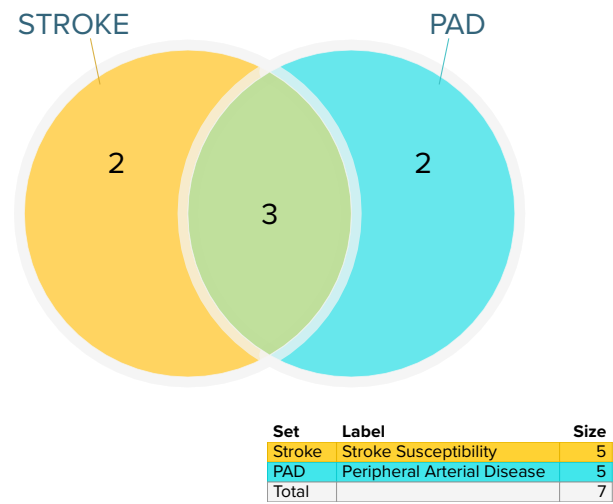


Figure 3.20. Venn diagram depicting genes associated with susceptibility to two diseases. The diagram shows the number of genes in each region: 2, 3, and 2, but does not show the genes.

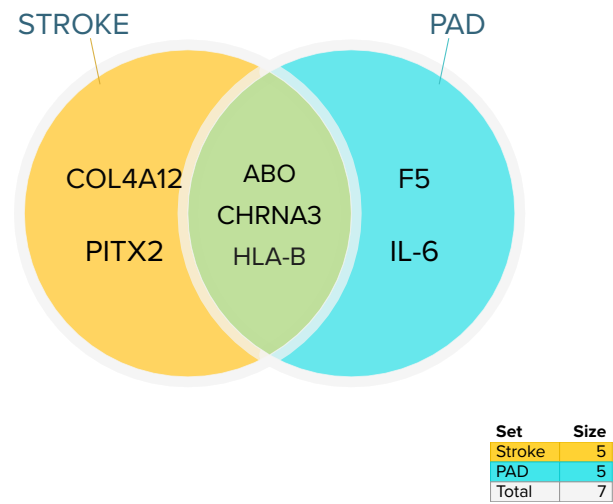


Figure 3.21. The same Venn diagram now showing genes as item labels inside the diagram, answering *the very next question*.

1. Items are associated with each Venn overlap region.
 - More than `max_items` items will not be displayed.
2. The type of item label is defined using argument `show_items`.
 - `"item": 'ACTB'`,
 - `"sign": '↑'`
 - `"sign item": '↑ ACTB'`
3. Label coordinates are arrayed across the Venn region.
 - `xratio` controls the x:y ratio when spacing points
 - `label_method` controls whether to use up/down offset, or row-wise column layout
4. The item font size is adjusted per region.
 - The adjustment uses the number of labels, and the area of the region relative to the total area.
5. Font colors are adjusted for contrast.
 - `make_color_contrast()` determines whether to use light or dark text for each background color.
6. Minor "jittering" is applied to each label
 - The effect color, size, and angle of labels are altered to help distinguish each label.
 - `jitter_cex=0.04`
 - `jitter_color=0.07`
 - `jitter_degrees=0`
7. Items are sorted by sign, then by name.
8. Items are rendered using `marquee` (Pedersen and Mitáš, 2025).

Note that no special effort is made to prevent overlaps between labels, nor to prevent labels from extending outside the region. These features may be implemented in future.

3.4.2 Venn Memes

`venn_meme()` provides a streamlined approach to displaying items. The data input process was described in [Data Import](#), in the section [Items](#).

The `venn_meme()` function itself calls `venndir()` with some pre-defined default settings, otherwise it can be customized the same way as with `venndir()`.

[Figure Boosting](#), for many cases, is easily carried out with `venn_meme()` alone. provides the most convenient starting point.

A simple example of a 3-way Venn meme is shown in [Figure 3.22](#).

```
bix <- list(
  "Biology",
  "Computer\nScience",
  "Stats",
  "Computational\nBiology",
```

```

"Data\nScience",
"Biostatistics",
"Bioinformatics")
venn_meme(bix,
  fontfamily="Futura")

```

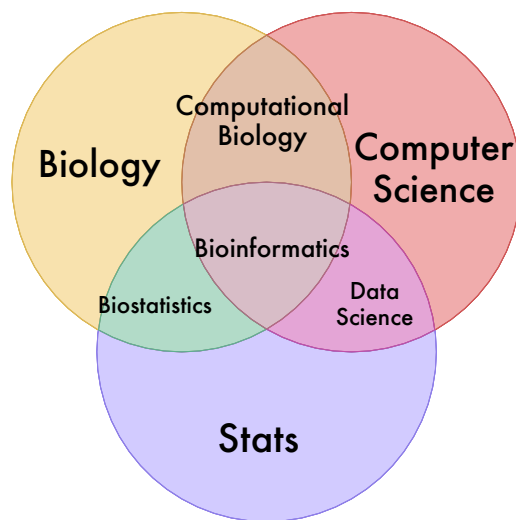


Figure 3.22. Venn diagram created using `venn_meme()` showing 'Biology', 'Computer Science', and 'Stats' coming together to represent the field of Bioinformatics.

Another fun example involves the perils of social media, and being a sports fan. This example also assigns custom "white" colors for `innerborder` and `outerborder` to give it a clean look.

A white border can be a nice change from the default dark border, since it does not conflict with dark text.

```

doom <- list(
  "short-lived\nhappiness",
  "prolonged\nsuffering",
  "sudden\nrage",
  "eating\ntoo much\nspicy\nfood",

```

```

    "stubbing\nyour toe,\ntwice",
    "doom-\nscrolling\nsocial\nmedia",
    "being\na sports\nfan"
)
venn_meme(doom,
  outerborder="white", innerborder="white",
  fontfamily="Optima", item_cex_factor=0.9,
  set_colors=c("gold", "royalblue", "firebrick3"))

```

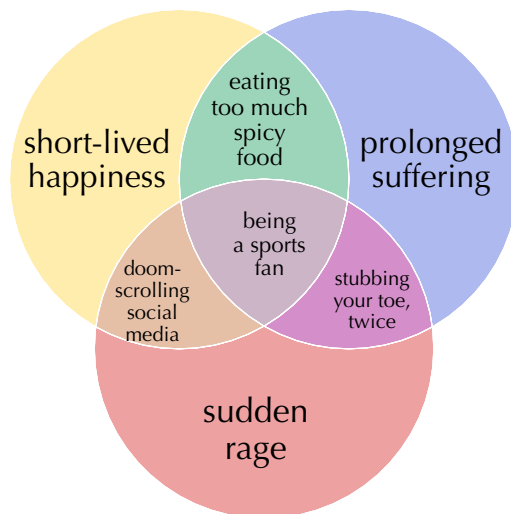


Figure 3.23. Venn meme created with elements of happiness and rage, doom-scrolling social media, and being a sports fan. It also demonstrates use of a thin white border.

3.4.3 Item Label Basics

By default, labels are arranged, sized relative to each region, and displayed. Items are displayed by adding "i" to the argument `show_labels`, for example `show_labels="Ni"`. See [Label Content](#) for a more thorough description.

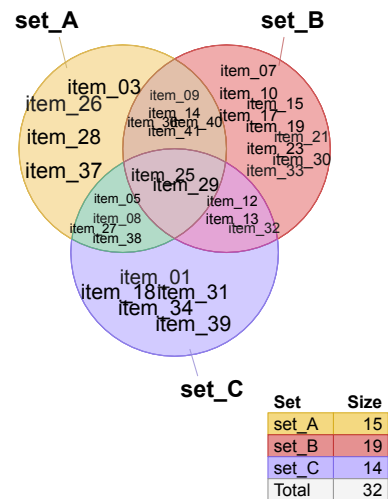


Figure 3.24. Venn diagram showing item labels using default settings.

3.4.3.1 Items inside, counts outside

When items are displayed inside, counts cannot also be displayed inside. However, counts may be displayed outside by adding UPPERCASE "C" to argument show_labels.

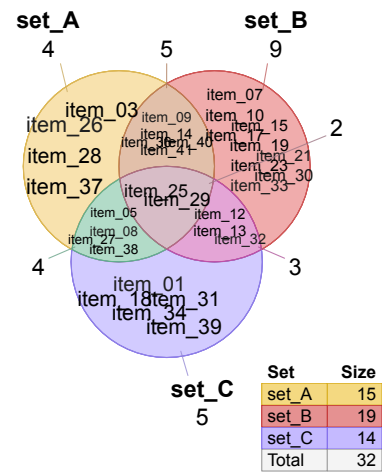


Figure 3.25. Venn diagram showing item labels inside, and count labels outside.

3.4.3.2 Signed Items

When the input `setlist` contains signed data, the items by default will display the sign and item together. This behavior is defined by the argument `show_items="sign item"`.

Conversely, item labels will not include the sign, if either: * the input `setlist` does not contain signed data, or * `overlap_type="overlap"` is defined, which ignores signed data.

```
setlist_signed <- make_venn_test(n_items=41, do_signed=TRUE)
venndir(setlist_signed,
  item_cex_factor=0.7,
  x_inset=grid::unit(-1, "lines"),
  show_labels="Ni")
```

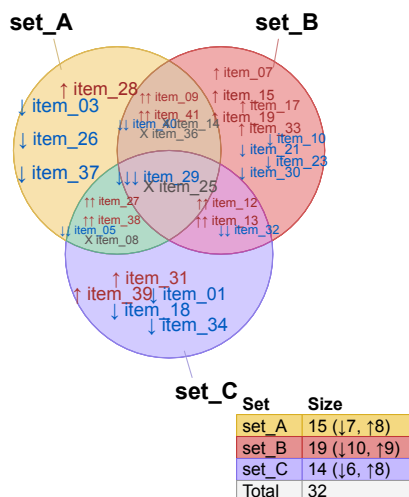


Figure 3.26. Venn diagram with signed data, showing item labels inside each region. The items now have directional arrows beside them, and the item labels utilize color to indicate the directionality.

Note that signed item labels are colorized consistent with the sign, as defined by argument `curate_df`. For a detailed discussion of signs, and how to customize the sign and corresponding color, see [Custom Signs](#).

3.4.3.3 Signs only

When there is a large number of items, one alternative is to display the sign. This option is effective partly because the color is also used, and has the effect of "filling the space" with proportional amount of color.

To display only the sign for each item, use `show_items="sign"`.

When showing only the sign, it may be useful to increase the font size. The argument `item_cex_factor` adjusts item fonts in each region, and is multiplied by the automated sizing based on the number of items and area in each region.

```
setlist_signed <- make_venn_test(n_items=850, do_signed=TRUE)
venndir(setlist_signed,
  show_labels="Ni",
  show_items="sign",
  x_inset=grid::unit(-5, "lines"),
  item_cex_factor=1.2)
```

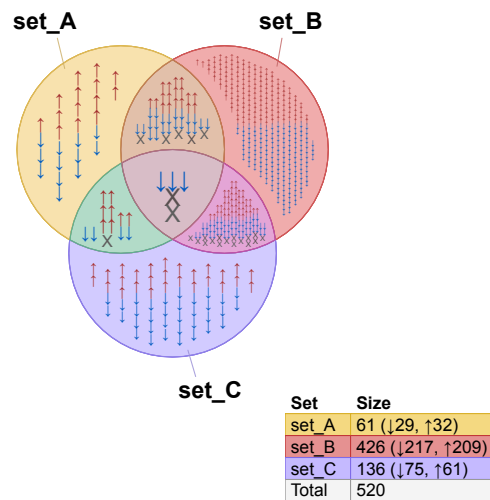


Figure 3.27. Venn diagram showing only the signs of the items filling each overlap region.

3.4.4 Item layout adjustments

There are three main points for adjusting the layout of item labels.

1. `xratio`: Defines the width-to-height ratio, default 1.1.
 - Useful to increase `xratio` for wider labels, or to enforce "column" type layout.
2. `label_method`: Determines whether labels alternate up/down per row.
 - `label_method="offset"` (default) uses a slight 'offset' on each row, resulting in somewhat triangular label placement. This approach is more effective at packing labels into the available space.
 - `label_method="columns"` does not apply an offset, which helps align items in neatly organized rows. This option is useful when also

using a higher value for `xyratio`.

3. `item_buffer`: Controls the buffer zone around the region border.

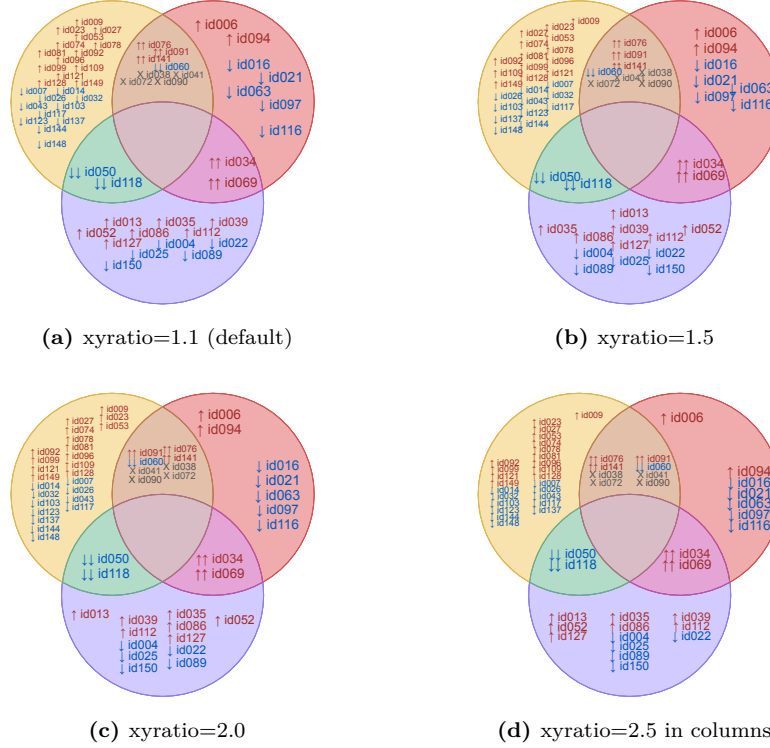


Figure 3.28. Four Venn diagrams, showing examples using different values for 'xyratio' and 'label_method'. Higher 'xyratio' values tend to become more column-oriented.

3.4.5 Item buffer

Two arguments are used during item labeling, both of which are motivated to help place typical text labels inside the respective Venn overlap region:

1. The argument `item_buffer` is used to impose a reasonable distance between item labels and the border of the region to be labeled. Units are relative to the fraction of area that would shrink the region to zero.
2. The argument `width_buffer` adds a small buffer to the left and right sides of each region, which is intended to correct for wide text labels.

The default `item_buffer` = `-0.15` is negative in order to shrink the region by this relative amount, and `-0.15` shrinks the region by 15% relative to the available buffer size for the region.

Note that `item_buffer = -0.15` (15% buffer) is a slightly different size for each region.

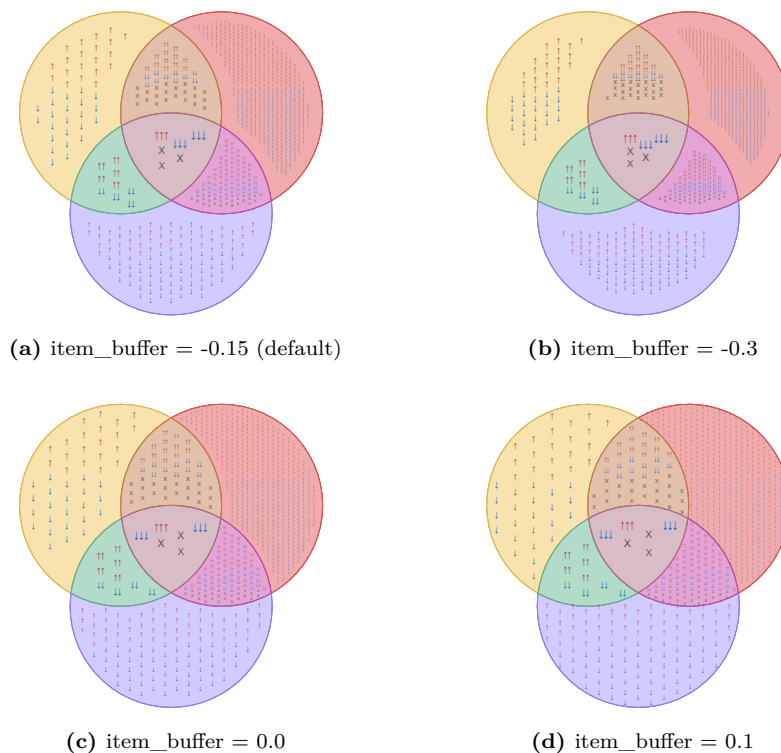


Figure 3.29. Four Venn diagrams showing the effects of adjusting 'item_buffer'.

3.5 Fonts and Font Sizes

If the most challenging aspects of Venndir were the intricacies of placing labels, surely the next big challenge was ensuring any chosen font rendered correctly.

Particularly for signed count labels, the default output uses Unicode up and down arrows: $\uparrow$ and $\downarrow$. These Unicode characters have caused the most heartache, for various reasons related to Unicode font support and computer platform locale.

Most* font rendering issues are handled smoothly by the combination of `marquee`², `systemfonts`³, and `textshaping`⁴. These R packages strive to "make

²<https://marquee.r-lib.org>

³<https://systemfonts.r-lib.org>

⁴<https://textshaping.r-lib.org>

it work” by finding each character glyph in the specified font, or by using a suitable font substitution.

Particularly for PDF output, it may be necessary to embed the font into the PDF file using `grDevices::embedFonts()` or `grDevices::embedGlyphs()`. Using `cairo_pdf()` should embed fonts automatically, but has rare exceptions.

End Result

Most fonts should 'just work' with enough flexibility to keep the focus on effective and artistic choices, not the annoyances.

(This text was rendered by *marquee*, using the 'Optima' font.)

Figure 3.30. Most fonts 'just work' thanks to R package *marquee*, with enough flexibility to keep the focus on effective and artistic choices, and away from any annoyances.

3.5.1 Font Sizes

The default font size used in Venndir is 16 points, and is adjusted with the argument `font_cex`. Default `font_cex=c(1, 1, 0.75)` therefore defines values 16, 16, 12, which are applied in order as follows:

- Set label
- Count label
- Signed count label(s)

The text is defined with a fixed font size, and is therefore independent of the size of the figure. If the figure itself is drawn on a small graphics device, the font sizes will remain the same, so some care should be taken to select font sizes appropriate for the output figure dimensions.

Ideally, the output device dimensions should be defined with fixed units, such as inches ('in'), or centimeters ('cm'), so that the final figure is clearly defined.

The argument `font_cex` has specific behavior based upon the number of values supplied in the argument:

- **font_cex with one value:** It is multiplied by `c(1, 1, 0.75)` to be an easy adjustment to the typical defaults. Thus `font_cex=2` will double all font sizes.
- **font_cex with two values:** The second value is repeated to make a vector of three values. This vector is multiplied by `c(1, 1, 0.75)`, again as convenient way to adjust Set and Count labels proportional to their default values.

`font_cex=c(2, 1)` will double the Set label, keeping Count and Signed at their defaults.

- **font_cex with three values:** Three values are used directly.

The adjusted `font_cex` is multiplied by 16 to define the font point size. Several examples are shown in Figure 3.31.

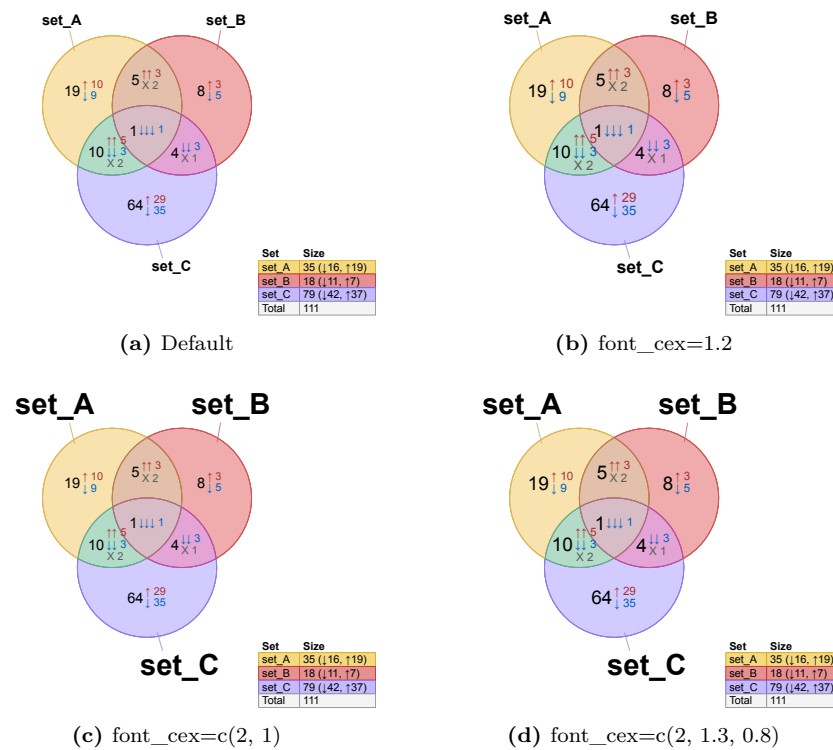


Figure 3.31. Venn diagram showing modifications to font sizes by adjusting the argument `font_cex`.

The last example (bottom-right panel) was created using the following code:

```
v <- venndir(make_venn_test(do_signed=TRUE),
  font_cex=c(2, 1.3, 0.8))
```

3.5.2 Overall Font Family

The overall font typeface used by `venndir()` is defined with the argument `fontfamily`, for which the default is `fontfamily="Arial"`.

See [Available Font Families](#) to review other fonts available.

Figure 3.32 uses the R 'serif' font, with `fontfamily='serif'`. .

```
setlist <- make_venn_test(n_sets=3, do_signed=TRUE)
v <- venndir(setlist,
  main="fontfamily='serif'",
  fontfamily="serif")
```

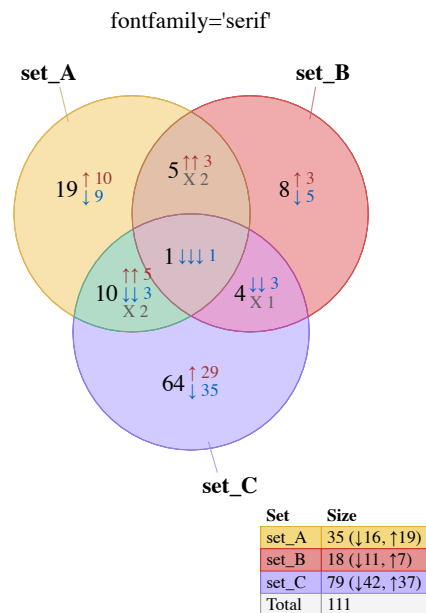


Figure 3.32. Venn diagram showing two sets, with all labels shown using a serif font, such as 'Times' or 'Times New Roman'.

Note that all the labels use 'serif' now, including in the legend.

Similarly, for Venn diagrams which show item labels, the item font is adjusted with `fontfamily`. This type of plot can be created using `venndir()` with argument `show_labels="Ni"`, to place set name 'N' outside, and item labels 'i' inside.

```
petlist <- list(
  Dog=c("Needs walks", "Barks", "Round eyes"),
  Cat=c("Freely roams", "Meows", "Thin pupils"),
  Dog&Cat=c("Furry", "Has claws", "Lovable"))
petsetlist <- overlaplist2setlist(petlist)

petv <- venndir(petsetlist,
  font_cex=1.5,
```

```

item_cex_factor=1.0,
show_labels="Ni",
draw_legend=FALSE,
show_segment=FALSE,
keep_item_order=TRUE,
item_buffer=0, xyratio=1.7,
fontfamily="Optima")

```

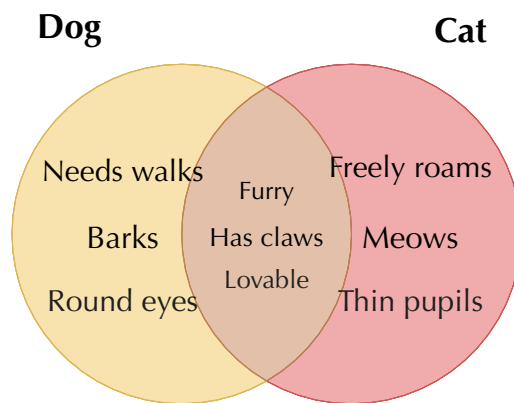


Figure 3.33. Venn diagram showing Two sets 'Dogs' and 'Cats' with some attributes unique to each, and shared by both. The item labels use the 'Optima' font.

The example above includes some other useful customizations, discussed in detail later in the book. For now, here is a brief description:

- `font_cex=1.5` enlarges the set name font.
- `draw_legend=FALSE` hides the color legend.
- `show_segment=FALSE` hides the line segments.
- `keep_item_order=TRUE` maintains the original order of item labels, therefore not sorting items alphabetically.
- `item_buffer=0` reduces the buffer zone of each Venn region before determining item label positions.
- `xyratio=2` increases the x:y ratio during item label placement, higher values tend to produce single column output.

Item labels can be displayed using `venn_meme()`.

```

petlist <- list(
  Dog=c("Needs walks", "Barks", "Round eyes"),

```

```

Cat=c("Freely roams", "Meows", "Thin pupils"),
"Dog&Cat"=c("Furry", "Has claws", "Lovable"))
venn_meme(petlist,
font_cex=2,
item_cex_factor=1.0,
item_buffer=0, xyratio=1.7,
fontfamily="Optima",
show_labels="Ni")

```

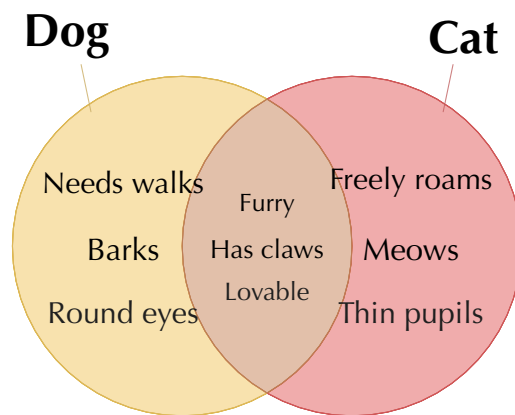


Figure 3.34. Venn diagram showing Two sets 'Dogs' and 'Cats', this time created using `venn_meme()`. The item labels use the 'Optima' font.

3.5.3 Individual Font Families

For more control over specific fonts, use argument `fontfamilies`. This argument expects a list with these elements:

- `overlap` - fontfamily to use for set or overlap labels
- `count` - fontfamily to use for main count labels
- `signed` - fontfamily to use for signed count labels

Any value not customized will use the default `fontfamily` value.

This example shows distinct fonts for each element, in fact it shows two fonts for "count" which corresponds to the **main counts**, and the percentage of counts, in order.

```

setlist <- make_venn_test(do_signed=TRUE)
venndir(setlist,
  show_labels="Ncps",
  main="Main Title (Times New Roman)",
  fontfamily="Times New Roman",
  y_inset=grid::unit(1, "lines"),
  expand_fraction=c(0.15, 0, 0, 0),
  fontfamilies=list(overlap="Optima",
    count=c("Arial Black", "Times"),
    signed="Arial Narrow"))

```

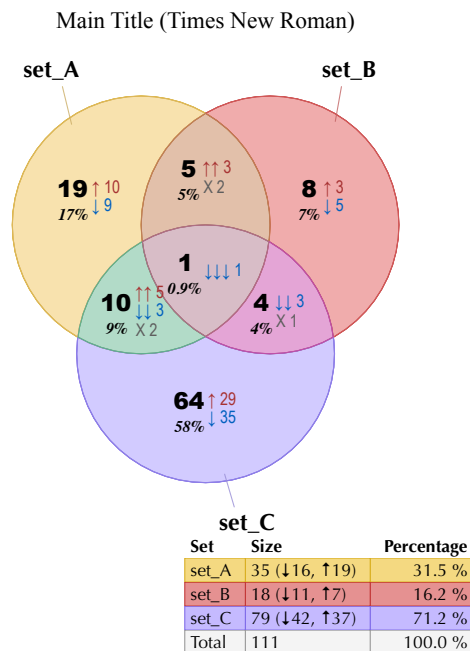


Figure 3.35. Venn diagram showing distinct font families used for each type of label: overlap, count, and signed.

The arguments to `assemble_venndir_label()` also include `fontfaces` which may also be customized to control whether each font is 'plain', 'bold', 'italic', or 'bold.italic' for example. These customizations are rather advanced, and are discussed in more detail later.

Table 3.3. Partial summary of 'Helvetica Neue' fonts via systemfonts.

name	family	style	weight
HelveticaNeue-UltraLight	Helvetica Neue	UltraLight	thin
HelveticaNeue-Thin	Helvetica Neue	Thin	ultralight
HelveticaNeue-Light	Helvetica Neue	Light	light
HelveticaNeue	Helvetica Neue	Regular	normal
HelveticaNeue-Medium	Helvetica Neue	Medium	medium
HelveticaNeue-Bold	Helvetica Neue	Bold	bold
HelveticaNeue-CondensedBold	Helvetica Neue	Condensed Bold	bold
HelveticaNeue-CondensedBlack	Helvetica Neue	Condensed Black	heavy
HelveticaNeue-UltraLightItalic	Helvetica Neue	UltraLight Italic	thin

3.5.4 Available Font Families

Before getting into details with fonts, some useful terminology should be defined. Thomas Pederson wrote an excellent overview of font terms specific to R in Typography and R⁵, as part of the **systemfonts** (Pedersen et al., 2025) package.

To paraphrase, the commonly used term "font" typically refers to a 'typeface', for example 'Arial' and 'Helvetica' are each typefaces. The typeface may have several variations that may include width (normal, bold, or thin), and style (normal, italic, or oblique). Each variation represents its own "font" within the typeface. Some typefaces such as 'Helvetica Neue' may have 18 or more individual fonts!

With these terms in mind, R refers to a typeface with either the term 'fontfamily' or 'family', and this convention is followed in Venndir.

The systemfonts⁶ R package (Pedersen et al., 2025) is used by Venndir, by way of the marquee⁷ R package (Pedersen and Mitáš, 2025).

A function `subset_systemfonts()` may be helpful to browse the available fonts on the system running R. The following example demonstrates how to summarize available fonts for 'Helvetica Neue', shown in Table 3.3.

```
subset_systemfonts(grepl("Helvetica Neue", family))
```

(fig:systemfonts-1) Partial summary of 'Helvetica Neue' fonts via systemfonts.

Note that it uses `grepl()` on the column 'family' to subset the set of all available fonts. To list all fonts, use: `subset_systemfonts()`.

⁵https://systemfonts.r-lib.org/articles/fonts_basics.html

⁶<https://systemfonts.r-lib.org>

⁷<https://marquee.r-lib.org>

3.5.4.1 Font family visualization

The results can be plotted by adding `do_plot=TRUE`, as a basic way of viewing the resulting fonts. Note that there are no provisions to prevent overlapping labels, so this approach is considered as a "quick check" to view only a few fonts.

```
ss <- subset_systemfonts(
  grepl("Helvetica", family),
  do_plot=TRUE)
```

Helvetica-Light	HelveticaNeue-UltraLight
Helvetica	HelveticaNeue-Thin
Helvetica-Bold	HelveticaNeue-Light
<i>Helvetica-LightOblique</i>	HelveticaNeue
<i>Helvetica-Oblique</i>	HelveticaNeue-Medium
<i>Helvetica-BoldOblique</i>	HelveticaNeue-Bold
	HelveticaNeue-CondensedBold
	HelveticaNeue-CondensedBlack
	<i>HelveticaNeue-UltraLightItalic</i>
	<i>HelveticaNeue-ThinItalic</i>
	<i>HelveticaNeue-LightItalic</i>
	<i>HelveticaNeue-Italic</i>
	<i>HelveticaNeue-MediumItalic</i>
	<i>HelveticaNeue-BoldItalic</i>

Figure 3.36. Graphical output as a "quick check" of fonts where that match the `subset_systemfonts()` output. These fonts show fonts where 'family' contains 'Helvetica'.

3.5.4.2 Automatic font substitution

Another helpful feature of `systemfonts` is that it will substitute a font, or a glyph within a font, when it cannot otherwise be found. In `Venndir`, this approach helps ensure suitable Unicode characters are used by default for example.

For example, the font family `'sans'` is replaced with the appropriate corresponding font, which is important because it differs on Windows, Mac, and linux architectures. The typical defaults, for example: 'Arial' on Windows; 'Helvetica' on MacOS; 'DejaVu Sans' on Linux.

Similarly, when trying to use 'Arial' on a machine which lacks 'Arial', `systemfonts` will substitute a suitable replacement, for example 'DejaVu Sans' or 'Helvetica'.

Table 3.4. Output from the font info function in systemfonts showing the font substitution for 'sans'.

path	index	family	style	italic	bold	monospace	weight	width
/System/Library/Fonts/Helvetica.ttc	0	Helvetica	Regular	FALSE	FALSE	FALSE	normal	normal

The font substitutions can be pre-defined, see `systemfonts::font_fallback()`.

To review the font substitution, one can use `systemfonts::font_info()` in the simple example below, shown in Table 3.4. There are many more options described in the systemfonts documentation.

```
data.frame(systemfonts::font_info("sans")[, 1:9])
```

3.6 Venn Legends

One of many lessons learned from creating hundreds of Venn diagrams: It is helpful to answer the most common questions upfront. One of the most frequent questions is "How many items are in A?" A straightforward answer can be provided in a table legend.

By default `venndir()` also calls `venndir_legender()` which adds a legend to each figure. This behavior can be skipped by using `draw_legend=FALSE`.

```
setlist <- make_venn_test()
v <- venndir(setlist)
```

The legend is simple, one column 'Set' lists each set name, and one column 'Size' lists the size of each set. The background color matches the Venn diagram, and the text is adjusted light or dark to maximize contrast.

The bottom row also includes 'Total' with the total number of unique items represented by the Venn diagram.

When the Venn diagram displays **signed data**, the legend will also include **signed counts**.

```
setlist_signed <- make_venn_test(do_signed=TRUE)
v <- venndir(setlist_signed)
```

Similarly, when the Venn diagram displays percentage values, a column is added 'Percent'.

```
v <- venndir(setlist_signed, show_labels="Ncps")
```

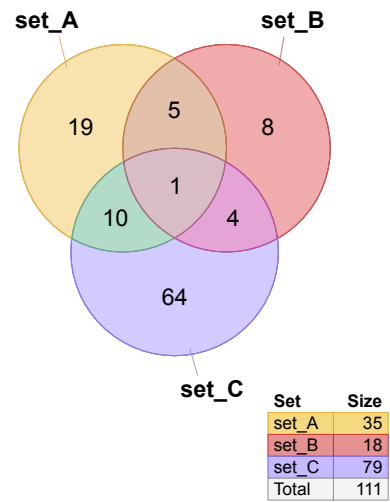


Figure 3.37. Default Venn diagram, showing a table legend in the bottom right corner.

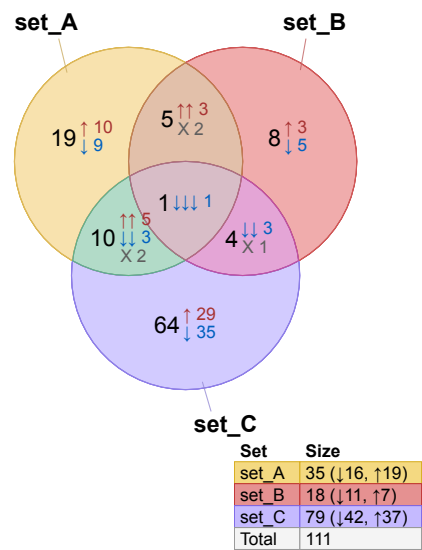


Figure 3.38. Venn diagram showing a table legend in the bottom right corner, this time the column 'Size' also includes counts tabulated by up and down directionality.

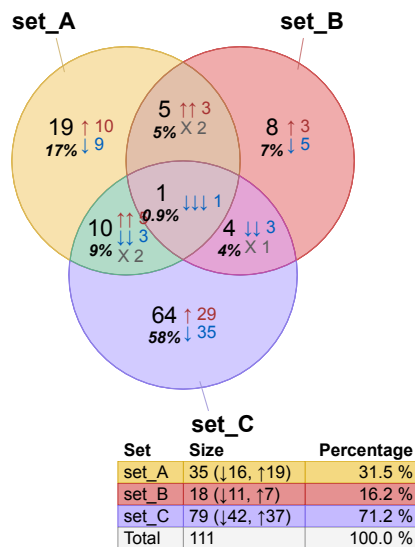


Figure 3.39. Venn diagram showing a table legend in the bottom right corner, now with new column 'Percent' since the Venn diagram also includes the percentage with each count.

3.6.1 Hide percent or signed labels

When percentage or signed labels are displayed in the Venn diagram, the default legend will also include these labels. However, for simplicity it may be preferred to hide these details from the legend, in order to display a cleaner legend. Several optional arguments can be used to hide specific components of the legend.

- `legend_signed=FALSE`
- `legend_percentage=FALSE`
- `legend_total=FALSE`

```
v <- venndir(setlist_signed,
  show_labels="Ncps",
  legend_signed=FALSE,
  legend_percentage=FALSE)
```

3.6.2 Custom legend labels

The legend labels may be customized in `venndir()` with the argument `legend_labels`. This argument defines the custom label in the `Venndir` object so it is used in subsequent plots.

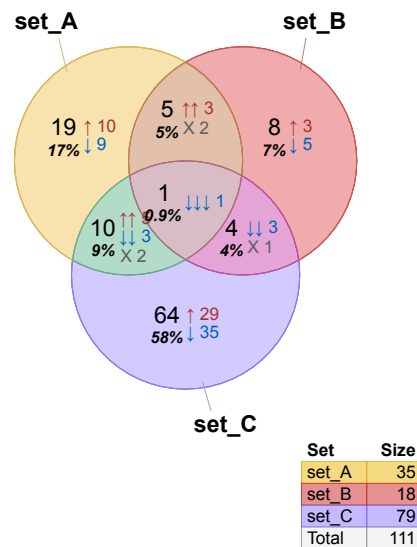


Figure 3.40. Venn diagram showing counts, percentage values, and signed counts. A Venn legend is shown which includes the 'Set' and 'Size' columns, with the 'Percentage' column not shown due to argument 'legend_percentage=FALSE'.

```
v <- venndir(setlist_signed,
  legend_labels=c("Example Set A",
    "Example Set B",
    "Example Set C"))
```

Note that `legend_labels` will only customize the labels in the table legend, and not on the Venn diagram itself. Sometimes it is necessary to customize the Venn diagram labels, with the argument `setlist_labels`.

A common reason to have distinct labels in the figure and legend is to impose line breaks in the figure which are not necessary in the table. At other times, it makes sense to have one detailed label and one simplified label.

```
v <- venndir(setlist_signed,
  expand_fraction=c(0.2, 0, 0, 0),
  setlist_labels=c("Example\nSet A",
    "Example\nSet B",
    "Example\nSet C"),
  y_inset=grid::unit(1, "lines"),
  legend_labels=c("Example Set A",
    "Example Set B",
```

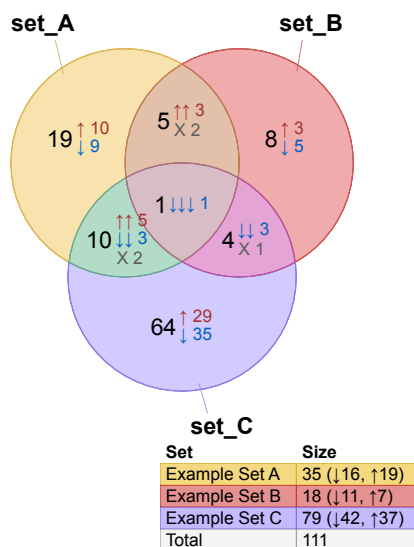


Figure 3.41. Venn diagram showing a table legend in the bottom right corner, with custom labels for each set, for example 'set_A' is renamed 'Example Set A' in the legend.

```
"Example Set C"))
```

3.6.3 Additional alias labels

The argument `alias` can be used to supply simple labels, which then pushes `legend_labels` into a new column 'Label'. The result may provide a helpful technique to associate extra details to the Venn diagram labels.

Note that `alias` must be named using the set names, matching the values in `names(setlist)`.

```
v <- venndir(setlist_signed,
  legend_headers=c(Set="Set",
    Size="Size", Percentage="Percentage",
    Sign="Sign", Label="Contrast"),
  alias=c(set_A="A",
    set_B="B",
    set_C="C"),
  setlist_labels=c("Set A",
    "Set B",
    "Set C"),
```

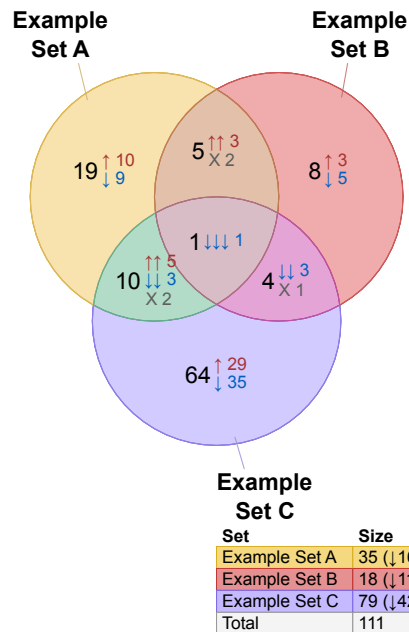


Figure 3.42. Venn diagram showing a table legend in the bottom right corner, with custom labels for each set. 'set_A' is renamed 'Example Set A' in the legend.

```
legend_labels=c(set_A="Dex - control",
  set_B="PGH - control",
  set_C="E2 - control"))
```

Figure 3.43 also uses the optional argument `legend_headers` to define custom headings in the legend, this adding 'Contrast' as a new header.

3.6.4 Separate Size and Sign

The default legend combines the set size together with **signed counts** when the `VennDir` data contains **signed setlist**. In other words, the label 30(↓16, ↑19) is displayed in one field.

Two optional arguments provide alternatives to the default:

- `combine_size=FALSE` The total size will not be combined with **signed counts**.

```
venndir(setlist_signed,
  combine_size=FALSE)
```

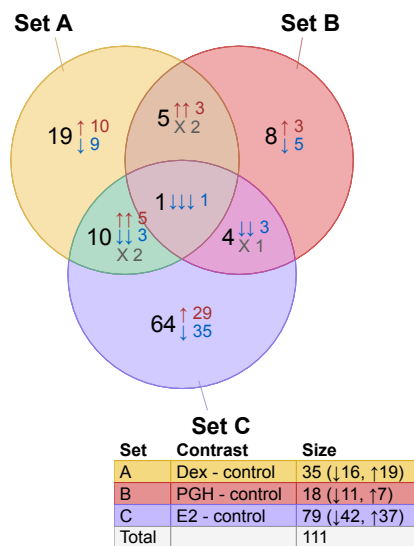


Figure 3.43. Venn diagram showing a table legend in the bottom right corner, with custom labels for each set. 'Set A' in the Venn diagram is indicated 'A' in the legend, with a new column 'Contrast' with descriptive information.

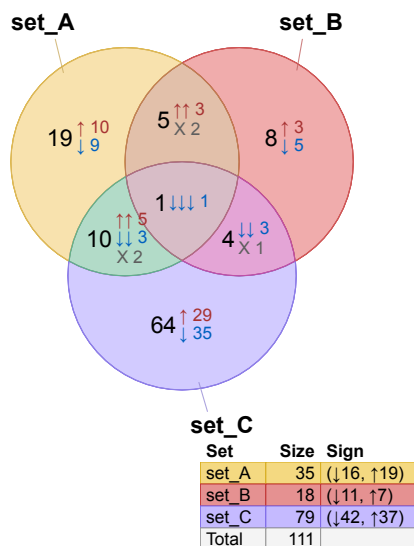


Figure 3.44. Venndir legend with separate columns for 'Size' and 'Sign'.

- `combine_signed=FALSE` The **signed counts** will not be combined and displayed in parentheses.

```
venndir(setlist_signed,
        combine_signed=FALSE)
```

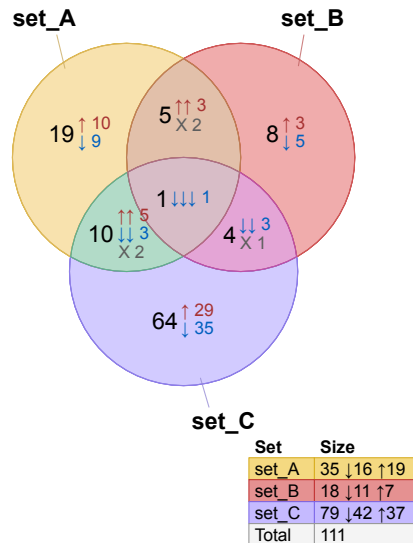


Figure 3.45. Venndir legend with one column 'Size', whose values not delineated by parentheses.

- `combine_size=FALSE`, `combine_signed=FALSE` The total size and **signed counts** will each appear in separate columns.

```
venndir(setlist_signed,
        combine_signed=FALSE,
        combine_size=FALSE)
```

3.6.5 Legend color style

The default legend uses the same color fill as the Venn diagram. However, the style can be customized using argument 'legend_color_style' to control both the fill and border colors.

Fill color

- "fill" uses the Venn `set_colors` after applying `poly_alpha`
- "nofill" uses no fill color, inheriting the plot background
- "greyfill" uses light grey fill color

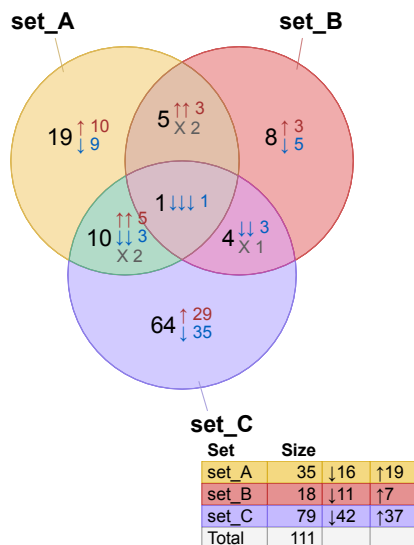


Figure 3.46. Venn diagram with legend that shows separate columns with 'Size', and two additional columns with signed counts which do not have a header label.

Border color

- "border" uses the Venn `set_colors` with no alpha transparency
- "noborder" uses no border
- "greyborder" uses medium-dark grey border
- "blackborder" uses black border

For example, to use no color fill, and black border:
`legend_color_style=c("nofill", "blackborder")`

```
v <- venndir(setlist_signed,
  legend_color_style=c("nofill", "blackborder"))
```

The header can be customized as well, using arguments: `header_color` for the text color, `header_bg` for background fill, and `header_border` for the header border.

3.6.6 Legend position

The legend itself is positioned using argument `legend_x` when called by `venndir()`, or `x` when called by `venndir_legender()`. The position recognizes the following character terms:

- x-axis position: 'left', 'center', 'right'
- y-axis position: 'top', 'center', 'bottom'

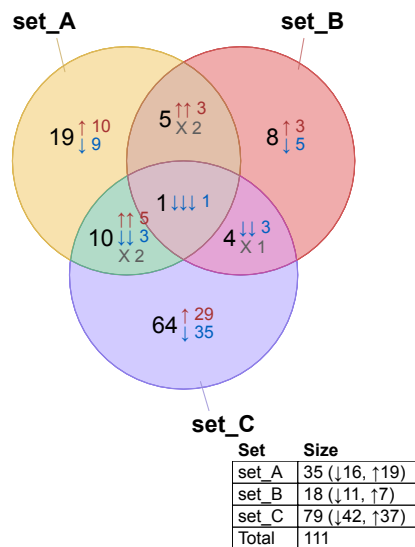


Figure 3.47. Venn diagram with a legend drawn using black border, and no background color.

The default position is "bottomright"

Some fine-tuning is available with two more arguments:

- `x_inset` - `grid::unit` object applied when `x` is 'right' or 'left', intended to apply an inset distance from the edge of the figure. The default is `grid::unit(2, "lines")` which moves the legend two character lines from the edge of the figure.
- `y_inset` - `grid::unit` object applied when `x` is 'top' or 'bottom', which is applied the same way as described for `x_inset`. The default is 2 character lines.

```
v <- venndir(setlist_signed,
  legend_x=c("bottomleft"))
```

For example, to move the legend closer to the bottom edge, and closer to the left edge, see the following example. Note the function `grid::grid.rect()` is used here to draw a draw box around the plot region.

```
v <- venndir(setlist_signed,
  x_inset=grid::unit(0, "lines"),
  y_inset=grid::unit(0, "lines"),
  legend_x=c("bottomleft"))
grid::grid.rect(gp=grid::gpar(fill=NA, lwd=2, col="grey"),
```

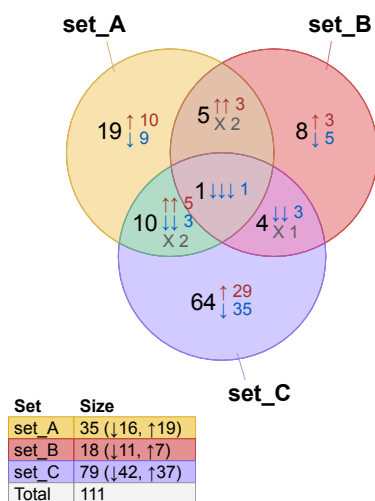


Figure 3.48. Venn diagram showing the legend positioned in the bottom left corner, using argument 'legend_x' inside the `venndir()` function.

```
vp=attr(v, "viewport")
grid::grid.rect(gp=grid::gpar(fill=NA, col="black"))
```

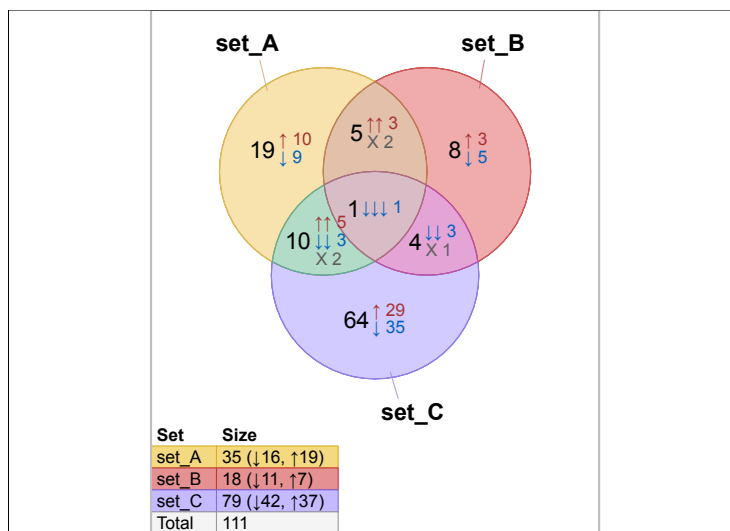


Figure 3.49. Venn diagram with legend in the bottom left corner, adjusted to the bottom edge, and moved more to the left of the figure. The grey box indicates the Venn figure region, the black box indicates the overall figure.

Note the x-axis position is not the exact left edge of the plot itself. The legend is drawn within the `Venndir` `grid` viewport, defined with fixed aspect ratio units `'snpc'` (see `grid::unit()`). As a result the `Venndir` viewport is roughly square, and if the graphics device is wider or taller than square, extra whitespace is imposed. The argument `x_inset` accepts negative values that can position the legend further left when necessary.

An alternative is to call `venndir()` with `draw_legend=FALSE`, then separately call `venndir_legender()` to create the legend. This process draws the legend in the overall viewport.

In Figure 3.50, the legend is positioned on the exact bottom-left edge of the plot, shown by the grey line drawn with `grid::grid.rect()`.

```
v <- venndir(setlist_signed,
  draw_legend=FALSE)
grid::grid.rect(gp=grid::gpar(fill=NA, lwd=2, col="grey"),
  vp=attr(v, "viewport"))
grid::grid.rect(gp=grid::gpar(fill=NA, col="black"))
vl <- venndir_legender(v,
  x_inset=grid::unit(0, "lines"),
  y_inset=grid::unit(0, "lines"),
  x=c("bottomleft"))
```

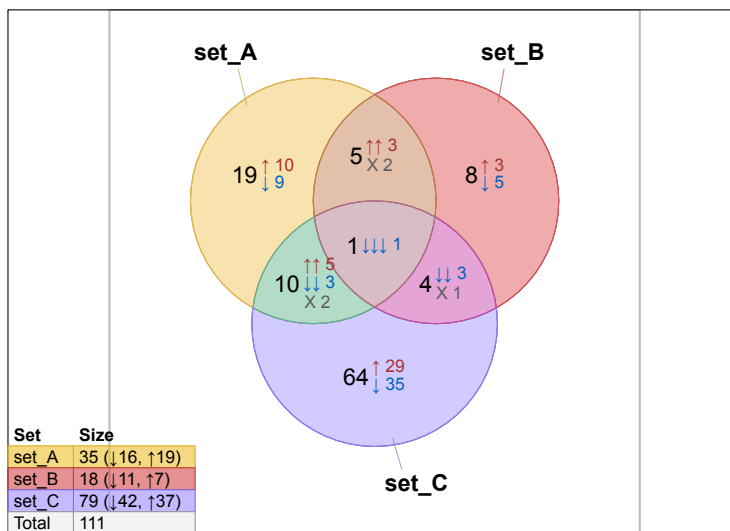


Figure 3.50. Venn diagram with legend in the bottom left corner, this time with the legend touching the exact bottom left edge of the overall figure. The grey box indicates the Venn figure region, the black box indicates the overall figure.

Altogether, the use of `venndir()` with argument `expand_fraction` should

provide detailed control over the position of the Venndir diagram. When necessary, the legend can be drawn separately with `venndir_legender()` to afford exact control over the position of the legend.

In future the plot method used for Venndir may be updated to use 'npc' coordinates instead of 'snpc', which may enable the legend to be freely positioned in the encompassing viewport, not the Venndir viewport. For now, this update is lower on the Todo list.

3.6.7 Legend font size

The legend font size can be adjusted with `legend_font_cex` when called by `venndir()`, otherwise it uses `font_cex` when called using `venndir_legender()`.

```
v <- venndir(setlist_signed,
  legend_font_cex=1.3,
  x_inset=grid::unit(-3, "lines"),
  y_inset=grid::unit(0, "lines"),
  legend_x=c("bottomleft"))
```

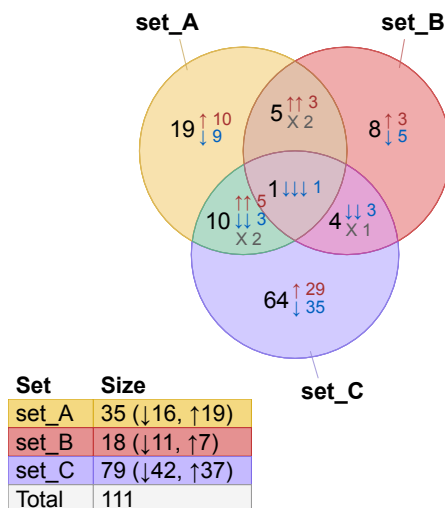


Figure 3.51. Venn diagram with legend drawn in the bottom left corner, using font size adjusted to 1.3 times the default font size with argument 'font_cex'.

3.6.8 Legend custom fontfamily

The legend fontfamily will use the `fontfamily` defined in the `Venndir` object (for example `v@metadata$fontfamily`). For most cases, supplying a custom `fontfamily` to `venndir()` will suffice, since it will apply the same custom font to the figure and to the legend.

It is possible to use a custom font specifically for the legend.

This process requires calling `venndir_legend()` independently, therefore the legend should not be drawn by `venndir()`.

The steps are described and demonstrated below:

1. Call `venndir()` with `draw_legend=FALSE`, assign output to a variable, in this case `v`.
2. Call `venndir_legend()` using the `Venndir` object `v` and custom `fontfamily`.

```
v <- venndir(setlist_signed,
  fontfamily="sans",
  draw_legend=FALSE)
vl <- venndir_legend(v,
  x="bottomright",
  x_inset=grid::unit(0, "lines"),
  y_inset=grid::unit(0, "lines"),
  font_cex=1.3,
  fontfamily="serif")
```

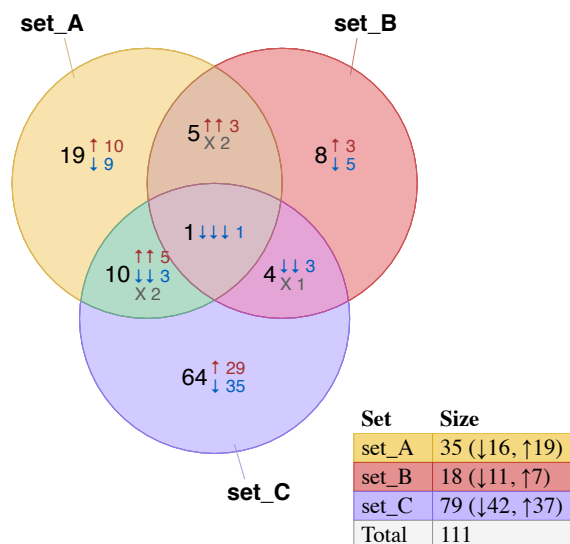


Figure 3.52. Venn diagram with legend drawn in the bottom right, using sans font for the Venn diagram labels, and serif font for the legend.

3.7 Figure Scale and Position

The Venndir figure is automatically sized to fit within the graphics device, and will re-adjust even when the figure is resized in RStudio or an R graphics window. However, it can be helpful to customize the scale and position further.

Common reasons to adjust scale and position:

- The outside labels extend beyond the page, usually when text labels contains multiple lines.
- The Venndir figure, or the outside labels overlap the legend.
- The figure could be resized to fit specific output dimensions.

Venndir uses the Venn circles and visible labels to create a square "bounding box". It is square to keep the aspect ratio 1:1, so circles remain circular. If the graphics device is not square, there will be blank space around the Venndir figure. Sometimes it helps to "zoom" the Venndir figure to fill the remaining space.

The argument `expand_fraction` adjusts the whitespace "buffer" around the Venndir diagram.

- `expand_fraction` above 0 makes the whitespace larger, shrinking the Venn circles.
- `expand_fraction` below 0 makes the whitespace smaller, enlarging the Venn circles.
- The font sizes are not adjusted.

The `expand_fraction` accepts four values, applied in clockwise order from the bottom:

`'bottom', 'left', 'top', 'right'`.

Values are recycled to length=4, so `expand_fraction=0.1` is applied to all four sides.

When the legend is shown, the Venndir figure is shifted up 0.18.

```
setlist <- make_venn_test()
venndir(setlist)
venndir(setlist, draw_legend=FALSE)
venndir(setlist, main="Main title")
venndir(setlist, draw_legend=FALSE, expand_fraction=-0.1)
venndir(setlist, sets=1:2)
venndir(setlist, sets=1:2, expand_fraction=c(0.1, 0, 0, 0) - 0.1)
```

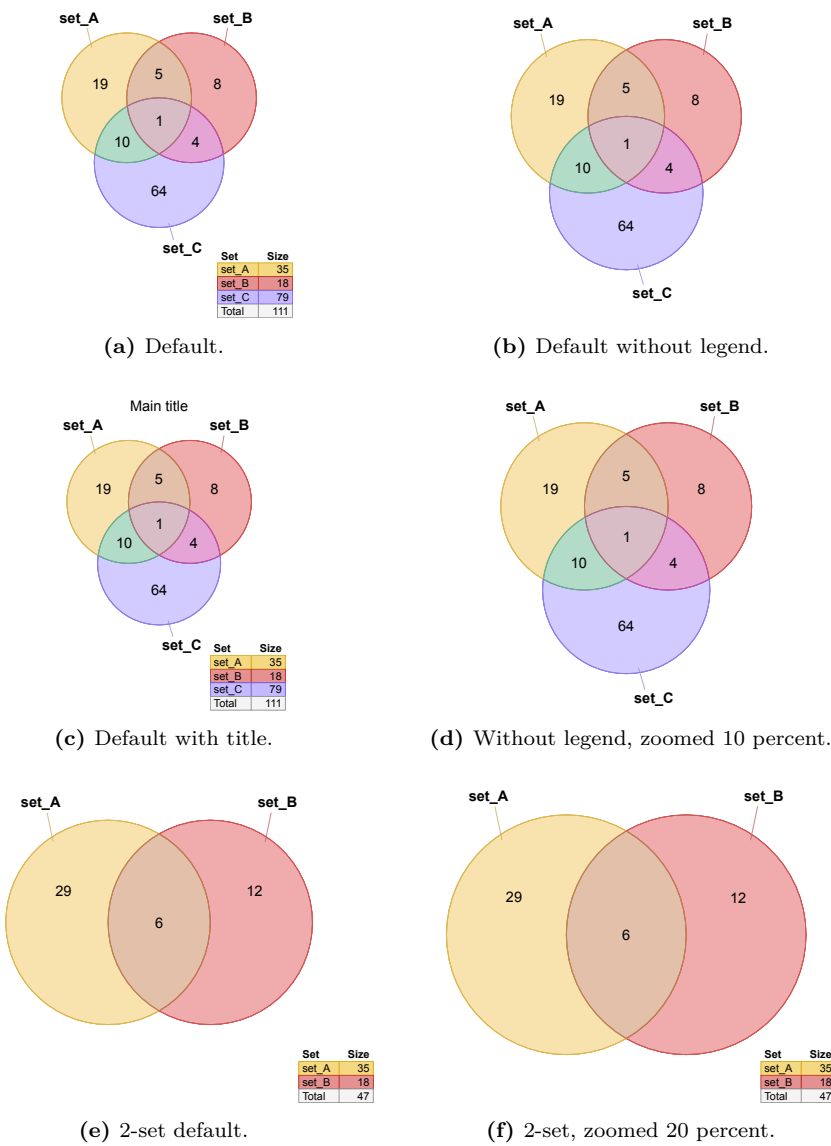



Figure 3.53. Venndir figure showing different values for 'expand_fraction'.

Tips:

The `expand_fraction` argument rules:

- When user-defined values are provided, or if they are stored in the metadata of the `Venndir` object, `expand_fraction` is used without any adjustments.
- When `expand_fraction` is not defined, these default adjustments are applied:
 1. `0.07` is added to the top margin for each line in the main title.
 2. If the legend is shown, using `draw_legend=TRUE`, and the legend position `legend_x` includes 'top' or 'bottom', `0.2` is added to the corresponding 'top' or 'bottom' margin. It will not adjust the 'left' or 'right' margin.
 3. The top and bottom margins are reduced slightly.
- The `expand_fraction` is stored in the returned `Venndir` object metadata when the plot is prepared, for example:
 - calling `plot()` or `render_venndir()`, or
 - calling `venndir()` with the default `do_draw=TRUE`.

3.7.1 Text Venn

Another convenient format for `Venndir` output is text Venn, intended for R console output. This output is convenient for analysis work on remote servers where interactive graphics are not available or not as convenient.

The core function `textvenn()` mimics the input and arguments of `venndir()` with some key limitations:

1. The input can only contain up to three sets.
2. The display only supports Venn layout, not proportional.
3. Output is a `data.frame` and not `Venndir` object.

The output follows the typical arrangement of text labels for a Venn diagram, without the encompassing circular lines. Colors and directional arrows are represented if the console text output support these features, otherwise `unicode=FALSE` will display '^' for "up", and 'v' for "down".

The basic 2-way `textvenn()` output follows:

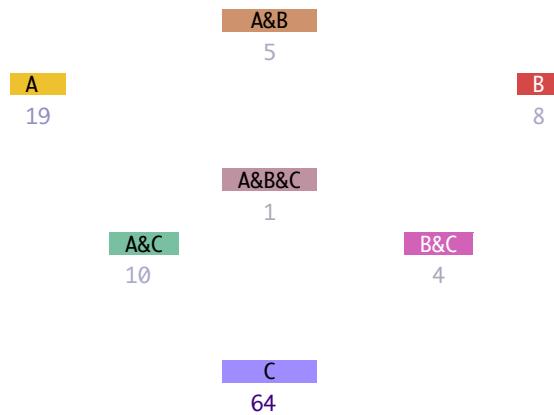
```
setlist <- make_venn_test(set_names=LETTERS[1:3])
textvenn(setlist, sets=c(1, 3))
```

A 24	A&C 11	C 68
--	--	--

The argument `sets` is described in [Consistent Set Colors](#), and is used to include a subset of sets while maintaining consistent color assignment to each set.

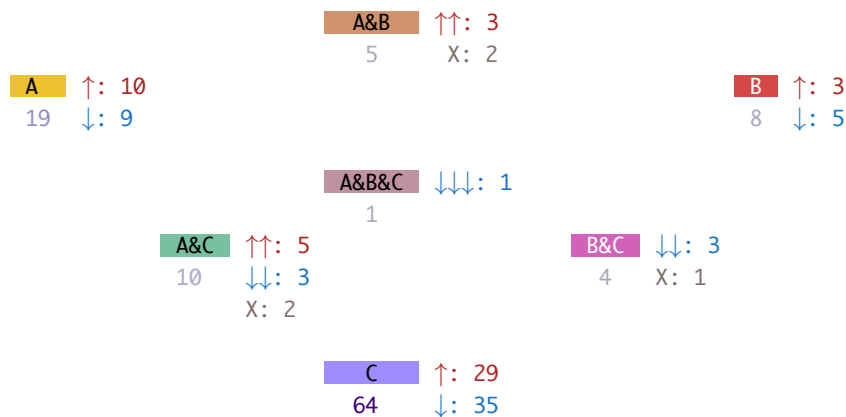
The 3-way `textvenn()` output follows:

```
textvenn(setlist)
```



These examples show only the Venn overlaps without directionality. However, if given a [signed setlist](#) it will also include directional tabulations:

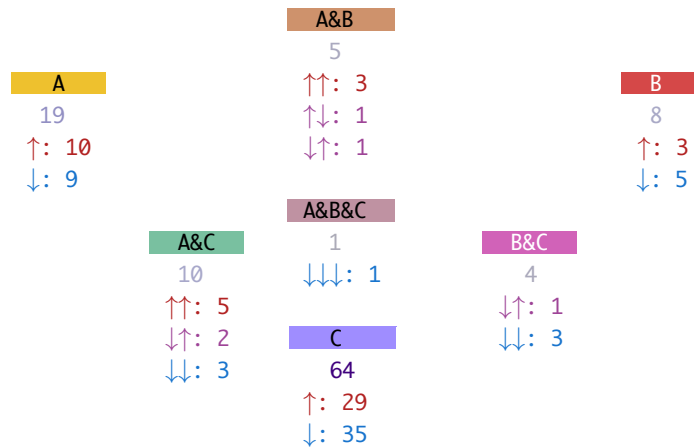
```
setlist3 <- make_venn_test(set_names=LETTERS[1:3],
  do_signed=TRUE)
textvenn(setlist3)
```



The `template` argument described in [Signed Label Placement](#) is also recognized. For example, `template='tall'` may improve the layout especially when using `overlap_type='each'`, an option for tabulating directionality that

is described in [Overlap Type](#).

```
textvenn(setlist3,
  overlap_type="each",
  template="tall")
```



3.7.1.1 Text Venn Items

A useful byproduct of `textvenn()` is that it returns a `data.frame`, invisibly, which means it can be assigned to a variable but otherwise is not printed to the output device. The `data.frame` includes the count summary for each overlap, in addition to the 'items' in each set.

It is straightforward to use `textvenn()` to show the number of shared and unique items, then use the 'items' column to answer the *most common next question*: "What are those?"

The example below captures `textvenn()` output into a variable `tv`.

```
tv <- textvenn(setlist, sets=c(1, 3), do_plot=FALSE)
```

```
##      A      A&C      C      FALSE
##     24     11     68     FALSE
```

The items can be printed or used for subsequent analysis:

```
tv$items
```

```
## $`A|1 0`
## [1] "item_014" "item_170" "item_043" "item_200" "item_091" "item_185"
## [7] "item_092" "item_137" "item_099" "item_072" "item_026" "item_007"
```

```

## [13] "item_198" "item_186" "item_164" "item_078" "item_081" "item_199"
## [19] "item_103" "item_117" "item_076" "item_109" "item_180" "item_023"
##
## $`C|0 1`
## [1] "item_041" "item_175" "item_060" "item_016" "item_116" "item_094"
## [7] "item_006" "item_086" "item_192" "item_039" "item_159" "item_034"
## [13] "item_004" "item_013" "item_069" "item_127" "item_052" "item_022"
## [19] "item_089" "item_160" "item_025" "item_035" "item_168" "item_112"
## [25] "item_030" "item_140" "item_189" "item_121" "item_110" "item_158"
## [31] "item_064" "item_142" "item_067" "item_151" "item_122" "item_079"
## [37] "item_085" "item_136" "item_051" "item_106" "item_098" "item_157"
## [43] "item_017" "item_046" "item_054" "item_174" "item_194" "item_161"
## [49] "item_024" "item_113" "item_107" "item_135" "item_102" "item_005"
## [55] "item_070" "item_146" "item_021" "item_055" "item_075" "item_083"
## [61] "item_150" "item_048" "item_077" "item_132" "item_056" "item_111"
## [67] "item_131" "item_068"
##
## $`A&C|1 1`
## [1] "item_195" "item_050" "item_118" "item_196" "item_153" "item_090"
## [7] "item_190" "item_143" "item_032" "item_182" "item_074"

```


Chapter 4

Deeper Customizations

While a Venn diagram may seem simple on the surface, real-world datasets rarely cooperate. At a certain point, those 'minor' details -- directionality, labeling, scaling, overlap type -- start to shape what the figure reveals, and what it obscures.

Every feature in Venndir has a story — some short, some strangely epic — that drove its creation.

4.1 Venndir Borders

The underlying geometry of each polygon is represented as a `JamPolygon` object (Ward 2025), a new S4 object designed for geometries used in Venndir.

An important distinction from typical polygons in R is that the objects are capable of describing three borders:

1. **'border'**: Line along the exact edge of the polygon.
2. **'innerborder'**: Line drawn on the inside edge.
3. **'outerborder'**: Line drawn on the outside edge.

`JamPolygon` objects allow adjacent borders to co-exist without rendering two borders on top of each other.

Figure 4.1 illustrates four examples of borders, showing the problem being addressed with `innerborder`. The left object has two halves labeled A and B, with borders gold and purple, respectively. When using `'border'` the gold border is covered by the purple border down the center line between A and B.

Instead, the next object with C and D shows the effect of using `'innerborder'`, with the respective gold and purple borders both visible. Also notice the object is not enlarged by adding the border, the line width extends *inside* the polygon.

The next panel shows E and F, with 'outerborder' wrapping around them both. The outerborder extends outside the object, and in fact the outerborder does not extend inside the object at all.

The right panel shows G and H, and includes each type of border, drawn in order: innerborder, outerborder, border. The 'border' was configured as a thin line, otherwise it could potentially cover both the innerborder and outerborder.

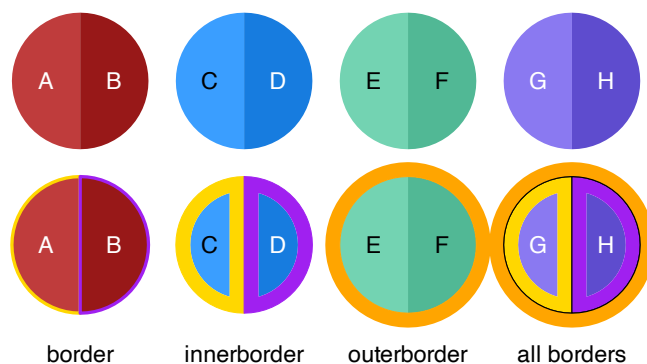


Figure 4.1. Four standard objects are displayed on the top row. The bottom row shows the effect of adding distinct borders: border, innerborder, outerborder, and all borders.

4.1.1 Venndir Rendering Steps

Venndir draws borders in a specific order to provide a layering effect, followed by other graphics elements such as labels and connecting line segments.

1. Venn Sets

- The **outerborder** is drawn using **set_colors** darkened slightly.
- Other borders are not drawn: **fill** color, **border**, and **innerborder**.

2. Venn Overlaps

- Colors from the overlapping sets, from **set_colors**, are blended.
- The **fill** color is drawn after applying **poly_alpha**.
- The filled region does not include the **innerborder.lwd** if the **innerborder** is defined.
- The **innerborder** is drawn using the **fill** color, darkened slightly, and with no alpha transparency.
- Other borders are not drawn: **border** and **outerborder**.

3. Count Labels

- The font colors are adjusted for contrast, relative to the background colors behind each label.

4. **Item Labels** (if enabled)
5. **Line Segments** (if enabled)
6. **Venndir Legend** (if enabled)
7. **Main Title** (if enabled)

The order that Venndir polygons are rendered, as described in (1) and (2) above, is determined by the order of polygons present in the Venndir 'jps' S4 slot. For example, if `v` is a Venndir object, the `JamPolygon` entries are accessed using `v@jps`.

The order in 'jps' can be modified, but all entries are required for the Venndir object to be valid. For example, `Highlight Venn Overlaps` re-orders 'jps' to place the highlighted overlap region last, thereby preventing `border` and `outerborder` from being overdrawn by other entries in 'jps'.

Additional `JamPolygon` entries may be added to `v@jps`, however the new names must not conflict with existing `names(v@jps)`.

Additional notes:

- When plotting a Venndir object, the ellipses '...' can be used to provide custom border settings, and are described in `Customize Venndir borders`.
- By default, Venndir objects do not use `border`, they use `outerborder` for sets, and `innerborder` for overlap regions.
- The overlap regions use colors that are blended based upon `set_colors` defined for the Venndir object.
- Default borders are quite subtle, but the effect is more distinctive when using larger values for `innerborder.lwd` and `outerborder.lwd`.
- It is usually best to use the same numeric value for both `innerborder.lwd` and `outerborder.lwd`, due to the way the borders are layered.

4.1.2 Customize Venndir borders

The border options can be customized directly when plotting the Venndir object, adding named arguments to any function that produces a Venndir plot:

```
* venndir(...) * render_venndir(Venndir, ...) * plot(Venndir, ...)
```

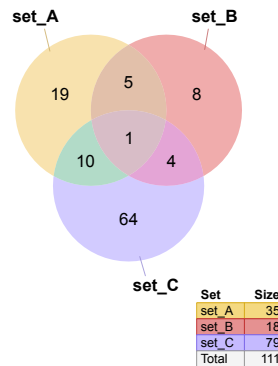
A few common alternatives are shown in Figure 4.2.

```
v1 <- venndir(make_venn_test(),
  innerborder=NA, outerborder=NA)
v2 <- venndir(make_venn_test(),
  border="grey40", border.lwd=1,
  innerborder=NA, outerborder=NA)
```

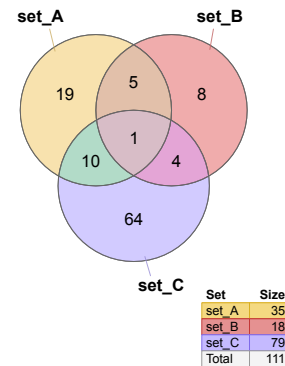
```

v3 <- vennDir(make_venn_test(),
  innerborder="white", innerborder.lwd=1,
  outerborder="white")
v4 <- vennDir(make_venn_test(),
  label_style="lite box", innerborder.lwd=1,
  outerborder="white", outerborder.lwd=2)

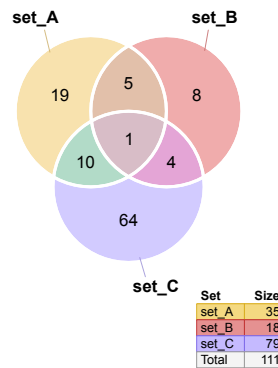
```



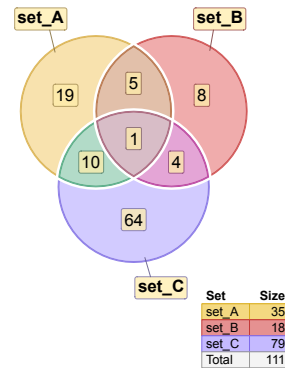
(a) Venndir shown with no borders.



(b) Venndir shown with thin grey border.



(c) Venndir shown with white border.



(d) Venndir with beveled effect.

Figure 4.2. Venndir diagrams demonstrating common border styles.

These options are useful for global changes to borders, but are not ideal to modify a specific region. For those customizations, see [Modify Venn Overlaps](#).

4.2 Modify Venn Overlaps

Each overlap region in the `Venndir` object can be customized to some extent. The `Venndir` object itself can be edited, but this process is tedious and not intended to be user-friendly. Instead `modify_venndir_overlap()` provides the most common modifications.

For a simple way to highlight a `Venndir` overlap, see [Highlight Venn Overlaps](#). This alternative offers a simple variation of `modify_venndir_overlap()`.

As described in [Venndir Rendering Steps](#), **Sets** are drawn as open circles, then each **Overlap** is drawn as a color-filled region. **Labels** are then rendered last. Each overlap may have options defined related to the region and the label. For example `fill` is applied to the polygon background color, however `label.fill` is applied to the corresponding label. The full set of recognized parameters will be described later.

Overlap regions have specific names using the set or sets involved. * `'set_A'` refers to the region unique to `'set_A'` with no other overlapping sets. * `'set_A&set_B'` refers to the region where the two sets `'set_A'` and `'set_B'` overlap, with no other overlapping sets.

The overlap name is provided with argument `overlap_set`. See the example [4.3](#)

```
# default Venndir
v <- venndir(make_venn_test(do_signed=TRUE),
  do_plot=FALSE)
plot(v,
  main="Default Venndir")

# modified Venndir
v_mod <- modify_venndir_overlap(v,
  overlap_set="set_A&set_B",
  params=list(
    fill="orange",
    alpha=0.8,
    fontsize=c(22, 16, 16),
    innerborder="royalblue",
    innerborder.lwd=5)
)
plot(v_mod,
  main="Modified 'set_A&set_B'")
```

This customization is fairly common, and straightforward:

- fill color was changed to `'orange'`.
- alpha was defined 0.8, to be applied to fill.
- innerborder color was changed to `'royalblue'`.

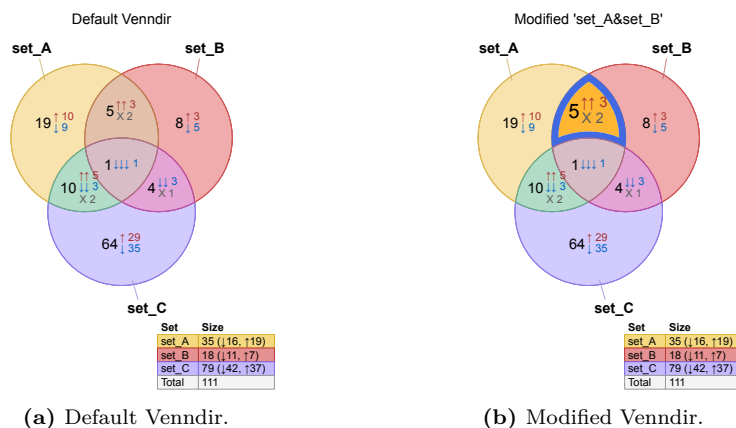


Figure 4.3. Venndir example showing a default Venn diagram, and a modified diagram (right) affecting the overlap region for 'set_A&set_B'. Changes were made to the fill color, innerborder, innerborder.lwd, and fontsize.

- innerborder.lwd line width was set to 5.
- Label fontsize values were increased.

Note that fontsize is applied to each individual label in the order it is drawn.

4.2.1 Overlap Attributes

The recognized Venndir overlap attributes are shown in Table 4.1.

(fig:overlap-attr-table) List of recognized region attributes, used to customize the Venndir figure.

Table 4.1. List of recognized region attributes, used to customize the Venndir figure.

Attribute	Description
Region Attributes	
fill	R color used to fill the overlap region, after also applying ‘alpha’. The default is derived from the set_colors for the sets involved.
alpha	Numeric alpha transparency applied to the ‘fill’ color, where alpha=0 is fully transparent, and alpha=1 is opaque.
border	R color for the on-edge border. The on-edge border is not recommended for most figures. Use border=NA to draw no border, which is default.
border.lwd	The numeric line width for the ‘border’ when defined.
innerborder	R color for the inner-boundary border, recommended in most customizations. When innerborder=NA, or innerborder.lwd=0 the innerborder is not drawn.
innerborder.lwd	The numeric line width for the ‘innerborder’. When innerborder=NA, or innerborder.lwd=0 the innerborder is not drawn.
outerborder	R color for the outer-boundary border, not recommended in most customizations. When outerborder=NA, or outerborder.lwd=0 the outerborder is not drawn.
outerborder.lwd	The numeric line width for the ‘outerborder’. When outerborder=NA, or outerborder.lwd=0 the outerborder is not drawn.
Label Attributes	
fontsize	Numeric font size in points for each overlap label as defined in the Venndir object.
label.color	Text color for each overlap label as defined in the Venndir object.
label.fill	Background label fill color with optional alpha transparency. When label.fill=NA, no fill is applied. The label.fill also affects the label.color, using fill and label.fill together as relevant with ‘make_color_contrast()’ to ensure contrasting text.
label.border	Optional border color around the label group. Use NA for no border.
label.count	Position or visibility for each count label as defined the Venndir object. Values are ‘outside’, ‘inside’, or ‘none’.
label.overlap	Position and visibility for the set overlap label. Currently only Set labels can be visible. Values are ‘outside’, ‘inside’, or ‘none’.

4.3 Customize Any Label

VennDir labels are customized using `modify_venndir_overlap()` as described in [Modify Venn Overlaps](#). Some label adjustments are best illustrated with specific examples.

The recognized attributes are described in [Overlap Attributes](#).

The overall font sizes used in a VennDir object are defined with argument `font_cex` when first calling `venndir()`, as described in [Font Sizes](#). The font point sizes are stored in the VennDir object, and may be modified.

The most common adjustments involve the font size, color, and optional border and label fill.

The figures in [Figure 4.4](#) show:

- the default VennDir
- customizations to one overlap count label
- moving this overlap label outside the figure.

```
# default VennDir
v <- venndir(make_venn_test(do_signed=TRUE),
  main="Default VennDir")
# modified VennDir
v_mod <- modify_venndir_overlap(v,
  overlap_set="set_A&set_C",
  params=list(fill="royalblue",
    alpha=1,
    fontsize=c(20, 16, 12, 12, 12),
    label.border="navy",
    label.fill="royalblue3",
    label.color=c("white", "pink", "skyblue"),
    innerborder="royalblue4",
    innerborder.lwd=5))
plot(v_mod, main="Modified 'set_A&set_C'")
# move the label outside
v_mod1 <- modify_venndir_overlap(v_mod,
  overlap_set="set_A&set_C",
  params=list(label.count="outside"))
plot(v_mod1,main="Outside 'set_A&set_C'")
```

A short list of techniques follows:

- To move the overlap count labels outside, set `label.count='outside'`.
- To move the overlap count labels inside, set `label.count='inside'`.
- To move an overlap Set label outside, set `label.overlap='outside'`.
- To move an overlap Set label inside, set `label.overlap='inside'`.
- To change label colors, define `label.color` for each label, in order.

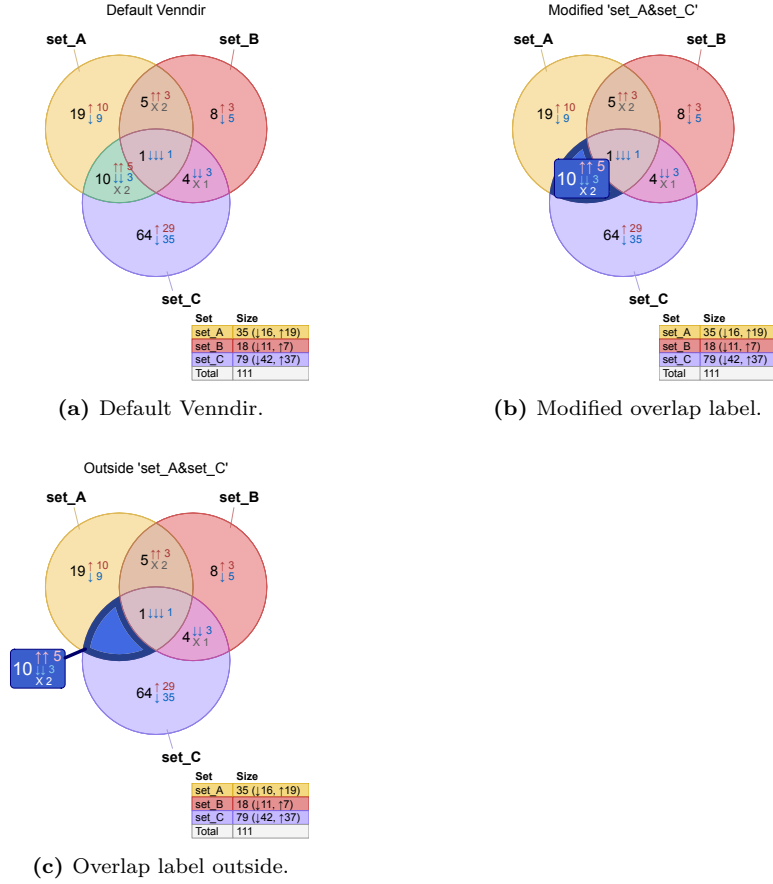


Figure 4.4. Venndir (left) and modified Venndir with customized label attributes (right), and how to move the label outside (bottom).

4.4 Automatic Text Contrast

In the previous section Figure 4.4, the overlap fill defined by `'label.fill'` was set to blue. Text colors were also manually changed using `label.color` to improve visual contrast. This step may be automated by using `make_color_contrast()`.

Given a set of input text colors in argument `x`, and background colors in `y`, the function returns a vector of colors to contrast with the background. The purpose is to retain some color saturation, as opposed to returning either white or black, while also prioritizing visual clarity.

An optional argument `bg` can be used to define the canvas background color, which is useful when the colors in `y` have partial transparency, and would also

partly show the canvas color.

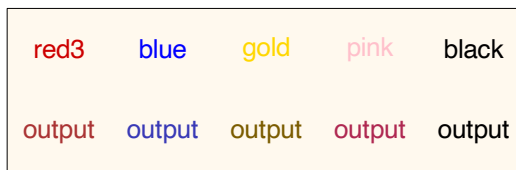
Two additional arguments to `make_color_contrast()` may be relevant: `L_lo=40` and `L_hi=95`, which control the output color darkness and lightness, respectively. Adjustments may be passed through `venndir()` using `'...'` ellipses to modify the color saturation, for example: `venndir(setlist, L_hi=85)`.

The argument `do_plot=TRUE` is used to visualize the input and output colors for review.

```
x <- c("red3", "blue", "gold", "pink", "black");
xc1 <- make_color_contrast(x, y="royalblue3", do_plot=TRUE);
xc2 <- make_color_contrast(x, y="#FF9EE", do_plot=TRUE);
```



(a) Blue background.



(b) Off-white background.

Figure 4.5. Two examples with blue background (top) and off-white background (bottom). In each plot, the top row labels indicate the input colors, the bottom row labels are the output colors adjusted to improve contrast.

Putting these techniques together, the example in Figure 4.6 uses `modify_venndir_overlap()` together with `make_color_contrast()` to ensure text labels have visual contrast.

```
v <- venndir(make_venn_test(do_signed=TRUE), do_plot=FALSE)

new_bg <- "darkgreen";
current_colors <- c("black", "firebrick", "dodgerblue3", "purple4");

v2 <- modify_venndir_overlap(v,
```



```

overlap_set="set_A&set_C",
params=list(
  label.fill=new_bg,
  label.border="black",
  label.color=make_color_contrast(current_colors, new_bg),
  label.count="outside",
  innerborder=new_bg,
  innerborder.lwd=3)
)
plot(v2)

```

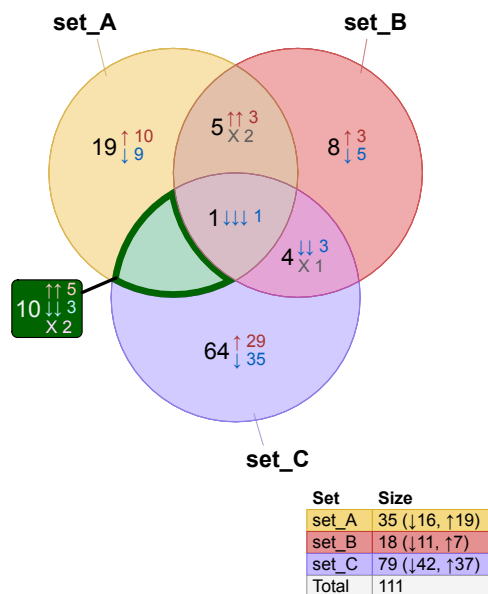


Figure 4.6. Venn diagram with modifications to the label for 'set_A&set_B'.

The modifications made in this figure:

- new_bg was defined as 'darkgreen'.
- new_bg was applied to the label.fill and the innerborder.
- innerborder.lwd was increased to 3.
- label.border was changed to black, to contrast with dark green.
- label.count was set to 'outside' to place count labels outside.
- current_colors were defined to match the existing label text colors.
- label.color was defined using make_color_contrast(), with arguments current_colors and new_bg.

4.5 Highlight Venn Overlaps

A convenient alternative to the previous section [Modify Venn Overlaps](#) is to apply a highlight with `highlight_venndir_overlap()`. This step applies a few options together.

The default settings:

- gold border
- yellow fill color
- red
- `fill = 'yellow'`
- `innerborder = 'gold'`
- `border = 'red'`

The example in [4.7](#) shows the effect of highlighting one overlap region, 'set_B&set_C'. The third panel demonstrates how to move the count labels outside with a background fill color. The line segment color is defined by `border` with default 'red', however if 'label.border' is defined, that color is slightly darkened and used for the line segment as well.

```
# default Venndir
v <- venndir(make_venn_test(do_signed=TRUE), do_plot=FALSE)
plot(v,
     main="Default Venndir")
v_mod <- highlight_venndir_overlap(v,
    border="red",
    "set_B&set_C")
plot(v_mod,
     main="Highlighted set_B&set_C")
v_mod2 <- modify_venndir_overlap(v_mod, "set_B&set_C",
    params=list(label.count="outside",
               label.fill="gold"))
plot(v_mod2,
     main="Highlighted set_B&set_C outside")
```

The overlap region is named using each set involved, separated by the ampersand '&', for example the overlap of 'set_B' and 'set_C' is named 'set_B&set_C'. The overlap is provided with argument `overlap_set`.

Other arguments are intended to provide convenient shortcuts for commonly used attributes, with sensible default values:

- 'innerborder' - the border shown on the inside edge of the overlap region.
- 'innerborder.lwd' - the width of the innerborder, with default 2 to make this border much wider than the default.
- 'outerborder', 'outerborder.lwd' - the corresponding outerborder

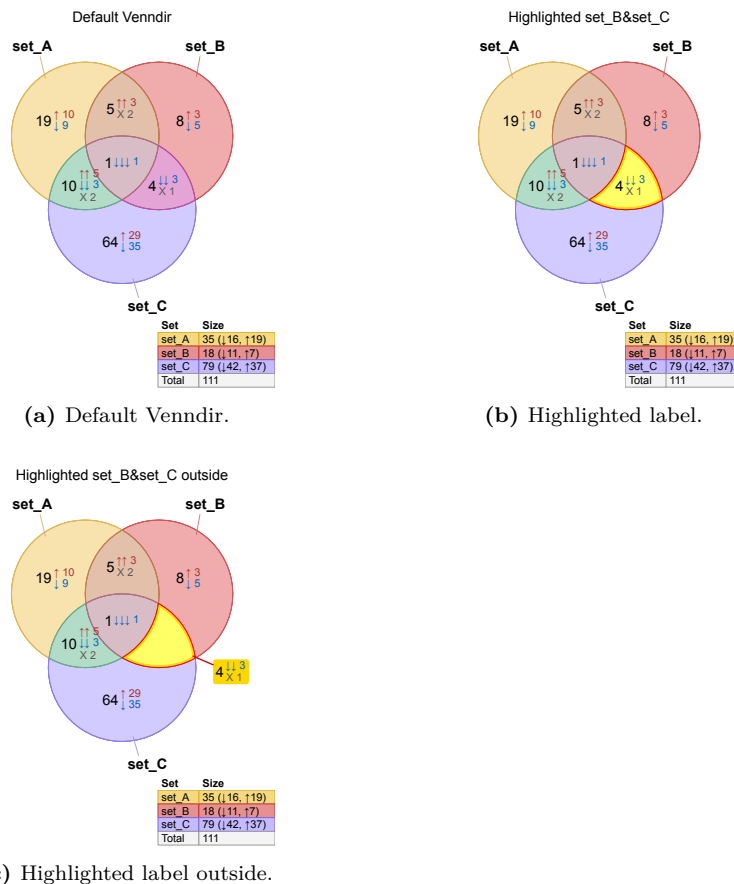


Figure 4.7. Example Venn diagrams showing default (left), highlighted (right), and highlighted-labeled alternatives.

displayed on the outside edge of the overlap region. By default the `outerborder` line width is zero.

- `'border'`, `'border.lwd'` - the color and width of the border, placed on the edge of the polygon itself. The default uses 'red' with line width 1. The main effect is to apply this color and line width to the line segment when the overlap label is outside.
- `reorder` - a logical value indicating whether to re-order the `Venndir` polygons. The default is `TRUE`, which causes the highlighted regions to be drawn last. This subtle change ensures that the `outerborder` remains visible, and is not over-drawn by other polygon regions in the `Venndir` object.

The argument `reorder` is a unique enhancement for

`highlight_venndir_overlap()`, and is the only convenient method for re-ordering `Venndir` polygons.

4.6 Rotate the Venndir

The `Venndir` figure can be rotated using degrees, on scale of 0 to 360, where 180 will rotate the figure exactly halfway. Currently the rotation must occur when calling `venndir()`, so that all polygons are rotated before the `Venndir` object is created.

The default 3-way Venn diagram has 60-degree symmetry, so rotations with multiples of 30 degrees tend to work best.

```
v1 <- venndir(make_venn_test(),
  main="rotated 0{.sup o}",
  rotate_degrees=0)
v2 <- venndir(make_venn_test(),
  main="rotated 60{.sup o}",
  rotate_degrees=60)
v3 <- venndir(make_venn_test(),
  proportional=TRUE,
  main="rotated 0{.sup o}",
  rotate_degrees=0)
v4 <- venndir(make_venn_test(),
  proportional=TRUE,
  main="rotated 90{.sup o}",
  rotate_degrees=90)
```

As indicated in the bottom-left panel (c), sometimes the legend overlaps the placement of a set label. See [Nudge the Legend](#) for alternatives.

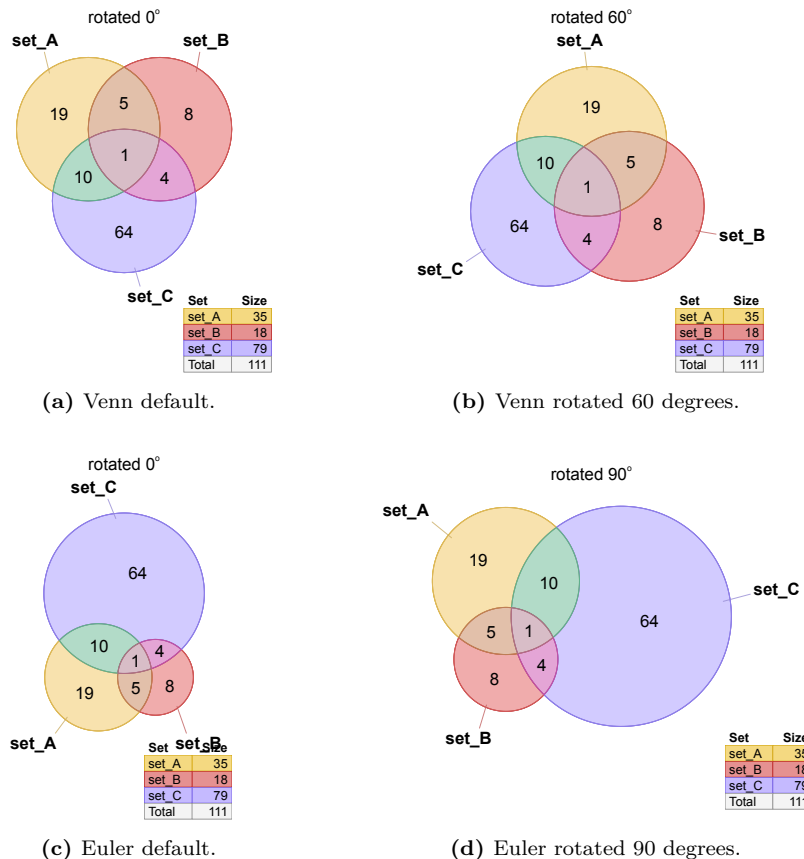


Figure 4.8. Several examples of rotated Venn and Euler diagrams.

4.7 Nudge the Legend

The legend placement is defined by `legend_x` when calling `venndir()`, `render_venndir()`, or `plot()`; or defined by `x` when calling `venndir_legend()` separately. This argument takes `character` string with combinations of 'top', 'bottom', 'left', and 'right'. Two words can be used, for example the default `legend_x="bottomright"` places the legend in the bottom-right corner of the `grid` viewport.

The `Venndir` viewport is defined to be square, to maintain 1:1 aspect ratio, and to make sure circles maintain their shape. As a result, sometimes there is whitespace on the left/right sides, or top/bottom edges.

The legend is 'inset' from the exact outer edge using arguments `x_inset` and `y_inset`, and these arguments can be passed through `'...'` ellipses from

`venndir()` to modify the legend. The default places the legend '2 lines' inside the viewport border, using 2 character lines of text as the unit. Any valid `grid::unit` can be provided, however a good starting point is to adjust by units of 'lines'.

```
v3a <- venndir(make_venn_test(),
  proportional=TRUE)
v3b <- venndir(make_venn_test(),
  x_inset=grid::unit(-5, "lines"),
  proportional=TRUE)
```

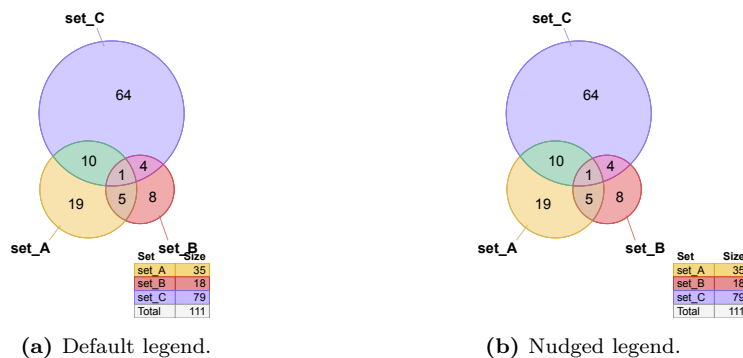


Figure 4.9. Two Euler diagrams showing default legend (left) and legend nudged 5 character lines (right).

4.8 Hidden Overlaps

Euler diagrams may offer improvements to standard Venn layout, by giving a visual indication of the relative area within each set and the corresponding overlaps. However, Euler diagrams are only *determinant* for two sets, exactly correct. With more than two sets, a Euler diagram provides the best *approximation* of the overlap sizes.

Some overlaps in a Euler diagram cannot be represented at all, due to limitations of 2-dimensional geometry.

`Venndir` objects retain all set overlaps, including those which cannot be displayed in a Euler diagram.

Four strategies mitigate the limitations:

1. **Discover** hidden counts using `warnings(Venndir)`.
2. **Recover** hidden overlaps using `overlaplist(Venndir)`.
3. **Adjust** the Euler diagram, for example see [Nudge Venndir Circles](#).

4. Optimize Euler modeling options with `eulerr::eulerr_options()`.

Data from Bisogno et al¹ help demonstrate how to workaround the issue. **Overlap counts** were imported to create a `setlist`.

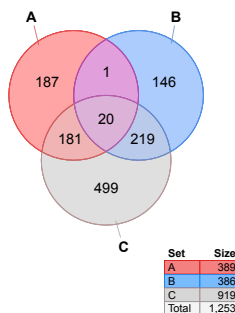
```
overlaps <- c(A=187, B=146, C=499,
  `A&B`=1,
  `A&C`=181,
  `B&C`=219,
  `A&B&C`=20);
setlist <- counts2setlist(overlaps)
```

The resulting `setlist` is shown in Figure 4.10 as a Venn diagram (left), and Euler diagram (right). It uses custom `set_colors` to match the published figure.

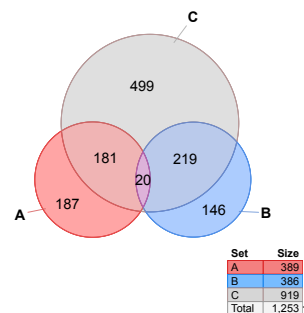
An astute observer may notice that 1 is missing in the Euler diagram.

This may be a good time to point out that the Euler diagram can also be rotated, see [Rotate the Venndir](#).

```
bisogno_colors <- c(A="firebrick2", B="dodgerblue", C="#BBBBBB")
v <- venndir(setlist,
  set_colors=bisogno_colors)
ve <- venndir(setlist,
  proportional=TRUE,
  set_colors=bisogno_colors)
```



(a) Venn diagram.



(b) Euler diagram.

Figure 4.10. Venn diagram using test data from Bisogno et al, as Venn (left) and Euler (right) diagrams. Notice the footnote in the bottom-right corner of the Euler diagram.

¹[doi:10.1126/sciadv.abc3851](https://doi.org/10.1126/sciadv.abc3851)

Discover hidden counts with `warnings(ve)`:

```
warnings(ve)
```

```
## Warning message:
```

```
## In "A&B" : 1
```

The output means there is one overlap 'A&B' with 1 item which is not displayed.

Recover overlaps using `overlaplist(ve)`. The output `list` provides a convenient way to summarize size of overlap by using `lengths()`, shown below.

```
ol <- overlaplist(ve)
lengths(ol)
```

```
##      A      B      C  A&B  A&C  B&C A&B&C
##    187    146    499     1    181    219     20
```

Adjust is shown in the next section **Nudge Venndir Circles**, which allows moving the Euler circles to create the geometry for overlap labels.

Optimize is beyond scope of Venndir, and is discussed in the R package `eulerr` documentation² ([Larsson, 2024](#)).

4.8.1 Nudge Venndir Circles

The example in **Hidden Overlaps** Figure 4.10 illustrates a weakness in using proportional Euler diagrams, while also showing a straightforward alternative. The Venn or Euler circles used can be nudged, prior to being used for visualization.

In the example, the overlap 'A&B' is not represented in the Euler diagram returned by `eulerr`, therefore the label 1 cannot be displayed. Two possible solutions:

1. Move A and B closer to each other so they overlap outside C.
2. Move C higher so that A and B overlap outside C.

The argument `circle_nudge` should be passed to `venndir()` in order for the geometry to be adjusted before the Venn overlap labels are assigned to the corresponding overlap regions. This argument should be a `list`, where each element is named using the set name. Each element should be a `numeric` vector with two values, the `x` and `y` coordinate adjustment.

The example in Figure 4.11 moves set 'A' to the right, and set 'B' to the left.

²10.32614/CRAN.package.eulerr


```

ve <- venndir(setlist,
  proportional=TRUE,
  set_colors=bisogno_colors)
vn <- venndir(setlist,
  circle_nudge=list(A=c(1, 0), B=c(-1, 0)),
  proportional=TRUE,
  set_colors=bisogno_colors)

```



Figure 4.11. Venn diagram showing the default Euler (left) with footnote, and nudged circles (right) with no footnote indicated. The overlap for 'A&B' now shows 1 count.

4.9 Nudge specific labels

Venndir automatically places labels which aims to cover the most common 85% scenarios. For all other cases, labels can be re-positioned using `nudge_venndir_label()`.

- For each overlap, the label can be displayed 'inside' or 'outside'.
- The position defined for 'inside' or 'outside' can be adjusted with `nudge_venndir_label()`.

As such, a label position is identified by two criteria:

1. **Overlap:** each set name, separated by ampersand '&', for example 'set_A&set_B'
2. **Location:** 'inside' or 'outside' the Venn diagram.

The components of each label, and placement 'inside' or 'outside', are stored in the Venndir object as described in [Venndir Labels](#). Individual changes can be made by following [Customize Any Label](#).

Instead, `nudge_venndir_label()` adjusts the coordinate position used when placing each label 'inside' or 'outside'.

```
setlist <- make_venn_test(100, 3, do_signed=TRUE)
vo <- venndir(setlist,
  overlap_type="each",
  label_style="lite box",
  main="Default venndir")
vo2 <- nudge_venndir_label(vo,
  set="set_A&set_C",
  label_location="inside",
  x_offset=-0.04)
plot(vo2, main="'set_A&set_C' moved left")
```

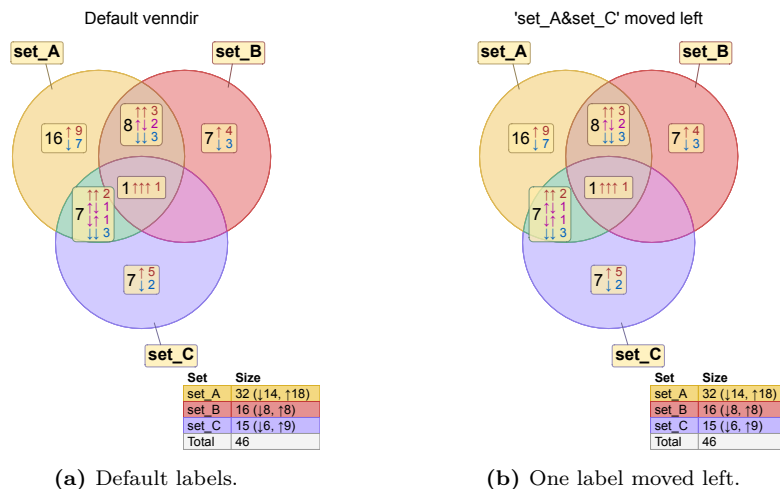


Figure 4.12. Venndir example with default label placement (left), and with 'set_A&set_C' moved to the left (right).

The changes were defined by these arguments:

- The overlap: `set="set_A&set_C"`
- The location: `label_location="inside"`
- The x position: `x_offset=-0.04`, with default units relative to the total size of the figure.

An alternative approach may be more convenient, using argument `offset_list`. The argument is a list, named by overlap, containing a two-value numeric vector with the x,y offset values. This style is more convenient when adjusting multiple labels together. Note that `label_location` is still required.

Figure 4.13 shows the effects after nudging labels 'set_A&set_C' and 'set_C' in one step.

```
vo3 <- nudge_venndir_label(vo,
  label_location="inside",
  offset_list=list(
    "set_A&set_C"=c(-0.04, -0.02),
    "set_C"=c(0, 0.05)
  ))
plot(vo3, main="Two labels moved")
```

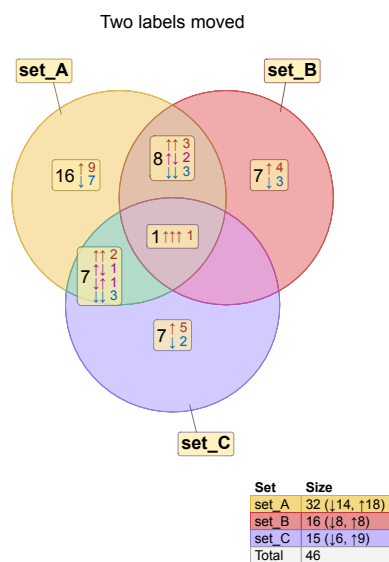


Figure 4.13. Venndir after moving labels 'set_A&set_C' and 'set_C' using `offset_list`.

4.10 Venndir Markdown Support

By virtue of using **marquee** (Pedersen and Mitáš, 2025), several labels in Venndir can be customized using **markdown** syntax, specifically following CommonMark³.

The main features are:

- support for font styles (**bold**, *italic*, underline)
- forced new-lines by ending a line with two spaces, or adding '\n'
- inline **marquee** styles

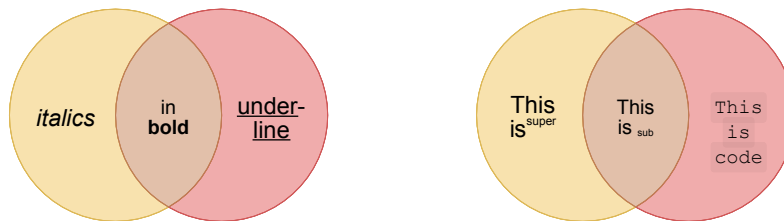
³<https://commonmark.org/help/>

- inline images
- inline R graphics objects

Other styles are recognized by **marquee**, such as `{.super }` and `{.sub }`.

For example, using `venn_meme()`:

```
venndir::venn_meme(c(
  "*italics*",
  "_under_-\\n_line_",
  "in\\n**bold**"))
venndir::venn_meme(jitter_color=0, jitter_cex=0,
  c(
    "This\\nis{.sup super}",
    "\\n`This`\\n`is`\\n`code`",
    "This\\nis {.sub sub}"))
```



(a) Markdown.

(b) Inline style.

Figure 4.14. Venn Meme showing markdown formatted items (left), and inline styles (right).

4.10.1 Inline styles

The **marquee** R package supports inline styles, using this syntax `'{.style text}'`, where 'style' is the name of a pre-defined style from `marquee::style()`, and 'text' is any valid text or markdown text, or even other inline styles.

Venndir defines `marquee::classic_style()` for use by any labels, which supports a large number of the most common HTML-like styles, including: base, body, ul, ol, li, hr, h1, h2, h3, h4, h5, h6, cb, p, qb, em, str, a, code, u, del, img, sub, sup, out

Argument 'marquee_styles' for `venndir()` accepts an optional list with

additional styles, which are added to the set of recognized styles. The marquee style specification includes *very many* customizations: font family, weight, width, size, color, lineheight, background, border, padding, and many more. Any detailed customization not already provided in Venndir should be attempted by creating a new marquee style, then applying it as an inline style.

Creating a new style is straightforward, shown below. Each style inherits attributes of the existing style. For example, it continues using the same font family, size, and color until any attribute is modified. The examples below change the font family, force it to have normal weight/width (not bold), and enlarge by 1.5 relative to the existing font size.

```
ms <- list(
  chalk=marquee::style(
    family="Chalkduster",
    weight="normal", width="normal",
    size=marquee::relative(1.5)),
  cursive=marquee::style(
    family="Above The Sky",
    weight="normal", width="normal",
    size=marquee::relative(1.5)),
  gothic=marquee::style(
    family="AcademyEngravedLetPlain",
    weight="normal", width="normal",
    size=marquee::relative(1.5)))

# create labels
il <- split(LETTERS, rep(letters[1:3], c(10, 10, 6)))
names(il) <- c("{.chalk A-J}",
  "{.gothic K-T}", "{.cursive U-Z}")
# apply inline markup around each set of labels
il[[1]] <- paste0("{.chalk ", il[[1]], "}")
il[[2]] <- paste0("{.gothic ", il[[2]], "}")
il[[3]] <- paste0("{.cursive ", il[[3]], "}")
# draw the rest of the owl
vm <- venn_meme(il, item_buffer=-0.05, marquee_styles=ms,
  expand_fraction=0.02,
  fontfamily="Chalkduster",
  show_labels="Ni", draw_legend=TRUE,
  legend_headers=c(Set="{.cursive Sets}", Size="{.chalk Sizes}"),
  main="{.chalk Custom} {.gothic Text} {.cursive Styles}")
```

Inline styles could be useful for other common operations:

- Slightly larger or smaller font size.
- Custom font family.

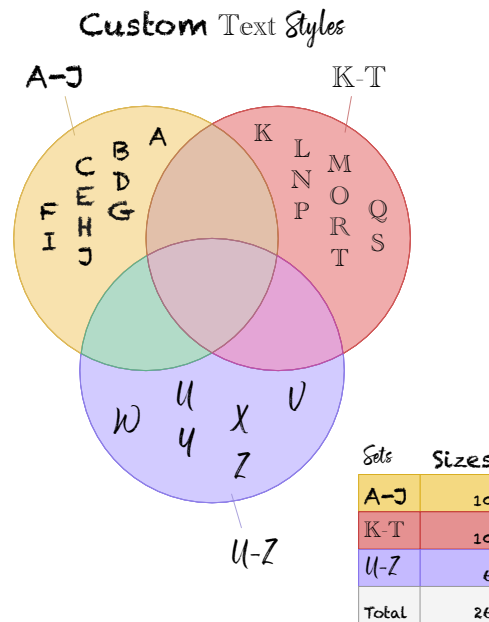


Figure 4.15. Example using inline styles to customize every conceivable field in Venndir: set name, main title, item labels, legend setlist, legend headers, legend count font.

- Left- or right-aligned text.
- Custom font color.
- Color fill background, border, border radius.

4.10.2 Inline Images

Inline images can be added using markdown syntax, which looks like this:

```
![[Image]](path/to/image)
```

The `path/to/image` can be the path to a local file, a web URL address, for example:

```
![[Image]](https://github.org/logo.png)
```

Further, inline R graphics can be added, in this form, where `Robject` is the name of an R graphical object, such as a `ggplot2` or `grid` object.

```
![[Image]](Robject)
```

Figure 4.16 shows a simple Venn diagram with an image used in the center, using the Venndir transparent logo. The function `marquee::marquee_glue()` was used as a convenient way to form the image tag. The example also shows

how to change the color of item labels, for example: '{.gold3 Venn}' to make the word 'Venn' dark-gold.

```
img <- system.file(package="venndir",
  "images", "venndir-transparent.png")
imgtag <- marquee::marquee_glue("![{img}]")

vm <- venn_meme(c("{.gold3 Venn}", "{.#6799AC dir}", imgtag),
  item_cex_factor=c(1.5, 1.5, 7),
  innerborder.lwd=1, outerborder.lwd=1,
  poly_alpha=0.5,
  set_colors=c("#EEDD79", "#87B9CC"))
```

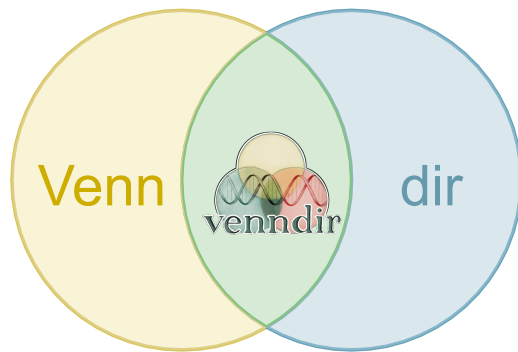


Figure 4.16. Venn diagram with image provided as an item label.

Similarly, a `ggplot2` or `grid` graphical object can be provided, and it will be displayed the same way as the image in Figure 4.16.

Each image or plot is sized relative to the line height of the corresponding text field. The example above used `item_cex_factor` to scale the center item to 7 times higher than normal.

In order to enforce a specific size, either adjust the corresponding font size with `item_cex` and `item_cex_factor`, or define an inline style with specific font size, or specific line height.

4.11 Patchwork with Venndir

The **patchwork** (Pedersen, 2024) R package is widely used to assemble multi-panel R graphics, due to its ability to include plots from **base R**, **ggplot2**, and **grid** graphics. Venndir uses **grid** and is compatible with patchwork.

There are two key steps:

1. Call `venndir()` with arguments:
 - `do_plot=TRUE` which tells Venndir to create the graphical objects which will be drawn. By default it is already `TRUE`.
 - `do_draw=FALSE` which tells Venndir not to draw these objects. This argument is optional, to prevent drawing the figure.
 - capture the result into a variable:


```
v <- venndir(..., do_draw=FALSE)
```
2. Use the Venndir 'gTree' graphical objects:
 - `attr(v, "gtree")` contains the Venndir objects
 - Use patchwork syntax:


```
wrap_elements(attr(v, "gtree"))
```

Figure 4.17 shows two Venn diagrams side-by-side.

```
library(patchwork)
suppressPackageStartupMessages(library(ggplot2))

setlist <- make_venn_test(do_signed=TRUE)
v1 <- venndir(setlist, sets=c(1, 2),
  expand_fraction=c(0.25, 0, -0.25, 0),
  do_draw=FALSE)
v2 <- venndir(setlist, sets=c(1, 3),
  expand_fraction=c(0.25, 0, -0.25, 0),
  do_draw=FALSE)

pw3 <- wrap_elements(attr(v1, "gtree")) +
  wrap_elements(attr(v2, "gtree")) +
  plot_annotation(tag_levels='a')
pw3
```

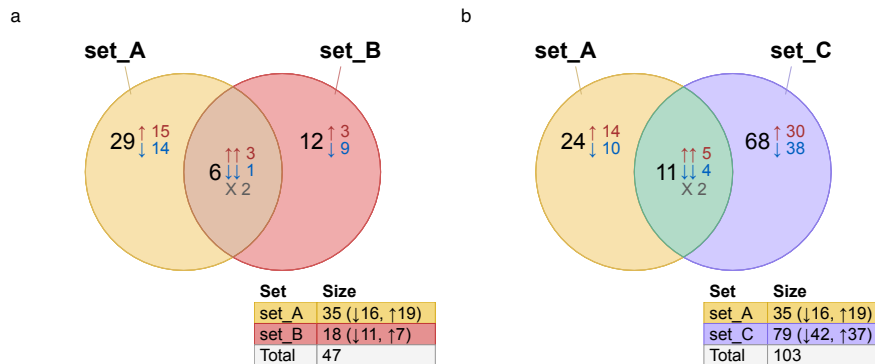



Figure 4.17. Two Venndir figures combined using the **patchwork** R package.

Tips:

- The Venndir font sizes are absolute points, and should be adjusted accordingly.
- The Venndir circles will adjust to the available output size, while the fonts will remain the same size.
- It works well to decide a target canvas size for the combined figure output, usually in actual inches or centimeters, then define an appropriate font size in that context.
- For more details on the Venndir **grid** data, see [Venndir Graphics Objects](#).

4.12 Custom Signs

One of the defining features of a Venndir figure is the use of ↑ up and ↓ down arrows to indicate signed counts. By default, these labels are also colored red and blue, respectively.

It may be useful to customize the colors or the symbols associated with each sign, and this definition can be customized in `venndir()` using an advanced argument `curate_df`.

4.12.1 Review Sign Curation

The default `curate_df` can be reviewed with `get_venndir_curate_df()`, the output is shown in [Table 4.2](#).

Table 4.2. Default table used to curate sign data when generating signed count labels.

from	sign	color	hide_singlet
-1	↓	dodgerblue3	FALSE
1	↑	firebrick	FALSE
0			FALSE
concordant	=	dodgerblue3	TRUE
agreement	=	dodgerblue3	TRUE
mixed	X	grey45	FALSE

Table 4.3. Values recognized overlap type, with corresponding description.

overlap_type	Description
'overlap'	Sign information is ignored. Overlap counts are tabulated without sign information.
'concordance'	Sign information is tabulated into three general subsets: (1) all agree up, (2) all agree down, and (3) 'mixed'. For (1) and (2) the original signs are retained, keeping '1 1 1' or '-1 -1 -1'. All values with mixed signs, such as '1 -1' are converted to 'mixed'. When hide_singlet=TRUE then single-set counts ignore the sign.
'each'	Signed is tabulated for every overlap in maintained without modification.
'agreement'	Similar to 'concordance', except that agreement up and agreement down are combined and labeled 'agreement'.

```
get_venndir_curate_df()
```

There is a bit to unpack in how this simple `data.frame` is used to create the labels shown by Venndir.

First, the input `setlist` is passed to `signed_overlaps()` to produce a complete summary of overlaps, and associated overlap signs. An important argument is `overlap_type` which defines how the overlap signs are summarized. The recognized values are shown in Table 4.3.

To summarize:

- Overlaps are tabulated by combinations of sets. Within each overlap, counts are also tabulated by the directionalities using the following rules:
- `overlap_type='overlap'` will not tabulate values by signs.

- `overlap_type='concordance'` will tabulate values by signs such as '1', '-1', '1 1', '-1 -1', and 'mixed'.
- `overlap_type='each'` will tabulate values by signs such as '1', '-1', '1 1', '-1 -1', '1 -1', and '-1 1'.
- `overlap_type='agreement'` will tabulate values by signs such as 'agreement', and 'mixed'.

Using `overlap_type='each'` is recommended to retain all combinations of the sign.

For each overlap, the tabulated signs are used to create a curated "sign prefix", for example '1 1' is converted to '↑↑', and this prefix is appended to the actual counts. For example the signed label might be '↑↑ 12'.

Consider the tabulated sign '1 -1':

- The tabulated sign '1 -1' is split by whitespace into components '1' and '-1'.
- The first component '1' is matched to 'from' in `curate_df`:

from	sign	color	hide_singlet
1	↑	firebrick	FALSE

- The value in column 'sign' (↑) is added to the sign.
- The color 'firebrick' is added to the sign colors.

- The next component '-1' is matched to 'from' in `curate_df`:

from	sign	color	hide_singlet
-1	↓	dodgerblue3	FALSE

- The value in column 'sign' (↓) is added to the sign.
- The color 'dodgerblue3' is added to the sign colors.
- The sign prefix is '↑↓'.
- The colors 'firebrick', 'dodgerblue3' are blended to purple.
- The complete signed count label is '↑↓ counts', colored purple.

Additional comments on processing:

- If the tabulated sign is 'agreement' the signed prefix will be '≡', colored blue.
- If the tabulated sign is 'mixed' the signed prefix will be 'X'.

4.12.2 Customize the Symbols

An optional argument for `venndir()` is `curate_df`, which provides a mechanism to customize the visual sign that appears in the Venn diagram.

To illustrate the process, in this example the goal is to use custom symbols to indicate agreement and disagreement: 'Checkmark' to indicate agreement, and 'Ballot X' to indicate disagreement.

Tips:

Unicode characters are described in Wikipedia Unicode^a.

Formats described as 'U+2191' should be used in R in this form:

"\u2191" in order to produce: ↑ * **Up arrow** is Unicode U+2191:

↑ * **Down arrow** is Unicode U+2193: ↓ * **Checkmark** is Unicode

U+2713: ✓ * **Ballot X** is Unicode U+2717:

^ahttps://en.wikipedia.org/wiki/List_of_Unicode_characters

The following steps were used to create Figure 4.18:

1. Retrieve the default `curate_df`.
2. Edit the 'agreement' row, and update column 'sign'.
3. Edit the 'mixed' row, and update columns 'sign', and 'color'.
4. Call `venndir()` with arguments `overlap_type='agreement'` and `curate_df=curate_df`.

```
curate_df <- get_venndir_curate_df()
curate_df[curate_df$from %in% "agreement",
  "sign"] <- "\u2713";
curate_df[curate_df$from %in% "mixed",
  "sign"] <- "\u2715";
curate_df[curate_df$from %in% "mixed",
  "color"] <- "red";

v <- venndir(make_venn_test(do_signed=TRUE),
  overlap_type="agreement",
  innerborder="white", outerborder="white",
  poly_alpha=0.8,
  curate_df=curate_df)
```

4.12.3 More Advanced Signs

In principle, and named `list` is recognized as a value-list, whose names are items, and values are 'signs'. These 'signs' are typically -1 and 1 to indicate directionality, *however any sign values may be used*.

Any sign values not recognized by `curate_df`, as described in [Customize the Symbols](#), remain unchanged and un-colored.

Figure 4.19 shows the effect of changing the signed list values from '1' and '-1', to strings: 'up' and 'dn'. Because 'up' and 'dn' are not defined in `curate_df`,

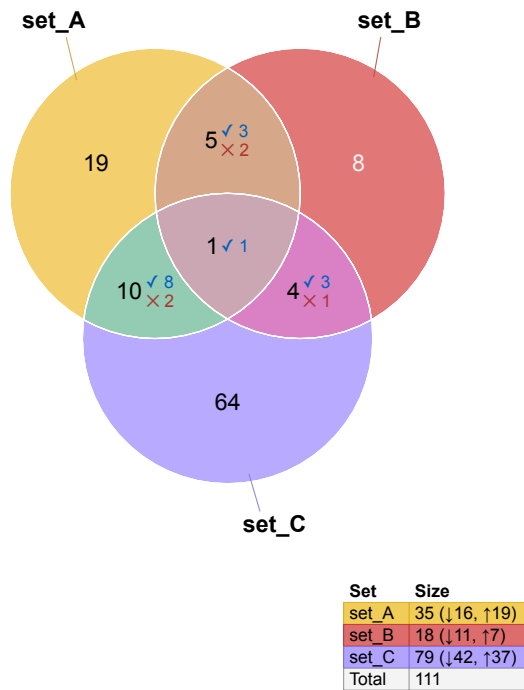


Figure 4.18. Venn diagram using 'Checkmark' and 'Ballot X' to indicate agreement and disagreement, respectively.

they remain unchanged in the output `Venndir`.

```
setlist <- make_venn_test(n_sets=2, do_signed=TRUE)
setlist2 <- lapply(setlist, function(i){
  i[] <- c("1"="up", "-1"="dn")[as.character(i)]
  i
})
v <- venndir(setlist2,
  overlap_type="each")
```

The figure shows combinations that include 'upup', 'dnup', and 'dnup'. Notice even the legend tabulates the signed counts using the sign values 'up' and 'dn'.

Further, the sign values 'up' and 'dn' can be assigned to Unicode character as shown in [Customize the Symbols](#). The value can be a string, or even [markdown](#), described in [Venndir Markdown Support](#). The custom value is also shown in the legend.

Figure 4.20 demonstrates how to customize `curate_df`, matching the sign

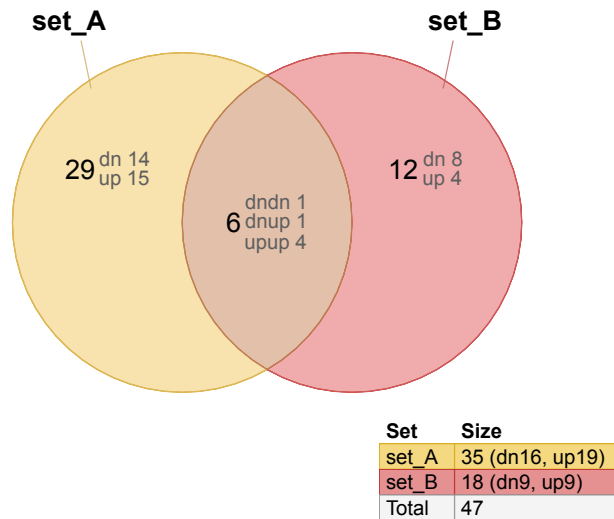


Figure 4.19. Venndir using custom values 'up' and 'dn' instead of 1 and -1.

values 'up' and 'dn', and converts them to new Unicode characters.

```
curate_df <- get_venndir_curate_df()
du_match <- match(c("-1", "1"), curate_df$from)
curate_df[du_match, "from"] <- c("dn", "up")
curate_df[du_match, "sign"] <- c("\u21E9", "\u21E7")
#
v <- venndir(setlist2,
  curate_df=curate_df,
  overlap_type="each")
```

Although directional sign is most often 'up' and 'down', Venndir can recognize more than two signs. The following example demonstrates three signs instead of two. In the simplest form, it uses three letters: 'a', 'b', and 'c'.

```
setlist <- make_venn_test(n_items=150, n_sets=3, do_signed=TRUE,
  set_names=c("Set 1", "Set 2", "Set 3"))
set.seed(123)
setlist2 <- lapply(setlist, function(i){
  i[] <- c("1"="up", "-1"="dn")[as.character(i)]
  i[] <- sample(letters[1:3],
    replace=TRUE, size=length(i))
```

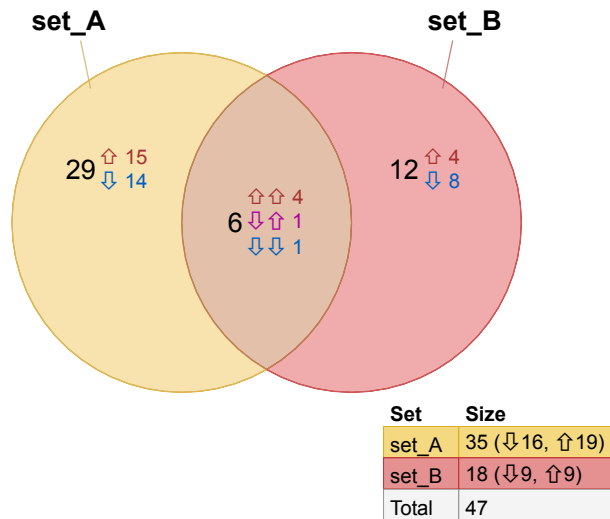


Figure 4.20. Venndir showing the custom signs 'up' and 'down' used to produce Unicode signs as before.

```
i
})
v <- venndir(setlist2,
  sets=1:2,
  overlap_type="each")
```

These signs are converted to custom symbols: ↑, ↓, Δ.

```
curate_df <- get_venndir_curate_df()
# define a, b, c
curate_df[1, "from"] <- "a"
curate_df[2, "from"] <- "b"
curate_df[3, "from"] <- "c"
# define Unicode symbols
curate_df[1, "sign"] <- "\u2191"
curate_df[2, "sign"] <- "\u2193"
curate_df[3, "sign"] <- "\u2206"
# define colors
curate_df[1, "color"] <- "firebrick"
curate_df[2, "color"] <- "royalblue"
```

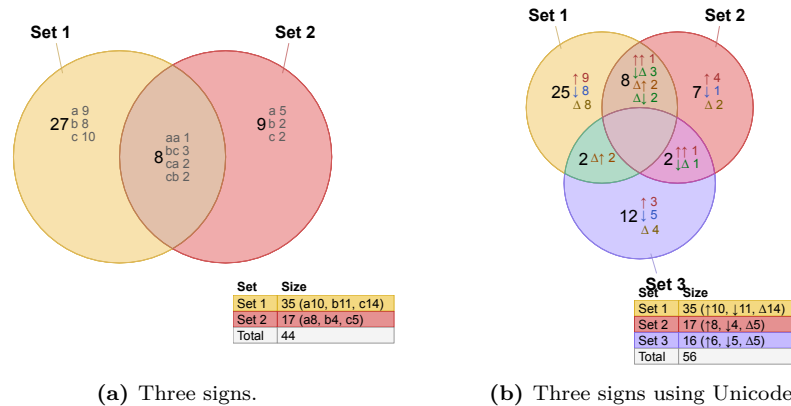


Figure 4.21. Venndir using three signs: 'a', 'b', and 'c' (left), which are converted to Unicode symbols (right).

```
curate_df[3, "color"] <- "gold"

v <- venndir(setlist2,
  curate_df=curate_df,
  overlap_type="each")
```

Figure 4.21 shows the three-sign example using letters (left), and Unicode symbols (right).

4.12.4 Custom Sign Label Colors

In some scenarios, colorized labels may not be appropriate, or custom colors may be preferred. Custom signed colors can be defined by editing `curate_df` as described in [Custom Signs](#) to this point.

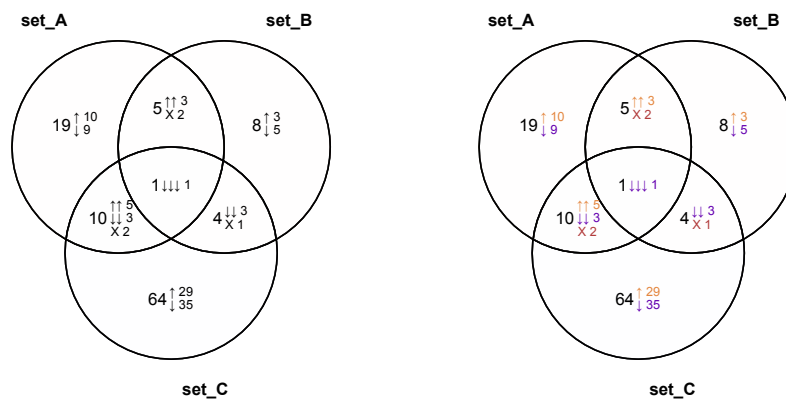
```
# black text labels
curate_df <- get_venndir_curate_df()
curate_df$color <- "black"
v1 <- venndir(make_venn_test(do_signed=TRUE),
  show_segments=FALSE,
  draw_legend=FALSE,
  curate_df=curate_df,
  poly_alpha=0.01,
  border="black",
  border.lwd=2)
# custom color labels
```



```

k <- match(c("-1", "1", "mixed"), curate_df$from)
curate_df$color[k] <- c("purple4", "darkorange", "red3")
v2 <- venndir(make_venn_test(do_signed=TRUE),
  show_segments=FALSE,
  L_lo=65,
  draw_legend=FALSE,
  curate_df=curate_df,
  poly_alpha=0.01,
  border="black",
  border.lwd=2)

```



(a) Signed labels in black.

(b) Signed labels in custom colors.

Figure 4.22. Signed Venn diagrams with sign count labels in black (left), and custom colors.

The important step is to edit the column 'color' in `curate_df`. In the example above, the color contrast is adjusted also, with `make_color_contrast()`, and optional argument `L_lo=65` is passed to allow slightly brighter text on white background, which allows the orange font color saturation to be slighter more pronounced.

4.13 Venndir Graphics Objects

The `Venndir` object also stores the `grid` graphical objects (grobs) used for visualization, however the grobs are only created by calling `plot()` or `render_venndir()`. The output of this function is `Venndir` object with hidden attributes which contain grobs.

When calling `venndir()`, two options may be useful:

1. `do_plot=TRUE`: This argument causes `venndir()` to call `render_venndir()`, which then creates the relevant grobs.
2. `do_draw=FALSE`: This argument is passed to `render_venndir()`, so the grobs are not drawn to the graphical device.

The following example creates a `Venndir` object without creating a new plot, and which creates all the necessary grobs.

```
v <- venndir(make_venn_test(),
  show_labels="Ni",
  main="Plot Title",
  fontfamily="Arial",
  do_plot=TRUE,
  do_draw=FALSE,
  poly_alpha=0.4,
  item_cex_factor=0.6,
  legend_color_style=c("black", "blackborder"))
```

The relevant attribute names are described as follows:

- `'gtree'`: the single `grid` object that contains all `Venndir` grobs.
- `'grob_list'`: the list of each `grid` object, separated by type.
- `'viewport'`: the `grid` object representing the graphics viewport.
- `'adjx', adjy'`: the adjustment function used to convert `JamPolygon` coordinates to unit `'nspc'` scaled values between 0 and 1.

4.13.1 Venndir gtree

The `'gtree'` attribute contains a `grid:gTree` object, which can be drawn directly. The `gtree` object already contains the relevant `viewport`.

Figure 4.23 shows the plot after drawing the `'gtree'` attribute.

```
gtree <- attr(v, "gtree");
grid::grid.newpage();
grid::grid.draw(gtree);
```

A detailed list of all grobs within the `gtree` can be seen by calling `grid::grid.ls()`, however it is not listed here because it contains over 350 grobs!

4.13.2 Venndir grob_list

An alternative representation is the `'grob_list'` attribute, which is a list with each `Venndir` visual component. The exact elements may vary dependent upon which options were used.

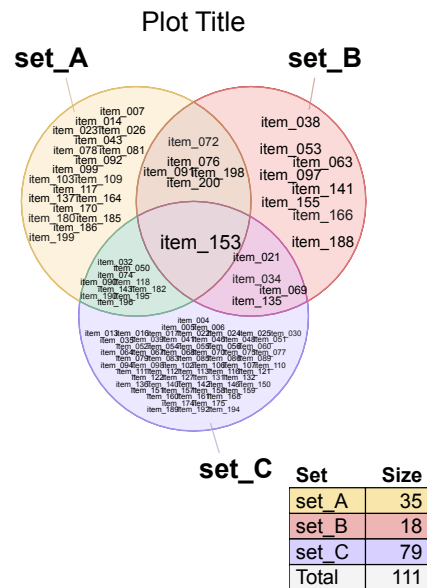


Figure 4.23. Venn diagram created by drawing the 'gtree' attribute object.

The purpose of inspecting `grid` grobs is mainly to allow direct manipulation of the grobs, for advanced customizations for example. Each individual component can be edited in detail.

The grobs by type are described below:

- 'jps': the `JamPolygon` object with polygons and border.
- 'item_labels': item labels, only if enabled by argument `show_labels`.
- 'labels': the `marquee` grobs used for all text labels.
- 'segments': the line segments connecting labels to polygons, where relevant.
- 'legend': the `grid` object representing the Venndir legend, if relevant.
- 'main_title': the title object, only if defined by argument `main`.

Figure 4.24 shows the plot after drawing each `grid` component from the 'grob_list' attribute.

```
grob_list <- attr(v, "grob_list");

grid::grid.newpage()
grid::grid.draw(grob_list$jps)
grid::grid.draw(grob_list$item_labels)
grid::grid.draw(grob_list$labels)
grid::grid.draw(grob_list$segments)
```

```
grid::grid.draw(grob_list$legend)
grid::grid.draw(grob_list$main_title)
```

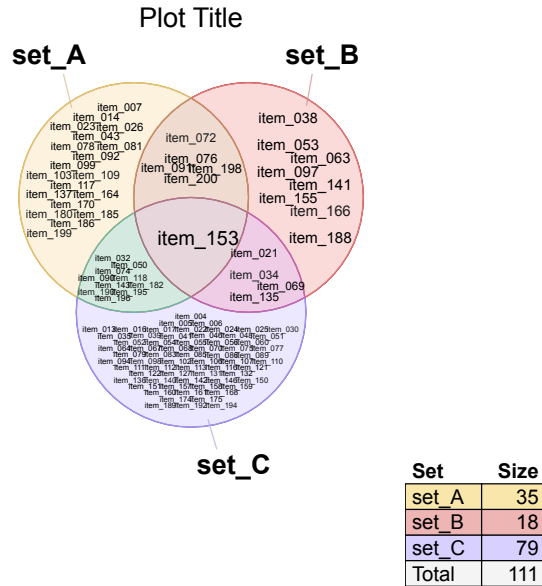


Figure 4.24. Venn diagram created by drawing each component in the 'grob_list' attribute.

4.13.3 Manipulate grob_list

The entries are named to be recognizable, for example the grob names in the 'jps' entry are listed with `grid::grid.ls()`:

```
grid::grid.ls(grob_list$jps)

## jps
## set_A|set:pathGrob:outer:1
## set_A|set:pathGrob
## set_B|set:pathGrob:outer:1
## set_B|set:pathGrob
## set_C|set:pathGrob:outer:1
## set_C|set:pathGrob
## set_A:pathGrob:inner
## set_A:pathGrob:inner:1
## set_A:pathGrob
## set_B:pathGrob:inner
```

```
## set_B:pathGrob:inner:1
## set_B:pathGrob
## set_C:pathGrob:inner
## set_C:pathGrob:inner:1
## set_C:pathGrob
## set_A&set_B:pathGrob:inner
## set_A&set_B:pathGrob:inner:1
## set_A&set_B:pathGrob
## set_A&set_C:pathGrob:inner
## set_A&set_C:pathGrob:inner:1
## set_A&set_C:pathGrob
## set_B&set_C:pathGrob:inner
## set_B&set_C:pathGrob:inner:1
## set_B&set_C:pathGrob
## set_A&set_B&set_C:pathGrob:inner
## set_A&set_B&set_C:pathGrob:inner:1
## set_A&set_B&set_C:pathGrob
```

Each polygon is named with substring up to the first ':' colon, after which are optional modifiers indicating the border applied, in order:

- Each polygon is drawn with outer border (if `outerborder` is defined),
- then polygon fill and border (if `border` is defined),
- then inner fill (if `innerborder` is defined),
- then inner border (if `innerborder` is defined).

As described in [Venndir Rendering Steps](#), a polygon is drawn for each full set, denoted with polygon names having '|set' at the end. Then each overlap region is drawn, where polygons are named by the overlap sets.

The example below is somewhat advanced, and demonstrates how to extract a specific polygon grob, use it to create a custom pattern fill, then reinsert the new grob into the `grob_list`.

First, extract one overlap region, and draw it to confirm:

```
grobname <- "set_C:pathGrob:inner"
use_grob <- grid::getGrob(grob_list$jps, gPath=grobname)
grid::grid.newpage();
grid::grid.draw(use_grob)
```

The next step is to define a grid pattern, in this case it requires the **gridpattern** R package, loading an image.

Another alternative is to define a color gradient, for example with `gridpattern::grid.pattern_gradient()`.



Figure 4.25. One overlap polygon is extracted from `grob_list` and drawn.

```
bg_file <- "images/water_bg.png"
bg_pat <- gridpattern::grid.pattern_image(filename=bg_file,
  type="tile", scale=1, draw=FALSE)
```

Now the grob `use_grob` is used to apply a clipping region to the pattern fill, so the pattern is only applied to this specific polygon region.

```
# clip background to the grob
clipped_name <- paste0(grobname, ".pattern")
clipped <- list(gridpattern::clippingPathGrob(
  clippee=bg_pat,
  clipper=use_grob,
  name=clipped_name));
names(clipped) <- clipped_name
```

Now draw the clipped background image:

```
grid::grid.newpage();
grid::grid.draw(clipped[[1]])
```

Finally, re-insert the background into the `grob_list`:

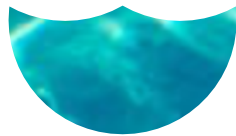


Figure 4.26. (ref:groblist-clip-draw)

```
grob_list_jps_list <- grob_list$jps$children;
grobmatch <- match(grobname, names(grob_list_jps_list));

if (grobmatch == 1) {
  grob_list_jps_list <- append(grob_list_jps_list,
    clipped,
    after=grobmatch - 1)
} else {
  grob_list_jps_list <- append(clipped,
    grob_list_jps_list,
    after=1)
}
# insert into grob_list
grob_list$jps <- grid::grobTree(
  children=do.call(grid::gList, grob_list_jps_list),
  name="jps")
```

Bonus points: Re-create the grob tree gTree object:

```
# make new gTree
jps_gtree <- grid::grobTree(
```

```
children=do.call(grid::gList, grob_list),
name="venndir.figure")
```

Drawing the `gTree` is sometimes more convenient than drawing each element in the `grob_list`:

```
grid::grid.newpage()
grid::grid.draw(jps_gtree)
```

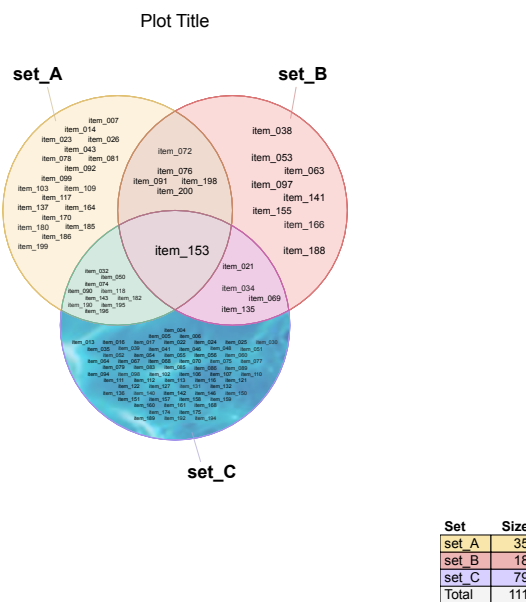


Figure 4.27. The custom Venndir figure is drawn, using pattern fill in one overlap region.

Note that the original polygon fill is drawn on top of the pattern fill, but this layer could be removed so the pattern is shown without interference. This change is left to the reader.

Chapter 5

Venndir Gallery

5.1 Figure Boosting

One of the best hobby activities while developing Venndir has been "figure sniping", which is loosely translated, "Can Venndir make that figure?"

There are two main motivations:

1. Can it be done? *A test of creativity, a test of wills.*
2. Can it be done better? *Dare we try?*

There are plenty of graphics tools with which someone could just create their own Venn diagram "by hand", such as Inkscape, Adobe Illustrator, Microsoft Powerpoint. For me, the fewer things I do "by hand" the better. To be completely frank with myself, the fewer things I don't *want* to do by hand the better. Also me: Sometimes I do what I *want* to do.

The first example was shown in Figure 3.21, to recreate part of the nice Venn diagram in (Salybekov et al., 2021) Figure 2.

5.1.1 Me - Electron

This example is straightforward, using a fun post on Reddit r/physicsmemes¹ (u/ScienceNerd42, 2022). The person "Me" and an "Electron" both start behaving differently when observed. This example demonstrates some of the basics of re-creating a figure, in order to make a similar figure, or to enhance the concept of the figure.

The labels are clear, the colors are easily approximated. The font looks like Times, so `fontfamily='serif'` should suffice. The center wording is best

¹https://www.reddit.com/r/physicsmemes/comments/v5adqq/you_and_i_arent_so_different/

represented as one item label, split across multiple lines using newline '\n' character.

Figure 5.1 shows the outcome, relatively quick and easy! Two more options are shown on the bottom row. The first (bottom left) customizes the innerborder and outerborder. The second calls `modify_venndir_overlap()` to highlight the center region.

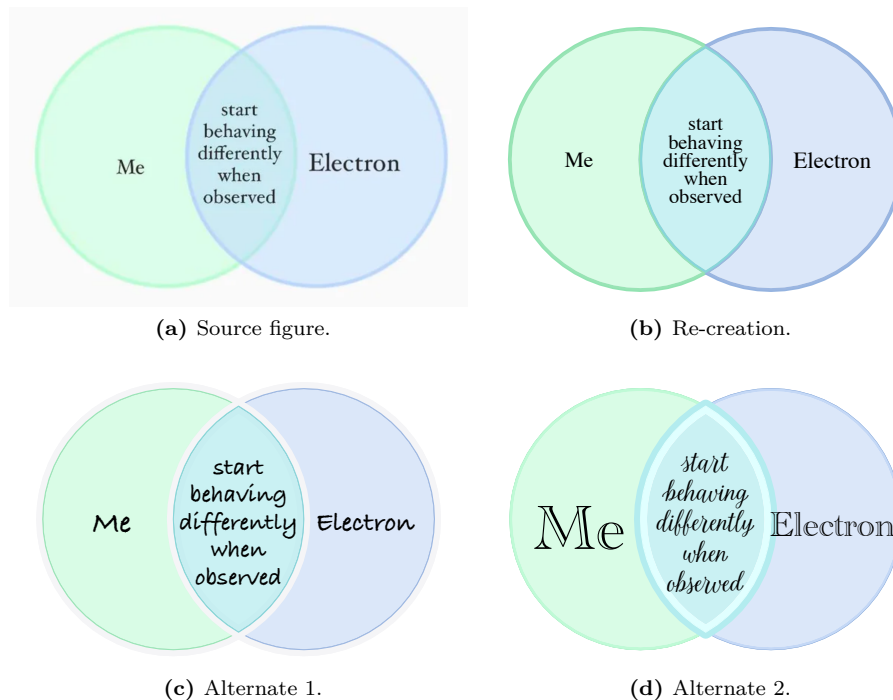


Figure 5.1. Target figure from r/physicsmemes (top left), re-created (top right), with two alternatives (bottom).

```
olist <- c("Me",
  "Electron",
  paste(collapse="\n",
    c("start", "behaving",
      "differently", "when",
      "observed"))
)
venn_meme(olist,
  outerborder.lwd=2, innerborder.lwd=2,
  poly_alpha=0.7,
  item_cex_factor=c(0.8, 0.8, 0.9),
```

```
fontfamily="serif",
set_colors=c("#BFFAD6", "#C4D8F6"))
```

5.1.2 eulerGlyphs

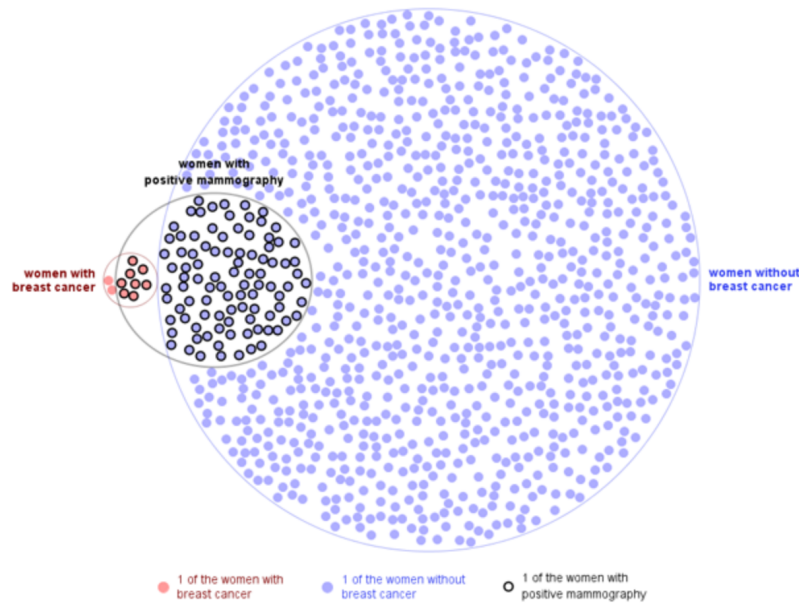


Figure 5.2. Target diagram from eulerGlyphs.

Figure 5.2 shows a fantastic figure created by eulerGlyphs² (Micallef et al., 2012), an application designed to create proportional Euler diagrams. The data represents breast cancer screening statistics³, and is a common reference dataset to study the visual perception of statistics.

A brief summary of the data follows:

- 10 out of 1,000 women age 40 have breast cancer.
- 8 of every 10 women *with* breast cancer got a positive test result.
- 95 of every 990 women *without* breast cancer got a positive test result.

Figure 5.3 shows the initial attempt, using `overlap_type="agreement"`, then visualizing items with only the sign, which for agreement uses '=' the equals sign. Items are rotated with `jitter_degrees` to provide some visual randomness.

²<https://www.eulardiagrams.com/eulerGlyphs/>

³[10.1109/TVCG.2012.199](https://doi.org/10.1109/TVCG.2012.199)

```

mammo_counts <- c(
  wob=895,
  wwbc=2,
  "wob&wwpm"=95,
  "wwbc&wwpm"=8)
mammo_list <- counts2setlist(mammo_counts)
mammo_labels <- c(
  wob="women without\nbreast cancer",
  wwbc="women with\nbreast cancer",
  wwpm="women with\npositive mammography")
mammo_colors=c("#AEAEFF", "#FF9D9D", "#896699")
v_mammo <- venndir(mammo_list,
  overlap_type="agreement",
  poly_alpha=0.3,
  set_colors=mammo_colors,
  setlist_labels=mammo_labels,
  xyratio=0.4,
  show_labels="Ni",
  show_items="sign",
  jitter_degrees=45,
  item_buffer=-0.01, width_buffer=0.05,
  item_cex=c(1, 1, 1, 1, 1),
  segment_distance=0.02,
  expand_fraction=c(-0.1, -0.2, -0.1, 0),
  rotate_degrees=180,
  draw_legend=FALSE,
  proportional=TRUE)

```

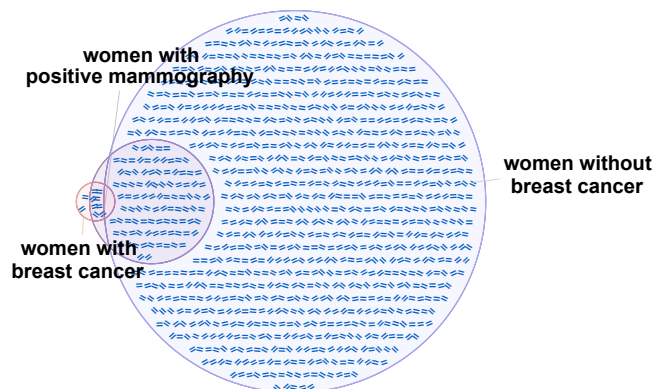


Figure 5.3. Initial attempt at re-creating the EulerGlyphs figure.

The first pass fills the space with '=' symbols, rotates the `eulerr` output, and placed the circles quite well. The argument `xratio=0.4` placed the '=' symbols closer together than default.

Another approach could improve the figure, using a the Unicode 'U+25CF' filled circle with the method described in [Customize the Symbols](#). This symbol would match the font color, which can be edited to match the source figure.

(In a pinch, the items themselves could be edited in the `Venndir` object: `v_mammo@label_df$items`. The items could be replaced with the Unicode symbol as one option.)

Figure 5.4 shows Unicode filled circles, and colors assigned to approximate the colors in `eulerGlyphs`. The set labels are nudged.

```
curate_df3 <- get_venndir_curate_df();
agg3 <- which(curate_df3$from %in% "agreement")
curate_df3[agg3, "sign"] <- "\u25CF";

# create a new Venndir
v_mammo3 <- venndir(mammo_list,
  do_plot=FALSE,
  circle_nudge=list(
    wwbc=c(-1.8, 0),
    wwpm=c(-1.6, 0)),
  innerborder.lwd=1, outerborder.lwd=1,
  overlap_type="agreement",
  poly_alpha=0.1,
  set_colors=mammo_colors,
  curate_df=curate_df3,
  setlist_labels=mammo_labels,
  xratio=0.6,
  fontfaces=list(overlap="plain"),
  show_labels="Ni",
  show_items="sign",
  segment_buffer=-0.05,
  jitter_cex=0, jitter_color=0,
  font_cex=0.8,
  item_buffer=-0.02,
  item_cex=c(1, 1, 1, 1, 1) * 1,
  segment_distance=0.02,
  rotate_degrees=180,
  draw_legend=FALSE,
  proportional=TRUE)

# edit the label colors
v_mammo3@label_df["wob", "color"] <- "blue1";
```

```

v_mammo3@label_df["wwbc", "color"] <- "red2";
v_mammo3@label_df["wob.agreement", "color"] <- mammo_colors[1];
v_mammo3@label_df["wob&wwpm.agreement", "color"] <- "royalblue";
v_mammo3@label_df["wwbc&wwpm.agreement", "color"] <- "#DD6666";
v_mammo3@label_df["wwbc.agreement", "color"] <- mammo_colors[2];

# nudge labels
v_mammo3n <- nudge_venndir_label(v_mammo3,
  label_location="outside",
  offset_list=list(wwbc=c(0.0, 0.03),
    wwpm=c(-0.07, 0.1),
    wob=c(0, -0.06)))

# visualize
plot(v_mammo3n,
  jitter_color=0, width_buffer=0.02,
  L_lo=80, L_hi=85, C_floor=50,
  expand_fraction=c(-0.1, -0.10, -0.1, -0.05),
  innerborder.lwd=0, outerborder.lwd=0.7)

```

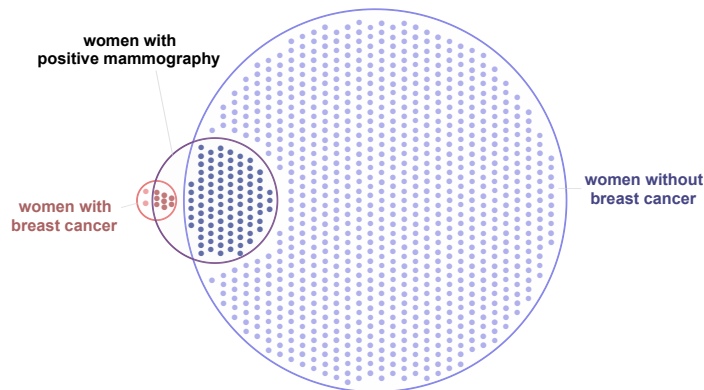


Figure 5.4. Second attempt at the EulerGlyphs figure. It already looks cleaner.

Both previous attempts showed "quick and easy" approximations, however the spirit of Figure Boosting is to re-create the image as closely as possible.

The eulerGlyphs figure used points colored to convey true breast cancer status, with black border to indicate a positive mammography test result. To mimic this effect requires using proper points.

The steps required:

1. Create the `Venndir` object without item labels.
2. Nudge the set labels, apply custom colors.
3. Extract the `JamPolygon` object.
4. Call `label_fill_JamPolygon()` for each overlap.
5. Render `grid::pointsGrob()` in the correct viewport.

Steps 1 and 2 are shown below:

```
# create a new Venndir
v_mammo4 <- venndir(mammo_list,
  do_plot=FALSE,
  circle_nudge=list(
    wwbc=c(-1.4, 0),
    wwpm=c(-1.6, 0)),
  innerborder.lwd=1, outerborder.lwd=1,
  overlap_type="agreement",
  poly_alpha=0.1,
  set_colors=mammo_colors,
  setlist_labels=mammo_labels,
  fontfaces=list(overlap="plain"),
  show_labels="N", keep_items=TRUE,
  segment_buffer=-0.05,
  font_cex=0.8,
  segment_distance=0.02,
  rotate_degrees=180,
  draw_legend=FALSE,
  proportional=TRUE)
# edit the label colors
k <- c("wob", "wwbc")
v_mammo4@label_df[k, "color"] <- c("blue", "red")
# nudge labels
v_mammo4n <- nudge_venndir_label(v_mammo4,
  label_location="outside",
  offset_list=list(wwbc=c(0.01, 0.045),
    wwpm=c(-0.09, 0.06),
    wob=c(-0.01, -0.06)))
```

The internal function `label_fill_JamPolygon()` defines coordinates for item labels inside a `JamPolygon`. The example iterates each overlap region that contains items, then stores item coordinates to use later. The point fill color and border are defined for each region as well.

```
# JamPolygon
v_items <- jamba::rmNULL(v_mammo4@label_df$items)
v_colors <- mammo_colors[c(1, 2, 1, 2)];
```

```

v_borders <- c(NA, NA, "black", "black")
v_buffers <- c(0.01, -0.2, 0, -0.15)
xy <- jamba::rbindList(lapply(seq_along(v_items), function(i){
  which_jp <- match(gsub("[I].+", "", names(v_items)[i]),
    names(v_mammo4@jps))
  xy <- label_fill_JamPolygon(jp=v_mammo4@jps[which_jp],
    width_buffer=0.01,
    buffer=v_buffers[i], xyratio=0.5,
    labels=seq_along(v_items[[i]]))$items_df;
  xy$color <- v_colors[i];
  xy$border <- v_borders[i];
  xy;
})))

```

Finally, the item coordinates are used with `grid::pointsGrob()` with some visual noise added by `rnorm()` for visual flair.

Figure 5.5 shows the result from the final steps, drawing the points in the correct viewport.

```

# plot the Venndir
v_mammo4p <- plot(v_mammo4n,
  expand_fraction=c(-0.1, -0.10,
    -0.1, -0.05))
# extract the viewport adjustments
vp <- attr(v_mammo4p, "viewport");
adjx <- attr(v_mammo4p, "adjx");
adjy <- attr(v_mammo4p, "adjy");
# create pointsGrob
set.seed(123);
pts <- grid::grid.points(
  x=adjx(xy$x + rnorm(1000)/6),
  draw=FALSE,
  y=adjy(xy$y + rnorm(1000)/6),
  pch=21,
  gp=grid::gpar(col=xy$border,
    fill=xy$color, cex=0.6),
  vp=attr(v_mammo4p, "viewport"),
  default.units="native")
# draw inside the viewport
grid::pushViewport(vp)
grid::grid.draw(pts)
grid::popViewport()

```

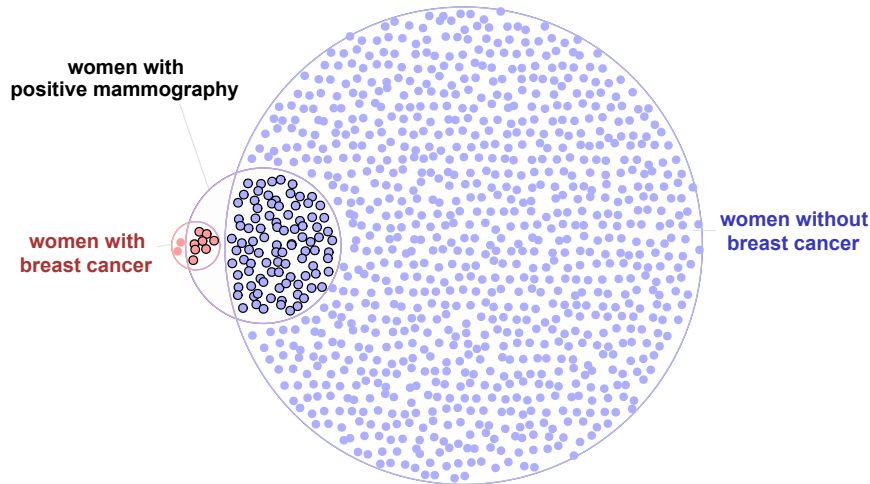



Figure 5.5. Third attempt at re-creating the eulerGlyphs figure.

Tips:

- The `Venndir` object must be plotted in order to define the viewport, since it depends upon the `expand_fraction` adjustments.
- The viewport must be pushed before drawing points.

The result turned out better than expected, and the workflow could be re-used for other datasets.

Can it be improved?

While creating the figure, the first question that arose:

”How many points are in each region?”

In truth, it took some effort to discover these values, despite being the focal point of the study (and the study about the study).

Figure 5.6 shows some potential improvement, labeling each section with the number of points.

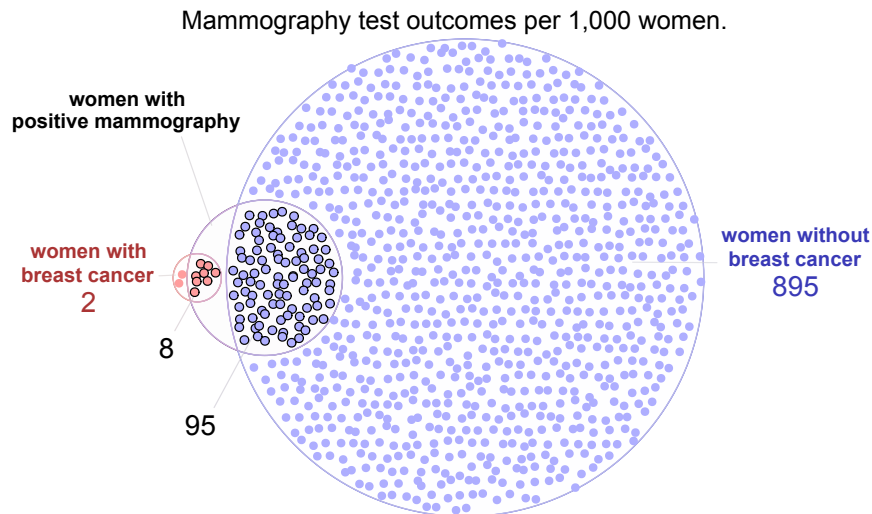


Figure 5.6. Update which labels the number of points in each region. The title reveals each point represents one woman tested per 1,000 in the study.

5.2 Venndir Case Studies

While many stylistic techniques were described earlier in the text, putting them together in an artistic way is not always straightforward.

The following case studies serve two purposes:

1. Illustrate some stylistic options while showing the associated techniques.
2. Demonstrate the utility of directional Venn diagrams.

In both cases, it often works best to use real world data.

5.2.1 Seehawer Kmt2 Gene Venns

Context

Many genomics studies compare results across experimental conditions using Venn or Euler diagrams—for instance, to identify genes affected by a perturbation such as a treatment, toxin exposure, or disease onset. These studies often aim to uncover the genetic basis of a perturbation, in hopes of revealing ways to mitigate or prevent its effects.

There are thousands of studies that identify “genes affected”, and many of them have associated direction of change. For example, “up” may be recognized as “up-regulated” or “increased function”, and “down” may be recognized as “down-regulated” or “decreased function.”

When two studies are compared, they often focus on the genes involved without regard to the direction of change. A common alternative is to compare only the concordant gene changes: For example, comparing up-regulated genes in two studies, and separately comparing down-regulated genes in two studies.

The underlying question is that the genes in each set represent the biological mechanism involved in the study, and that shared genes may therefore represent shared biology.

If the genes represent the process, surely the direction of change represents the effect on the process? Venndir proposes the important distinction that the concordance of direction represents whether the underlying process is also affected in the same way.

Direction is important in science and medicine, where it often means the difference between disease and treatment.

1. Overlap alone is not enough.
2. Directionality matters.

To date, no Venn software tool indicates overlap and directionality together.

Example Data

A scientific paper by Seehawer et al⁴ (Seehawer et al., 2024) studied brain cancer metastasis, the migration property of cancer cells associated with poor clinical prognosis.

They studied the effect of losing either of two genes, *Kmt2c* and *Kmt2d*, and they included two very different cell types: "168FARN" and "67NR".

Figure 5.7 shows the target figure. Each Venn diagram compares the genes affected in two cell types: "168FARN" and "67NR".

- The left panels focus on *Kmt2c*, the right panels focus on *Kmt2d*.
- The top panels compare only up-regulated genes, the bottom panels compare only down-regulated genes.

The authors concluded that the two cell lines indicated a shared genetic mechanism, even in very different cell types, due to the concordant gene changes shown in each Venn diagram. However, these figures did not indicate discordant changes.

Data for Extended Figure 4h were available via Supplemental Table 4, filtered for adjusted P-value below 0.1 and absolute log2 fold change of at least 0.6. The gene symbol and fold change sign were used in Venndir.

Figure 5.8 shows the Venndir re-creation of Seehawer Extended Figure 4h, with separate up- and down-regulated Venn diagrams (Seehawer et al., 2024). The Venndir output is nearly identical, differing only by 1 gene each in the 67NR-specific overlap - most likely a discrepancy by rounding error.

⁴<https://www.nature.com/articles/s41556-024-01446-3>

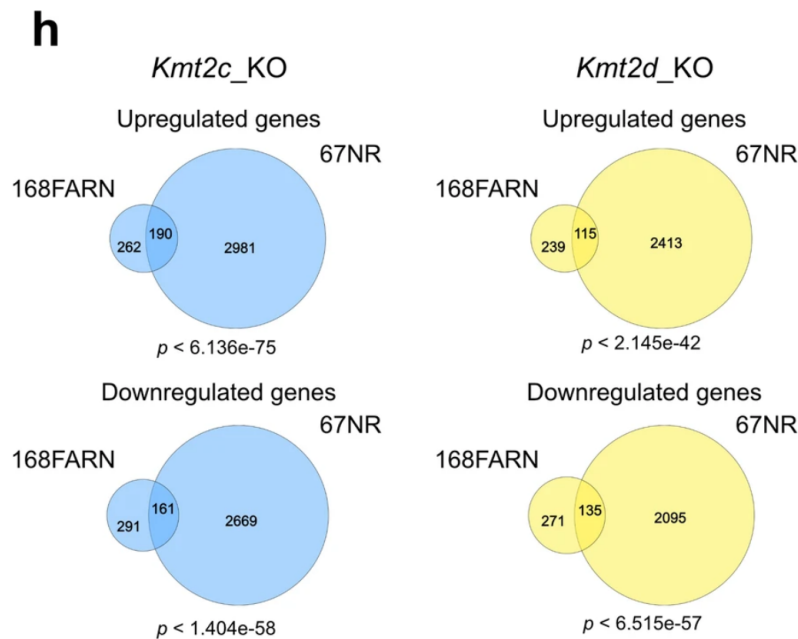


Figure 5.7. Target Extended Figure 4h from Seehawer *et al* 2024.

However, the *next question* is whether the shared gene changes are concordant or discordant in direction.

Figure 5.9 includes up- and down-regulated genes together, with *Kmt2c_KO* (left) and *Kmt2d_KO* (right).

Findings:

The first row of panels shows regulated genes without regard to direction.

- For *Kmt2c_KO* (left) and *Kmt2d_KO* (right), the shared genes represent a higher proportion of the Venn diagrams than in Seehawer.
- This increase in shared genes suggests that the increase is caused by discordant changes.

The second row summarizes the overlap direction by '**agreement**', described in **Overlap Type**, using '=' for agreement, and 'X' for disagreement.

- Nearly as many shared genes disagree in direction as those that agree.

The bottom row summarizes the overlap by '**concordance**', using arrows to indicate direction, and 'X' for discordant changes.

Conclusion:

Directionality provides important and context which was previously missing in

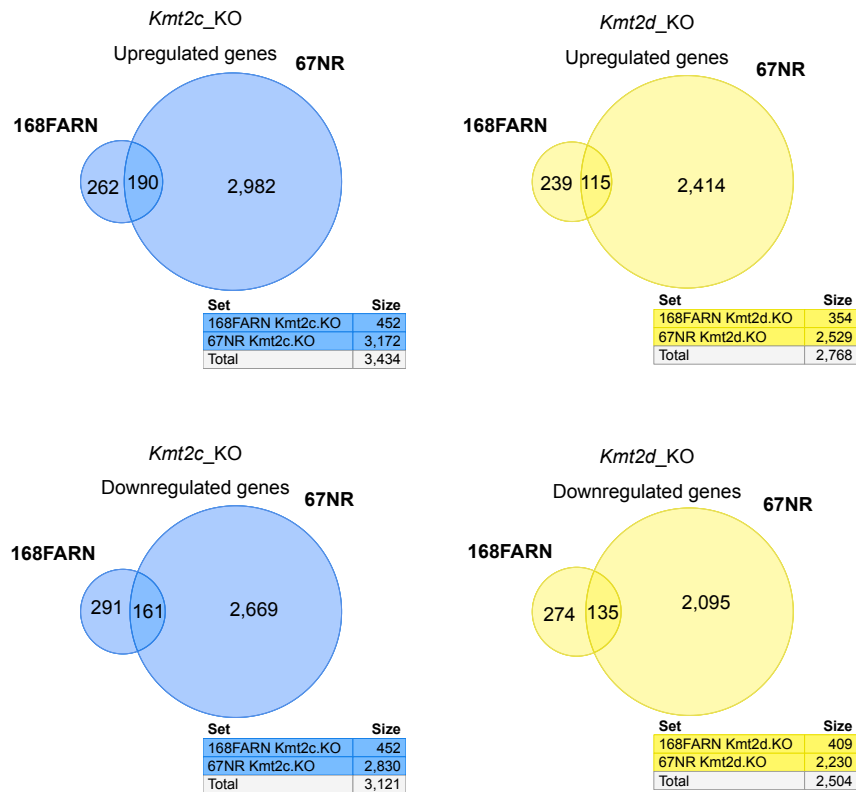


Figure 5.8. Venndir re-creation of Seehawer Ext. Fig 4h.

the Seehawer data (Seehawer et al., 2024). While it is true the two cell types shared some proportion of gene changes, the directional concordance was not particularly strong. That said, the "true answer" is also not known, the subset of concordant genes may represent the shared components, or maybe the discordant genes could play different functional roles in the two cell types. In any case, the concordance adds important context to the question and interpretation of supporting data.

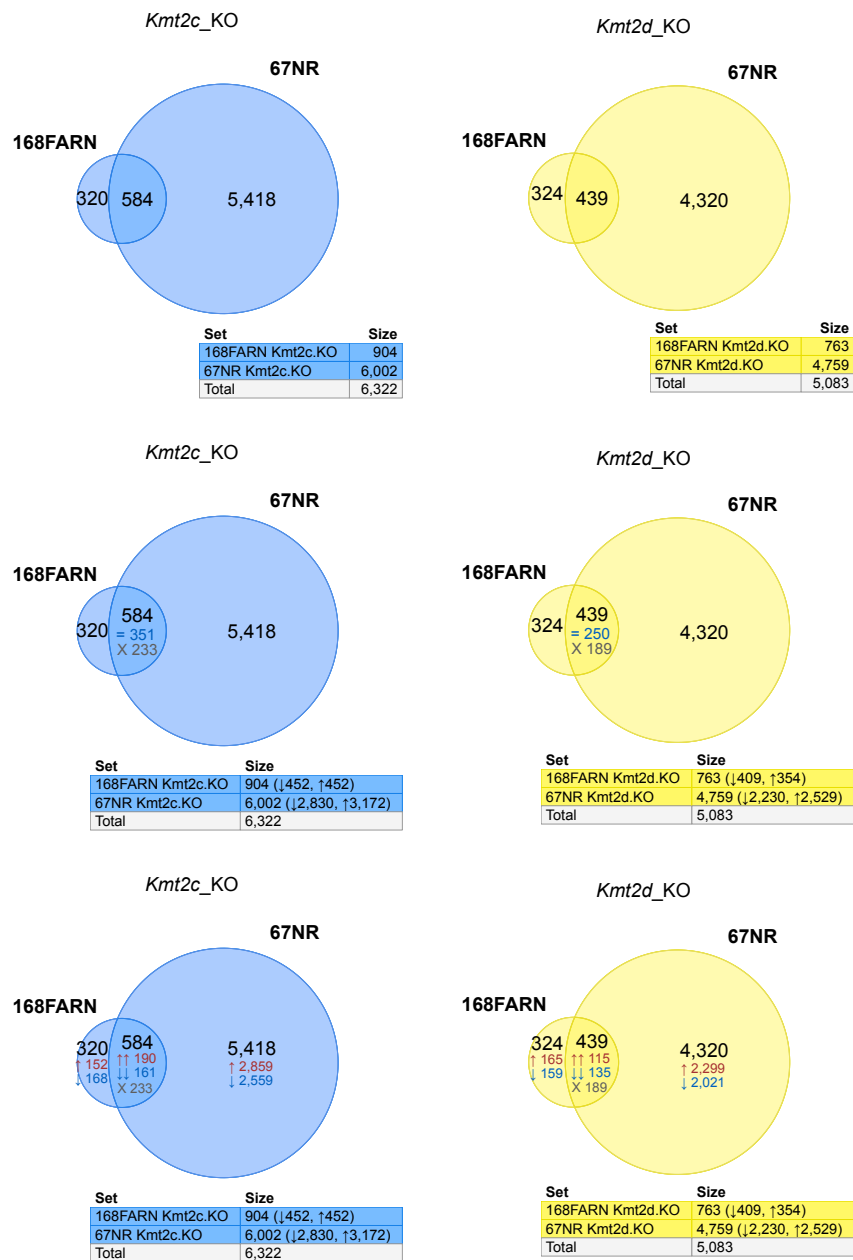


Figure 5.9. Venndir re-creation of Seehawer Ext. Fig 4h, using the complete set of up- and down-regulated genes together. The top row shows 'overlap', the second row shows 'agreement', and the third row shows 'concordance'.

Chapter 6

What's New by Version

6.0.1 v0.3.0 | 13 July 2025

- Edits to Preface, Introduction.

v0.2.0 | 06 July 2025 | venndir v0.0.58.900

New chapters and sections:

- Inserted Chapter 6: [What's New by Version](#)
- Inserted Chapter 7: [Glossary](#)
- Section 3.7.1: [Text Venn](#)
- Section 3.7: [Figure Scale and Position](#)
- Section 4.12.4: [Custom Sign Label Colors](#)
- Section 5.2.1: [Seehawer Kmt2 Gene Venns](#)

v0.1.0 | 26 June 2025 | venndir v0.0.56.900

- Initial release, six chapters:
 1. What is a Venn Diagram?
 2. Data Import
 3. Venndir Basics
 4. Deeper Customizations
 5. Venndir Gallery
 6. References

Chapter 7

Glossary

R A programming language and environment for statistical computing, data analysis, and graphical representation. R is widely used in scientific research due to its extensive ecosystem of custom packages with emphasis on reproducibility.

R package A bundled collection of R functions, data, documentation, and metadata that extends the base functionality of R. Packages can be installed from CRAN, Bioconductor, or GitHub, and are essential for tasks like data modeling, visualization, and scientific analysis workflows in bioinformatics.

Venn diagram A graphical representation of all possible logical relationships between a collection of sets. Each set is shown as a circle, and all possible intersections are depicted—even if some are empty. Venn diagrams are especially useful for illustrating complete overlap structures but can become cluttered with more than 3–4 sets.

Euler diagram A simplified set diagram that shows only the **actual** (non-empty) relationships between sets. Unlike Venn diagrams, Euler diagrams omit intersections that don't exist, making them more concise and readable when dealing with sparse or realistic data overlap.

intersection In set theory, the intersection of two or more sets refers to the elements shared by each set. In Venndir visualizations, the intersection contains the overlapping region shared between two or more sets.

union In set theory, the union of two or more sets contains the unique set of elements present in *any* of the sets. In Venndir visualization, the union represents the combined geometric region covering any sets involved.

subtraction In set theory, subtraction is also known as set difference, which yields the elements present in one set and *not* in another. For example, A

- **B** includes elements in set **A** that are absent from **B**.

In *venndir* visualization, the subtraction is used to represent a geometric region from one set that does not overlap any region of another set.

buffer region In geometry, a buffer region is zone created by expanding or shrinking a geometric shape by a fixed distance.

For expansion, it defines a region with fixed distance from any solid point in the geometric shape. It is used in *Venndir* to create "outer border", and to place text labels outside the Venn or Euler diagrams.

For shrinking, it defines a region with fixed distance inside the border of the geometric shape. It is used in *Venndir* to create "inner border", and to array multiple text labels inside specific regions.

agreement Property of two or more elements that all agree in direction or sign.

concordance Property of two or more elements that agree in direction or sign.

concordant Description for two or more elements that agree in direction or sign.

disagreement Property of two or more elements that do not all agree in direction or sign.

discordance Property of two or more elements that do not agree in direction or sign, equivalent to the term discordance.

discordant Description for two or more elements that do not agree in direction or sign.

vector In R, a one-dimensional sequence of elements of the same base type (e.g., *character* or *numeric*). In *Venndir*, a **vector** is used to represent elements of a *set*, sometimes also with directional sign.

set A collection of unique elements. In *Venndir*, a set represents a group of items where the items are typically genes, features, or observations that share a common label or condition. For example, the items may be differentially expressed genes (DEGs) from a statistical contrast.

In *Venndir*, a set is stored as a *vector* in one of two ways: 1. a *character* vector of items without vector names (*set*), 2. a *numeric* vector of signs, often but not strictly -1 and 1, where item names are stored as vector names (*signed set*).

signed data A dataset in which each element carries a direction label (e.g., "up" or "down", "positive" or "negative", "gain" or "loss"). This signature guides *Venndir* in assigning directional sign-aware overlaps and patterns.

Signed data may be used to create a **signed set** in Venndir.

signed set A specialized type of **set** used in Venndir to store elements with directional sign. Signs are stored as **vector** values, and items are stored as vector names.

Signed sets are required components in a **signed setlist**.

list In R, a **list** is an ordered collection of objects that may contain elements of **different types and structures**. A **list** is commonly used to store a collection of different objects, and may even contain other lists.

In Venndir, a **setlist** is a named **list** with multiple **sets**. The **list** names store the name of each set.

setlist In Venndir, a named **list** of **sets**, where set names are stored as **list** names. where each set is stored as a **vector**. The set names are stored in **list** names.

signed setlist A specialized form of **setlist** in which each element is a **signed set**. Each set therefore represents items as **vector** names, where values contain the directional sign, and names store the items.

Venn overlap A Venn overlap refers to a specific set theory comparison such as **intersection** and/or **subtraction**. Venn overlaps are defined when two or more **sets** are compared.

A property of a Venn overlap is that each item may be assigned to only one such Venn overlap.

For example, consider Venn overlaps which involve three sets 'A', 'B', 'C'. An item may be only present or absent in each of the three sets. Therefore, if an item exists in 'A&B' it is present in 'A' and 'B' and *not* in 'C'.

In Venn overlap terminology 'A' refers to the set of item present in 'A' that are not present in 'B' nor in 'C'.

main counts In Venndir, main counts refers to the number of items in a **Venn overlap**, without regard to directional sign. The items in a Venn overlap may separately be tabulated by sign to produce **signed counts**.

signed counts In Venndir, signed counts refers to the number of items in a **Venn overlap** tabulated by additional directional criteria. For example, overlap type 'each' will tabulate items based upon each combination of signs observed in the Venn overlap. For a Venn overlap of the intersection of 'A' and 'B', with sign values -1 and 1, there may be four tabulated values:

- '1 1'
- '1 -1'
- '-1 1'

- '1 1'

hidden overlaps A **Venn overlap** which cannot be represented in a Venn or proportional Euler diagram, usually due to geometric constraints. For three or more sets, proportional Euler diagrams are not guaranteed to have a geometric solution when using circle or ellipse shapes.

In Venndir, hidden overlaps are indicated by placing a footnote symbol †, also called a 'dagger', in the bottom-right corner of the figure.

matrix A type of data object in **R** called a **matrix** that is used to store two-dimensional data by row and column. A **matrix** in **R** can only store one data type in each cell.

incidence matrix A **matrix** encoding set membership. Rows represent elements, columns represent sets. The **matrix** values are **1** if the element is present, and either **0** or **NA** otherwise.

In Venndir, any non-empty value indicates the element (row) is present in the set (column), where "" empty string is considered an empty value.

signed incidence matrix An enhanced **incidence matrix** that represents set membership and directional sign. A signed incidence matrix encodes sign as the matrix value, usually values such as:

- 1 for positive or "up" direction,
- -1 for negative or "down" direction, and
- 0 or NA for absence of the element.

That said, Venndir will recognize any non-empty value as a sign, empty values include 0, NA, and ''.

data visualization Graphical representation of data intended to provide insight, pattern recognition, and communication. In Venndir, data visualization helps to summarize set relationships such as overlaps, directionality, and concordance of direction.

markdown A plain text style of writing that encodes special formatting by following a lightweight markup language.

Venndir supports the CommonMark specification, which is summarized by the CommonMark.org Summary¹. The main features used by Venndir:

- Font changes: **bold**, *italics*
- inline images
- lists (like this one)

¹<https://commonmark.org/help/>

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